

Rubinstein–Sarnak densities for weighted prime number races

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Abstract

For $m \in \mathbb{R}$ consider the weighted prime race $D_m(x) = \sum_{p \leq x, p \equiv 3 \pmod{4}} p^m - \sum_{p \leq x, p \equiv 1 \pmod{4}} p^m$ and its analogues for other moduli. The regime $m \leq 0$ has a developed literature: the critical $\frac{1}{2} \log \log x$ deflection at $m = -\frac{1}{2}$ (Aoki–Koyama; Sheth), the sign-change threshold (Shimada–Koyama), and, for $m < -\frac{1}{2}$, limits expressible through Meissel–Mertens constants for progressions (Languasco–Zaccagnini). We contribute the pieces this literature does not contain: (i) the first numerical Rubinstein–Sarnak logarithmic densities for a weighted race between primes in arithmetic progressions (for weighted prime-sum races against main terms, cf. Lamzouri’s Mertens-race density [13]), computed across the delicate range under the standard GRH + LI hypotheses by deterministic Fourier inversion — e.g. for modulus 4, $\delta(1) = 0.79763$, $\delta(2) = 0.69452$, $\delta(3) = 0.64585$, with the calibration $\delta(0) = 0.995928$ recovering the classical Rubinstein–Sarnak value to six decimals; (ii) the growing-weight regime $m > 0$ (we are not aware of prior Rubinstein–Sarnak density computations or weight-aspect density asymptotics for growing weights p^m , $m > 0$), with exact sieve computations to 10^{10} agreeing with the predicted densities to $\lesssim 4 \times 10^{-3}$ for modulus 4; (iii) an asymptotic dissolution law in the weight aspect, $\delta_q(m) - \frac{1}{2} \sim (2\pi \cdot 2(m + \frac{1}{2}) \log \frac{q(m+1/2)}{2\pi})^{-1/2}$, an analogue of the Fiorilli–Martin density asymptotics, proved here with an error term for the density of the limiting distribution of any real primitive character under GRH and $L(\frac{1}{2}, \chi) \neq 0$ — under LI(χ) it applies to the logarithmic density of the race itself for every such χ — and verified numerically in moduli 3 and 4. We complement the computations with an interpolation theorem, proved by the framework of Akbary–Ng–Shahabi and the method of Humphries and stated for an arbitrary real primitive character $\chi \pmod{q}$: under GRH(χ), LI(χ) and $L(\frac{1}{2}, \chi) \neq 0$ (the nonvanishing is unconditional for $q = 3, 4$), for every $m \in (-\frac{1}{2}, \infty)$ the logarithmic density $\delta_\chi(m)$ of the set of x where the weighted nonresidue count leads (for $\chi = \chi_4$, where the class 3 (mod 4) leads) exists and lies in $(\frac{1}{2}, 1)$, the map $m \mapsto \delta_\chi(m)$ is continuous, and $1 - \delta_\chi(m) \leq \exp(-1/(2C(\chi)(2m+1)^2))$ with $C(\chi) = \sum_{\gamma > 0} 2\gamma^{-2}$ evaluated in closed form ($C(\chi_4) = 0.1560296\dots$, $C(\chi_3) = 0.1134021\dots$), so $\delta_\chi(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$. Two further theorems complete the picture: the dissolution asymptotic holds *uniformly* over all real primitive characters and all $M = m + \frac{1}{2} \geq \delta_0$ (any fixed $\delta_0 > 0$, with effective constants; only the corner $m \rightarrow -\frac{1}{2}$ jointly with $q \rightarrow \infty$ is left open), and $m \mapsto \delta_\chi(m)$ is *strictly decreasing* for $m \geq 16.82$ ($q = 4$; 18.62 for $q = 3$), so beyond these thresholds Chebyshev’s bias dissolves monotonically. The constants $C(\chi)$, every variance σ_m^2 , the

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positivity of the central L -values, and the completeness of the zero lists underlying the numerics are machine-certified in ball arithmetic. This exhibits the classical density formulation ($m = 0$) and the weighted formulation of Aoki–Koyama ($m = -\frac{1}{2}$) as the two ends of one continuous, quantitatively controlled family — a conditional structural connection between, though not a logical inclusion between, the two definitions compared in Problem 12 of the comparative prime number theory problem list of Hamieh, Kadiri, Martin and Ng, which remains open.

1 Introduction

Chebyshev’s observation — that primes $\equiv 3 \pmod{4}$ predominate over primes $\equiv 1 \pmod{4}$ for most x — was quantified by Rubinstein and Sarnak [1]: under GRH and the linear independence (LI) of the ordinates of zeros of $L(s, \chi_4)$, the set $\{x : \pi(x; 4, 3) > \pi(x; 4, 1)\}$ has logarithmic density $0.9959\dots$. A separate line of work, initiated by Aoki and Koyama [2] and developed by Sheth [3] and Shimada–Koyama [4], studies *weighted* counts $\sum_{p \leq x, p \equiv a \pmod{q}} p^{-s}$ and locates a transition at $s = \frac{1}{2}$: for the mod-4 race the weighted difference obeys $\sum_{p \leq x} \chi_4(p) p^{-1/2} = -\frac{1}{2} \log \log x + O(1)$ (under DRH in [2], under GRH off a small exceptional set in [3]), so the weighted race is single-signed for large x , while for weights p^{-s} with $s < \frac{1}{2}$ sign changes persist [4]; Hayani [14] has recently obtained the density-one statement at $s = \frac{1}{2}$ under GRH alone, without linear independence. Humphries [5] established the analogous phase structure for weighted sums of the Liouville function, including the limit $\delta_\alpha \rightarrow 1$ as $\alpha \rightarrow \frac{1}{2}^-$. Problem 12 of the comparative prime number theory problem list [6] (posed by Koyama) asks how the weighted formulation relates to the classical logarithmic-density formulation.

Related work. The weighted-race literature just cited concerns constant or decaying weights, and recent activity stays on that side of the boundary. In the absolutely convergent regime (weights $p^{-\sigma}$, $\sigma > 1$; $m = -\sigma < -1$ in our normalization) Yamamuro [19], following Ay-mone [16], proves unconditional two-sided bounds on the tails $\sum_{n > x} \Lambda(n) \chi(n) n^{-\sigma} / \log n$ through infinitely divisible distributions of Riemann-zeta type — no GRH, no LI, and no logarithmic densities enter. Hayani [20] generalizes the Aoki–Koyama threshold $\frac{1}{2}$ to higher roots in prime *ideal* races, for weights $(N\mathfrak{p})^{-\alpha}$ with $\alpha \geq 0$, under GRH and partly unconditionally — asymptotic and sign statements, not Rubinstein–Sarnak densities, and only nonincreasing weights. For unweighted races, Bailleul, Hayani and Untrau [21] prove a quantitative Kronecker–Weyl theorem in the 1-Wasserstein metric under GRH and an effective linear-independence hypothesis, with rates of convergence to the limiting distributions and conditional bounds for generalized Skewes numbers; an effective, Wasserstein-quantified version of the interpolation theorem below is a natural question we do not pursue. Martin and Ng [23] obtain inclusivity of classical races under GRH plus the existence of sufficiently many self-sufficient zeros, a hypothesis substantially weaker than full LI; it is plausible that the qualitative parts of Theorem 1 (existence of the limiting distribution and $\frac{1}{2} < \delta(m) < 1$) survive a weakening of LI in their direction, but we do not prove this. Finally, Leung [22] proves, under RH and a quantitative form of LI, multivariate Gaussian limit laws with weak negative correlations for smoothed counts of primes in multiple short intervals, with short-interval race applications; the weights there are smoothings in the interval aspect, not powers p^m , and the contact point with this note is the outlook at the end of Section 7. In all of this literature the weight is constant or decaying; we are not aware of prior Rubinstein–Sarnak density computations, or weight-aspect density asymptotics, for growing weights p^m , $m > 0$.

This note treats the two formulations as endpoints of one family — a viewpoint anticipated in the s -parametrized theorems of Shimada–Koyama [4] and in Aymone’s races with a general weight exponent [16]. For $m \in \mathbb{R}$ set

$$D_m(x) = \sum_{\substack{p \leq x \\ p \equiv 3 \pmod{4}}} p^m - \sum_{\substack{p \leq x \\ p \equiv 1 \pmod{4}}} p^m,$$

so $m = 0$ is the classical race and $m = -s$ the Aoki–Koyama weighting; we allow $m > 0$; the related work reviewed above contains, to our knowledge, no prior treatment of the race with growing weights. Apart from the interpolation theorem of Section 2, our results are computational, at the standard of numerical work in this area (GRH + LI where densities are involved). Every headline number was computed twice: the sieve counts by fully independent implementations, the densities by deterministic inversion against Monte Carlo — two evaluations that share the same zero data and tail model and are independent only in the evaluation step; the shared tail model is controlled by the validated enclosures of Section 7 (Table 5: exact tail variance and explicit analytic error bounds, with quadrature error estimated rather than certified).

Results

1. Densities across the delicate range. For $m > -\frac{1}{2}$ the limiting logarithmic distribution of the normalized race exists by the framework of Akbary–Ng–Shahabi [7] (cf. Devin [8]) applied to the shifted explicit formula; the normalized race has the almost-periodic form $1 + \sum_{\gamma > 0} 2a_\gamma(m) \cos(\gamma u)$ with

$$a_\gamma(m) = \frac{2m + 1}{\sqrt{(m + \frac{1}{2})^2 + \gamma^2}},$$

γ the ordinates of $L(s, \chi_4)$, and the constant 1 normalizing the prime-square bias term. Deterministic Gil–Pelaez inversion with validated error bounds, cross-checked by Monte Carlo under LI (Section 7), gives Table 1. These are, to our knowledge, the first Rubinstein–Sarnak densities computed for a weighted race between primes in arithmetic progressions (for weighted prime sums against main terms, densities exist in the Mertens-race literature [13]; for analytic explicit-formula control of the log-weighted sum $\theta(x)$, cf. Platt–Trudgian [15]).

2. Exact computations, $m > 0$. By segmented sieve with exact 128-bit signed accumulators to $X = 10^{10}$: the mod-4 race first favours the residue class at $p = 5$ for every $m \geq 1$ (against Leech’s 26,861 for $m = 0$); lead changes to 10^{10} number 3,082 ($m = 0$), 125,624, 159,942, 199,540 ($m = 1, 2, 3$); the finite- x logarithmic measures at $X = 10^{10}$ — here and throughout, the finite- x logarithmic measure of the leading set is $(\log(X/3))^{-1} \int_3^X \mathbf{1}[D_m(t) > 0] \frac{dt}{t}$ for modulus 4 and $(\log(X/2))^{-1} \int_2^X \mathbf{1}[D_m(t) > 0] \frac{dt}{t}$ for modulus 3, the race being tracked from its first participating prime (3, resp. 2; conventions in Section 7) — are 0.79380, 0.69275, 0.64900 for $m = 1, 2, 3$, within $\lesssim 4 \times 10^{-3}$ of Table 1. For modulus 3 the weighted first flips occur at $p = 13$ ($m = 1$) and $p = 7$ ($m = 2, 3$), while the classical race, as expected, does not flip below 10^{10} .

3. Dissolution law. Substituting the zero-counting density $\frac{1}{2\pi} \log \frac{qt}{2\pi}$ into the variance of the limiting distribution gives

$$\mathrm{Var}_q(m) \sim 2\left(m + \frac{1}{2}\right) \log \frac{q(m + \frac{1}{2})}{2\pi}, \quad \delta_q(m) - \frac{1}{2} \sim \frac{1}{\sqrt{2\pi \mathrm{Var}_q(m)}} \quad (m \rightarrow \infty),$$

m	0	1	2	3	8	20	60	100
$\delta_4(m)$	0.995928	0.797630	0.694521	0.645851	0.571870	0.538455	—	0.513778
$\delta_3(m)$	0.999063	0.836880 [†]	0.723473	0.666554	0.578536	0.540768	0.519717 [†]	—

Table 1: Logarithmic densities of the set where the nonresidue class leads, moduli 4 and 3: midpoints of the validated enclosures of Table 5 (Section 7), rounded to six decimals; dashes mark uncomputed points. The enclosure widths are below 3×10^{-7} , so every sixth decimal is determined except the two marked [†], whose intervals straddle the rounding boundary: at $(3, 1)$ the value is 0.836880 or 0.836881, at $(3, 60)$ it is 0.519717 or 0.519718. The underlying computation is deterministic Gil–Pelaez Fourier inversion of the limiting distribution (511 resp. 537 zeros computed to 25 digits, unseen zeros Gaussianized with the exact tail variance of Remark 5), cross-checked by independent Monte Carlo with $2\text{--}4 \times 10^6$ samples per entry (binomial standard error $< 3.6 \times 10^{-4}$; largest deviation 2.2 standard errors over the 14 entries, consistent with binomial sampling), and at $(q, m) = (4, 1)$ by a single 2×10^7 -sample run, 0.797599 ± 0.000090 , consistent with the table entry (0.3 standard errors). The $m = 0$ column reproduces the classical values 0.995928 and 0.999063 (cf. [1, 9]) to all six decimals.

a weight-aspect analogue of the Fiorilli–Martin density asymptotics [9] (cf. Meng [10] for the k -aspect), now a theorem: Section 4 proves it with error term $O(1/(M \log M))$ and a second-order refinement for the density of the limiting variable of any real primitive character under GRH and $L(\frac{1}{2}, \chi) \neq 0$, and under LI(χ) for the logarithmic density of the race itself, by Theorem 1(ii) (Corollary 3 records the case $q = 4$). At $q = 4$, $m = 100$: deterministic 0.513778, Gaussian $\Phi(1/\sigma)$ 0.51379, the law 0.51380. At $q = 3$, $m = 60$: deterministic 0.519717[†] (dagger as in Table 1) against the law’s 0.51978 — agreement to 6×10^{-5} . The modulus ordering $\delta_3(m) > \delta_4(m)$ holds at every computed m . In the law it is a conductor effect: $\text{Var}_4(m) - \text{Var}_3(m) \sim 2(m + \frac{1}{2}) \log \frac{4}{3}$, and from the zero data this conductor term accounts for the entire variance gap at $m = 8, 20, 60$ (100–101%). The larger first zero $\gamma_1(\chi_3) = 8.04 > \gamma_1(\chi_4) = 6.02$ pushes in the same direction but is not the mechanism of the law: it contributes 23% of the gap at $m = 8$ and 0.2% at $m = 60$; only at small m ($0 \leq m \leq 3$, where the first zero carries 24–36% of the total variance) is it a comparable driver of the ordering.

4. The boundary $m \leq -\frac{1}{2}$, numerically. Confirming (not discovering) the theorems cited above: with $C^* = \frac{1}{2}(M - \log 2 - \frac{1}{2}) + K_3 - \log L(\frac{1}{2}, \chi_4) = -0.07153$ (M the Meissel–Mertens constant, K_3 the $k \geq 3$ prime-power contribution), the parameter-free prediction $D_{-1/2}(x) = \frac{1}{2} \log \log x + C^* + \varepsilon(x)$ gives 1.4968 ± 0.0172 at $x = 10^{10}$ against the exact value 1.50775; the fitted slope over $[10^6, 10^{10}]$ is 0.494. The fluctuation ε decays like $\sqrt{\Sigma}/\log x$ with $\Sigma = \sum_{\gamma} 2/\gamma^2 = 0.1560296$, visible decade-by-decade in the residuals — a small refinement of the qualitative picture. For $m < -\frac{1}{2}$ the limits are identified with differences of Meissel–Mertens constants for progressions: $\lim D_{-1} = M(4, 3) - M(4, 1) = 0.334981325299993\dots$, our Möbius-over-log L evaluation agreeing with Languasco–Zaccagnini [11] to all compared digits (the difference itself is OEIS A086239; the individual constants are A368645 and A368646). Notably, none of the races $m \in \{-1, -\frac{3}{4}, -\frac{1}{2}\}$ changes sign even once on $[3, 10^{10}]$ (minimum margin $1/\sqrt{3} - 1/\sqrt{5} \approx 0.13014$, attained by the $m = -\frac{1}{2}$ race at $x = 5$; the $m = -1$ margin there is $2/15$): in this range of weights Chebyshev’s bias is not merely predominant but, empirically, total from the first odd prime.

5. An interpolation theorem. Section 2 (Theorem 1) proves, for an arbitrary real

primitive character $\chi \bmod q$ under $\text{GRH}(\chi)$, $\text{LI}(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$ (the nonvanishing is unconditional for $q = 3, 4$): for every $m \in (-\frac{1}{2}, \infty)$ the normalized race has a limiting logarithmic distribution, the density $\delta_\chi(m)$ exists and lies in $(\frac{1}{2}, 1)$, the map $m \mapsto \delta_\chi(m)$ is continuous, and

$$1 - \delta_\chi(m) \leq \min \left\{ C(\chi) (2m + 1)^2, e^{-1/(2C(\chi)(2m+1)^2)} \right\}, \quad C(\chi) = \sum_{\gamma > 0} \frac{2}{\gamma^2},$$

with $C(\chi_4) = 0.1560296\dots$ and $C(\chi_3) = 0.1134021\dots$ (both evaluated in closed form), so $\delta_\chi(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$ — a weight-aspect instance of the highly biased phenomenon of Fiorilli [12] — matching, at $q = 4$, the single-sign regime established at $m = -\frac{1}{2}$ by [2, 3, 14]. (The theorem is stated for the log-weighted race; Proposition 1 transfers every statement verbatim to the unlogged race D_m .) The existence statement is an application of the Akbary–Ng–Shahabi framework [7], and the limit $\delta(m) \rightarrow 1$ is the arithmetic-progression analogue of Humphries’s theorem for the weighted Liouville function [5, Thm. 1.5], by a transposed form of his method; what we believe is new is the prime-race statement itself, the continuity of δ in the weight, and the explicit rate with a computable constant.

6. Uniformity and monotonicity. Section 5 proves the dissolution asymptotic uniformly over all real primitive characters and all $M = m + \frac{1}{2} \geq \delta_0$ (any fixed $\delta_0 > 0$, effective constants; the corner $m \rightarrow -\frac{1}{2}$ jointly with $q \rightarrow \infty$ is genuinely different and left open), placing the Fiorilli–Martin modulus aspect and the weight aspect of this paper inside one statement. Section 6 proves $\delta'_\chi(m) < 0$ for $m \geq 16.82$ ($q = 4$) resp. 18.62 ($q = 3$): beyond these thresholds Chebyshev’s bias dissolves *monotonically*; on the compact remainder monotonicity remains Conjecture 1, supported by the validated 30-point grid.

2 An interpolation theorem

Throughout this section χ is a *fixed* real primitive Dirichlet character mod $q \geq 3$, of parity $\chi(-1) = (-1)^a$ with $a \in \{0, 1\}$; the two moduli computed in this paper are χ_4 (the nontrivial character mod 4; $a = 1$) and χ_3 (the quadratic character mod 3; $a = 1$), and they serve as running examples. $\rho = \beta + i\gamma$ runs over the nontrivial zeros of $L(s, \chi)$ (listed with multiplicity), $\text{GRH}(\chi)$ denotes the Riemann hypothesis for $L(s, \chi)$, and $\text{LI}(\chi)$ the linear independence over $\mathbb{Q}$ of the multiset of positive ordinates γ (LI in particular forces all γ to be distinct, i.e. the zeros to be simple; we do *not* build the nonvanishing of ordinates into the LI convention). Alongside $\text{GRH}(\chi)$ we assume throughout the no-central-zero hypothesis

$$\text{NCZ}(\chi) : \quad L(\tfrac{1}{2}, \chi) \neq 0,$$

which under GRH forces every ordinate to be nonzero (a zero with $\gamma = 0$ would sit at $s = \frac{1}{2}$), so that the least positive ordinate $\gamma_1(\chi)$ is a strictly positive number — possibly small for large q ; the theorem is for fixed χ , and all implied constants in this section may depend on χ (equivalently on q) without further mention. For $q = 3, 4$ the hypothesis NCZ is unconditional, by grouped alternating series: for χ_4 , $L(\frac{1}{2}, \chi_4) = \sum_{k \geq 0} (-1)^k (2k + 1)^{-1/2} > 0$ by the alternating series test, while for χ_3 the Dirichlet series is not alternating (the vanishing terms $\chi_3(3k) = 0$ interleave), but grouping it in blocks of three gives $L(\frac{1}{2}, \chi_3) = \sum_{k \geq 0} [(3k + 1)^{-1/2} - (3k + 2)^{-1/2}] \geq 1 - 2^{-1/2} > 0$ (justification and numerics in Section 7: $L(\frac{1}{2}, \chi_4) = 0.667691457\dots$, $L(\frac{1}{2}, \chi_3) = 0.4808675\dots$). For general real primitive χ , NCZ is central nonvanishing — Chowla’s conjecture, known in every numerically checked case but open

in general; under GRH, $L(\frac{1}{2}, \chi) \geq 0$, so NCZ is then exactly $L(\frac{1}{2}, \chi) > 0$ (see the discussion in Section 7). For $m \in (-\frac{1}{2}, \infty)$ define the (θ -form) weighted race

$$R_{m,\chi}(x) = - \sum_{p \leq x} \chi(p) p^m \log p = \sum_{\substack{p \leq x \\ \chi(p)=-1}} p^m \log p - \sum_{\substack{p \leq x \\ \chi(p)=+1}} p^m \log p$$

(nonresidues minus residues; for $\chi = \chi_4$ this is the race of $p \equiv 3$ against $p \equiv 1 \pmod{4}$, for χ_3 of $p \equiv 2$ against $p \equiv 1 \pmod{3}$), and its normalization

$$E_{m,\chi}(x) = (2m+1) x^{-(m+1/2)} R_{m,\chi}(x).$$

Since χ is fixed we abbreviate $R_m = R_{m,\chi}$ and $E_m = E_{m,\chi}$. Write

$$a_\gamma(m) = \frac{2m+1}{\sqrt{(m+\frac{1}{2})^2 + \gamma^2}}, \quad X_m = 1 + \sum_{\gamma > 0} 2a_\gamma(m) \cos \theta_\gamma,$$

with θ_γ i.i.d. uniform on $[0, 2\pi)$; the series converges a.s. and in L^2 since $\sum_\gamma a_\gamma(m)^2 < \infty$ (the ordinates have density $\ll_q \log \gamma$). (X_m is the $X_{m,\chi}$ of Section 4.) Set

$$C(\chi) := \sum_{\gamma > 0} \frac{2}{\gamma^2} < \infty.$$

Under $\text{GRH}(\chi)$ and $\text{NCZ}(\chi)$ (in force throughout this section) this constant has a closed evaluation: the completed function $\Lambda(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$ is entire and satisfies $\Lambda(s, \chi) = \Lambda(1-s, \chi)$ — every real primitive character has root number $+1$, by Gauss's evaluation of the sign of the quadratic Gauss sum, $\tau(\chi) = \sqrt{q}$ for $a = 0$ and $i\sqrt{q}$ for $a = 1$ ([17, Chs. 2, 9]; see the proof of Proposition 3) — so $\Xi(z) := \Lambda(\frac{1}{2} + z, \chi)$ is even; since $\Xi(0) \neq 0$ (by NCZ; under GRH in fact $\Xi(0) > 0$), $\sum_\gamma \gamma^{-2} < \infty$, and, under GRH, the zeros of Ξ are exactly $\pm i\gamma$, the Hadamard factorization takes the genus-zero form $\Xi(z)/\Xi(0) = \prod_{\gamma > 0} (1 + z^2/\gamma^2)$, exactly as for the Riemann ξ -function [17, §12] (the argument is written out, at $q = 4$, in Proposition 2, and applies verbatim). Hence

$$C(\chi) = (\log \Lambda)''(\tfrac{1}{2}, \chi) = \tfrac{1}{4} \psi'(\tfrac{2a+1}{4}) + \frac{d^2}{ds^2} \log L(s, \chi) \Big|_{s=1/2}.$$

Example ($q = 4$). For $\chi = \chi_4$ ($a = 1$; $\gamma_1(\chi_4) = 6.0209\dots$),

$$C(\chi_4) = \tfrac{1}{4} \psi'(\tfrac{3}{4}) + \frac{d^2}{ds^2} \log L(s, \chi_4) \Big|_{s=1/2} = 0.15602964988964488\dots,$$

with $\psi'(\frac{3}{4}) = \pi^2 - 8G$ (G Catalan's constant), evaluated at 40-digit working precision (Corollary 2). Cross-check from the zero data: the first 511 ordinates give 0.152546 and the Riemann–von Mangoldt tail $\int_{\gamma_{511}}^\infty 2t^{-2} dN(t)$ with $dN(t) = (2\pi)^{-1} \log(2t/\pi) dt$ gives 0.003487, total 0.156033, agreeing with the exact value to 3.4×10^{-6} — within the expected $O(10^{-5})$ fluctuation of the zero-counting remainder. (Earlier drafts printed the cruder value 0.156033.)

Example ($q = 3$). For $\chi = \chi_3$ ($a = 1$; $\gamma_1(\chi_3) = 8.0397\dots$, from the 537 ordinates computed to 25 digits in Section 7),

$$C(\chi_3) = \tfrac{1}{4} \psi'(\tfrac{3}{4}) + \frac{d^2}{ds^2} \log L(s, \chi_3) \Big|_{s=1/2} = 0.11340215685275857\dots,$$

the second derivative being $(\log L)''(\frac{1}{2}, \chi_3) = -0.522067755\dots$; the value is computed by the same closed-form route as $C(\chi_4)$ and is recorded among the certified constants of Section 7. Cross-check from the zero data: the 537 ordinates give 0.110308 and the tail $\int_{\gamma_{537}}^{\infty} 2t^{-2} dN(t)$ with $dN(t) = (2\pi)^{-1} \log(3t/(2\pi)) dt$ gives 0.003098, total 0.113406, agreeing with the closed form to 4.1×10^{-6} .

Theorem 1. *Let χ be a real primitive character mod q . Assume $GRH(\chi)$ and $NCZ(\chi)$: $L(\frac{1}{2}, \chi) \neq 0$ (the latter unconditional for $q = 3, 4$, as noted above).*

- (i) *For each $m \in (-\frac{1}{2}, \infty)$ the function $u \mapsto E_m(e^u)$ possesses a limiting distribution ν_m : for every bounded continuous $f : \mathbb{R} \rightarrow \mathbb{R}$, $\lim_{Y \rightarrow \infty} \frac{1}{Y} \int_0^Y f(E_m(e^u)) du = \int_{\mathbb{R}} f d\nu_m$. Under $LI(\chi)$, ν_m is the law of X_m .*
- (ii) *Under $LI(\chi)$, ν_m is absolutely continuous with support $\mathbb{R}$; the logarithmic density $\delta_\chi(m)$ of the set $\{x \geq 2 : R_m(x) > 0\}$ exists, equals $\nu_m((0, \infty))$, satisfies $\frac{1}{2} < \delta_\chi(m) < 1$ for every $m \in (-\frac{1}{2}, \infty)$, and $m \mapsto \delta_\chi(m)$ is continuous on $(-\frac{1}{2}, \infty)$. (We write $\delta(m)$ for $\delta_\chi(m)$ when χ is understood; for $\chi = \chi_4, \chi_3$ these are the densities printed as δ_4, δ_3 in Table 1, and under $LI(\chi)$ they coincide with the quantities $\delta_q(m) = \mathbb{P}(X_{m,\chi} > 0)$ of Section 4.)*
- (iii) *Under $LI(\chi)$, for all $m \in (-\frac{1}{2}, \infty)$,*

$$1 - \delta_\chi(m) \leq C(\chi) (2m + 1)^2 \quad \text{and} \quad 1 - \delta_\chi(m) \leq \exp\left(-\frac{1}{2C(\chi) (2m + 1)^2}\right),$$

with no restriction on m ; the second bound is in fact the stronger of the two for every m . In particular $\delta_\chi(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$. Both bounds are informative only near the left endpoint: the (stronger) exponential bound drops below the trivial estimate $1 - \delta_\chi(m) < \frac{1}{2}$ of part (ii) precisely for $m < m_0(\chi) := \frac{1}{2}((2C(\chi) \log 2)^{-1/2} - 1)$, and the first bound is ≥ 1 for $m \geq \frac{1}{2}(C(\chi)^{-1/2} - 1)$; for $m \geq m_0(\chi)$ part (iii) asserts nothing beyond part (ii), and this theorem makes no claim about the rate at which $\delta_\chi(m)$ approaches $\frac{1}{2}$ as $m \rightarrow \infty$ (for that rate, see Section 4).

- (iv) *($\chi = \chi_4$.) Consequently the classical density formulation of Chebyshev's bias at $m = 0$ (logarithmic density $\delta(0) = 0.995928\dots$; 0.9959 in [1], six decimals as in [9] and re-derived here) is joined by the continuous family $\{\delta(m)\}_{m \in (-1/2, \infty)}$, with $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$, to the total-bias regime at the weighted threshold $m = -\frac{1}{2}$, where the unlogged race $D_{-1/2}(x) = \sum_{p \leq x} (-\chi_4(p)) p^{-1/2}$ equals $\frac{1}{2} \log \log x + O(1)$ and is eventually of one sign (under DRH [2]; under GRH off a set of finite logarithmic measure [3]; single-signed on a set of logarithmic density one under GRH alone [14]). None of these three endpoint results assumes LI . In the opposite direction the family now covers every positive weight; there the computed densities of Table 1 decrease toward $\frac{1}{2}$, in the manner of the dissolution law of the Results section, which this theorem does not prove (it is proved, with a rate, in Section 4).*

Example ($q = 4$). χ_4 is odd, $NCZ(\chi_4)$ is the alternating-series positivity $L(\frac{1}{2}, \chi_4) > 0$, and $C(\chi_4) = 0.15602964988964488\dots$ (in closed form above). The thresholds of (iii) are $m_0(\chi_4) = 0.57507\dots$ and $\frac{1}{2}(C(\chi_4)^{-1/2} - 1) = 0.76580\dots$, the Chebyshev bound falling below $\frac{1}{2}$ only for

$m < 0.395058\dots$; parts (ii) and (iv) recover the classical anchor $\delta(0) = 0.995928\dots$ of [1, 9] at $m = 0$ (Table 1).

Example ($q = 3$). The race R_{m,χ_3} pits $p \equiv 2$ against $p \equiv 1 \pmod{3}$; χ_3 is odd, $\text{NCZ}(\chi_3)$ is the grouped-series positivity $L(\frac{1}{2}, \chi_3) > 0$, and $C(\chi_3) = 0.11340215685275857\dots$ (Example above). Under $\text{GRH}(\chi_3)$ and $\text{LI}(\chi_3)$, parts (i)–(iii) give existence, continuity and the strict bounds for $\delta_3(m)$ on all of $(-\frac{1}{2}, \infty)$; the computed anchors are $\delta_3(0) = 0.999063$ (the classical mod-3 bias, re-derived to six decimals in Table 1) and, from the sieves of the introduction, the weighted first flips at $p = 13$ ($m = 1$) and $p = 7$ ($m = 2, 3$), the classical race $m = 0$ not flipping below 10^{10} .

Corollary 1 (mod-3 rate). *Assume $\text{GRH}(\chi_3)$ and $\text{LI}(\chi_3)$ ($\text{NCZ}(\chi_3)$ holds unconditionally). Then the logarithmic density $\delta_3(m) = \delta_{\chi_3}(m)$ of the set $\{x \geq 2 : R_{m,\chi_3}(x) > 0\}$ satisfies, for every $m \in (-\frac{1}{2}, \infty)$,*

$$1 - \delta_3(m) \leq \exp\left(-\frac{1}{2C(\chi_3)(2m+1)^2}\right), \quad C(\chi_3) = 0.11340215685275857\dots,$$

and likewise $1 - \delta_3(m) \leq C(\chi_3)(2m+1)^2$; in particular $\delta_3(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$. The exponential bound is informative (below $\frac{1}{2}$) precisely for $m < m_0(\chi_3) = 0.761048\dots$, the Chebyshev form for $m < 0.549891\dots$; at $m = 0$ the exponential bound gives $1 - \delta_3(0) \leq e^{-1/(2C(\chi_3))} = 0.01216\dots$, against the computed value 9.37×10^{-4} (Table 1).

Proof. Theorem 1(iii) with $\chi = \chi_3$; the numerical thresholds are evaluated from $C(\chi_3)$ as in the Example above. $\square$

Remark 1. *Part (iii) is an arithmetic-progression analogue of Humphries's theorem that $\delta(P_\alpha) \rightarrow 1$ as $\alpha \rightarrow \frac{1}{2}^-$ for weighted sums of the Liouville function [5, Thm. 1.5]. The mechanism is transposed rather than copied: in [5] the mean of the limiting distribution diverges while the variance stays bounded, whereas here the normalized mean is fixed at 1 while the variance collapses like $C(\chi)(2m+1)^2$; in both cases the bias-to-spread ratio diverges, and Humphries already remarks (after his Cor. 7.6) that Chebyshev's inequality suffices for the limit. We claim novelty only for the statement in the prime-race setting, the continuity assertion in (ii), and the explicit constant in (iii). Part (i) does not require LI; it is an application of the framework of Akbary–Ng–Shahabi [7] (cf. Devin [8]).*

We isolate the two number-theoretic inputs as lemmas.

Lemma 1 (truncated explicit formula, all truncations). *Let χ be a real primitive character mod q , of parity $\chi(-1) = (-1)^a$, and let $\rho = \beta + i\gamma$ run over the nontrivial zeros of $L(s, \chi)$, listed with multiplicity. Unconditionally, for all $x \geq 2$ and all $T \geq 2$,*

$$\psi(x, \chi) = - \sum_{|\gamma| \leq T} \frac{x^\rho}{\rho} + O_q\left(\frac{x \log^2(xT)}{T} + \log x\right). \quad (1)$$

No inequality between T and x is assumed; in particular $T > x$ is allowed. For even χ ($a = 0$) the trivial zero of $L(s, \chi)$ at $s = 0$ contributes a genuine $-\log x + O_q(1)$ term, absorbed by the $O_q(\log x)$ floor (cf. [17, Ch. 19]); this is the only point at which the parity alters the shape of the formula.

Proof. All implied constants depend at most on q (χ is fixed). We use five standard facts; each is a statement about a single point s or a single height t and contains no parameter x , so no hidden comparison of T with x can enter through them.

(F1) (*Perron kernel*). For $y > 0$, $c > 0$, $T \geq 2$ put $I(y, T) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} y^s s^{-1} ds$ and $\delta(y) = 1, \frac{1}{2}, 0$ according as $y > 1$, $y = 1$, $y < 1$. Then

$$|I(y, T) - \delta(y)| \leq y^c \min\left(1, \frac{1}{\pi T |\log y|}\right) \quad (y \neq 1), \quad \left|I(1, T) - \frac{1}{2}\right| \leq \frac{c}{\pi T}.$$

Proof of (F1). For $y < 1$ close the contour to the right: on the rectangle with right edge at $\operatorname{Re} s = A$ the integrand $y^s s^{-1}$ is analytic, and the right edge is $\ll y^A/A \rightarrow 0$ as $A \rightarrow \infty$; hence I equals minus the two horizontal rays from $c \pm iT$ to $+\infty \pm iT$, each of modulus $\leq \frac{1}{2\pi T} \int_c^\infty y^\sigma d\sigma = \frac{y^c}{2\pi T \log(1/y)}$. For $y > 1$ close to the left, picking up the residue 1 at $s = 0$; the rays to $-\infty \pm iT$ give the same bound with $\log y$. For the $\min(1, \cdot)$ alternative, replace the vertical segment by the arc of the circle $|s| = \sqrt{c^2 + T^2}$ through $c \pm iT$ on the side not containing 0 (for $y > 1$, the side containing 0, picking up the residue 1): on the arc $|y^s| \leq y^c$ and $|s| = \sqrt{c^2 + T^2}$, so the arc integral is $\leq y^c \cdot (\text{length})/(2\pi|s|) \leq y^c$. For $y = 1$ the odd part of the integrand integrates to zero and $I(1, T) = \frac{1}{\pi} \arctan(T/c) = \frac{1}{2} - \frac{1}{\pi} \arctan(c/T)$, with $\arctan(c/T) \leq c/T$. Nothing here involves any second parameter.

(F2) $-\zeta'/\zeta(\sigma) \leq (\sigma - 1)^{-1} + O(1)$ for $1 < \sigma \leq 3$ (simple pole of ζ at 1; [17, §17]).

(F3) $N(t+1, \chi) - N(t, \chi) \ll_q \log(t+2)$ for $t \geq 0$, where $N(t, \chi)$ counts zeros with $0 < \gamma \leq t$; unconditional Riemann–von Mangoldt for $L(s, \chi)$ [17, §16]. Zeros are symmetric under $\gamma \mapsto -\gamma$ (χ is real).

(F4) For $-1 \leq \sigma \leq 2$, $s = \sigma + it$, $\frac{L'}{L}(s, \chi) = \sum_{\rho: |\gamma - t| \leq 1} \frac{1}{s - \rho} + O(\log(|t| + 2))$ [17, §16]; and for $2 \leq \sigma \leq 3$, $|L'/L(s, \chi)| \leq \sum_n \Lambda(n) n^{-2} = O(1)$.

(F5) $\Lambda(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$ satisfies $\Lambda(s, \chi) = \Lambda(1-s, \chi)$: the root number of a real primitive character is +1 ([17, Chs. 2, 9]; Gauss's evaluation $\tau(\chi) = \sqrt{q}$ for $a = 0$, $i\sqrt{q}$ for $a = 1$ — for $q = 4$ the two-term sum gives $\tau(\chi_4)/(i\sqrt{4}) = 2i/2i = +1$). Taking logarithmic derivatives,

$$\frac{L'}{L}(s, \chi) = -\log \frac{q}{\pi} - \frac{1}{2} \psi\left(\frac{s+a}{2}\right) - \frac{1}{2} \psi\left(\frac{1-s+a}{2}\right) - \frac{L'}{L}(1-s, \chi).$$

If $\sigma \leq -1$ and either $|t| \geq 2$ or $\sigma + a$ is an odd (negative) integer, then: $\operatorname{Re}(1-s) \geq 2$, so $|L'/L(1-s, \chi)| = O(1)$ by (F4); $\operatorname{Re}(\frac{1-s+a}{2}) \geq 1$, so $|\psi(\frac{1-s+a}{2})| \ll \log(2 + |s|)$ by Stirling; and for $w = \frac{s+a}{2}$ the reflection $\psi(w) = \psi(1-w) - \pi \cot(\pi w)$ applies with $\operatorname{Re}(1-w) \geq 1$ and with $|\cot(\pi w)| \leq \coth(\pi)$ when $|\operatorname{Im} w| = |t|/2 \geq 1$, respectively $|\cot(\pi w)| = |\tanh(\pi t/2)| \leq 1$ when $\operatorname{Re} w$ is a half-odd-integer (which is the case when $\sigma + a$ is an odd integer). Hence on those sets $|L'/L(s, \chi)| \ll_q \log(2 + |s|)$.

The behaviour at $s = 0$ depends on the parity. For odd χ ($a = 1$), $L(0, \chi) \neq 0$: evaluating $\Lambda(0, \chi) = \Lambda(1, \chi)$ from (F5), $(q/\pi)^{1/2} \Gamma(\frac{1}{2}) L(0, \chi) = (q/\pi) \Gamma(1) L(1, \chi)$ with $L(1, \chi) \neq 0$ (Dirichlet); for χ_4 , $L(1, \chi_4) = \pi/4$ (Leibniz) gives the explicit value $L(0, \chi_4) = (4/\pi)^{1/2} (\pi/4) / \sqrt{\pi} = \frac{1}{2}$. Consequently, for odd χ , L'/L is regular at $s = 0$. For even χ ($a = 0$), $s = 0$ is a *simple* trivial zero of $L(s, \chi)$: $\Gamma(s/2)$ has a simple pole there while $\Lambda(0, \chi) = \Lambda(1, \chi) \neq 0$ is finite, so

$$\frac{L'}{L}(s, \chi) = \frac{1}{s} + b(\chi) + O(|s|) \quad (s \rightarrow 0)$$

for a finite constant $b(\chi) = O_\chi(1)$ (cf. [17, Ch. 19]). Moreover every nontrivial zero has $0 < \beta < 1$: there are no zeros with $\beta = 1$ (Dirichlet, de la Vallée Poussin, unconditional), hence, by (F5), none with $\beta = 0$ apart from the trivial zero at $s = 0$ in the even case, which is accounted for separately below.

Step 1 (Perron). Let $c = 1 + 1/\log x \in (1, 1 + 1/\log 2]$, so $x^c = ex$. On $\operatorname{Re} s = c$ the Dirichlet series $-L'/L(s, \chi) = \sum_n \Lambda(n) \chi(n) n^{-s}$ converges absolutely, so term-by-term application of (F1) on the finite segment gives

$$\psi(x, \chi) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \left(-\frac{L'}{L}(s, \chi) \right) \frac{x^s}{s} ds + O(R_1), \quad R_1 = \sum_{n \geq 2} \Lambda(n) \left(\frac{x}{n} \right)^c \min \left(1, \frac{1}{T |\log(x/n)|} \right) + \log x,$$

the final $\log x$ covering the possible term $n = x$ (where (F1) gives $\frac{1}{2}$ instead of the weight 1 used in ψ , a discrepancy $\leq \Lambda(x) \ll \log x$). Split R_1 : for $n \leq \frac{3}{4}x$ or $n \geq \frac{5}{4}x$ one has $|\log(x/n)| \geq \log \frac{5}{4}$, so these terms contribute $\ll T^{-1} x^c \sum_n \Lambda(n) n^{-c} = T^{-1} x^c (-\zeta'/\zeta(c)) \ll (x/T) \log x$ by (F2). For $\frac{3}{4}x < n < \frac{5}{4}x$, $n \neq x$: here $\Lambda(n) \ll \log x$, $(x/n)^c \ll 1$, and $|\log(x/n)| \geq |n - x|/(2x)$, so these terms contribute

$$\ll \log x \sum_{1 \leq k \leq x} \min \left(1, \frac{2x}{Tk} \right) \ll \log x \left(1 + \frac{x}{T} + \frac{x}{T} \log x \right) \ll \log x + \frac{x \log^2 x}{T},$$

where k runs over the possible values of $\lceil |n - x| \rceil$ (the at most two integers nearest x are covered by the $\min(\cdot) \leq 1$ bound). Hence $R_1 \ll x \log^2 x / T + \log x$, for every $T \geq 2$; when $T > x$ these bounds only become smaller.

Step 2 (choice of height T'). Let $G = \{\gamma : T - 1 < \gamma < T + 2\}$; by (F3), $\#G \leq A_1 \log(T + 2)$ for some $A_1 = A_1(q)$. With $r = 1/(4A_1 \log(T + 2))$ the union of the intervals $(\gamma - r, \gamma + r)$, $\gamma \in G$, has measure $\leq \frac{1}{2}$, so there exists $T' \in [T, T + 1]$ with $|T' - \gamma| \geq r$ for all $\gamma \in G$; for $\gamma \notin G$, $|T' - \gamma| \geq 1$ automatically. Thus $|T' - \gamma| \gg 1/\log(T + 2)$ for *every* ordinate γ , and by the $\gamma \mapsto -\gamma$ symmetry the same holds at height $-T'$.

Step 3 (contour). For $K \in \mathbb{N}$ set $U = 2K + 1 - a$ (so the line $\sigma = -U$ passes midway between consecutive trivial zeros in either parity) and integrate $(-L'/L)(s, \chi) x^s/s$ counterclockwise around the rectangle with vertices $c \pm iT'$, $-U \pm iT'$. The poles inside are: the nontrivial zeros with $|\gamma| \leq T'$ (residue $-x^\rho/\rho$ each, with multiplicity; $\rho \neq 0$ since $0 < \beta < 1$); the point $s = 0$; and the trivial zeros $s = -(2j + a)$ of $L(s, \chi)$ with $j \geq 1 - a$ and $2j + a \leq U$ (simple, from the poles of $\Gamma(\frac{s+a}{2})$; residue $+x^{-(2j+a)}/(2j + a)$ each). At $s = 0$: for odd χ the integrand has only the simple pole of $1/s$, with residue $-L'/L(0, \chi) = O_q(1)$ (using $L(0, \chi) \neq 0$); for even χ the pole of $1/s$ meets the simple trivial zero of L , and the Laurent expansion of (F5) gives a double pole with residue

$$\operatorname{Res}_{s=0} \left(-\frac{L'}{L}(s, \chi) \right) \frac{x^s}{s} = -\log x - b(\chi) = O_q(\log x).$$

Writing $r_0(x, \chi)$ for the residue at $s = 0$ just computed (in either parity $r_0(x, \chi) = O_q(\log x)$),

$$\frac{1}{2\pi i} \oint = - \sum_{|\gamma| \leq T'} \frac{x^\rho}{\rho} + r_0(x, \chi) + \sum_{\substack{j \geq 1-a \\ 2j+a \leq U}} \frac{x^{-(2j+a)}}{2j+a}.$$

Step 4 (horizontal segments, $-1 \leq \sigma \leq c$). On $s = \sigma \pm iT'$ with $-1 \leq \sigma \leq 2$, (F4) and Step 2 give $|L'/L(s, \chi)| \leq \sum_{|\gamma \mp T'| \leq 1} |T' - \gamma|^{-1} + O(\log T') \ll \log^2(T + 2) (\ll \log(T + 2) \text{ terms, each})$

$\ll \log(T+2)$); for $2 \leq \sigma \leq c$ (nonempty only if $x < e$) the $O(1)$ bound of (F4) applies. Hence these pieces contribute

$$\ll \frac{\log^2(T+2)}{T} \int_{-1}^c x^\sigma d\sigma \ll \frac{\log^2(T+2)}{T} \cdot \frac{x^c}{\log x} \ll \frac{x \log^2(xT)}{T}.$$

Step 5 (horizontal segments $-U \leq \sigma \leq -1$, and vertical segment $\sigma = -U$). By (F5), $|L'/L| \ll \log(2+|s|)$ on both. The horizontal parts contribute

$$\ll \frac{1}{T} \int_1^\infty x^{-v} \log(3+v+T) dv \ll \frac{1}{T} \left(\frac{\log(3+T)}{x \log x} + \frac{1}{x} \right) \ll \frac{\log(3+T)}{T} \ll 1,$$

uniformly in U , absorbed in $O(\log x)$. The vertical segment has length $2T'$, $|x^s/s| \leq x^{-U}/U$, so it contributes $\ll T' x^{-U} \log(U+T'+2)/U \rightarrow 0$ as $K \rightarrow \infty$ (here $x \geq 2 > 1$).

Step 6 (assemble; $U \rightarrow \infty$). Letting $K \rightarrow \infty$, the trivial-zero sum increases to $\sum_{j \geq 0} x^{-(2j+1)}/(2j+1) = \operatorname{artanh}(1/x) \leq \operatorname{artanh}(\frac{1}{2}) < 0.55$ for odd χ , and to $\sum_{j \geq 1} x^{-2j}/(2j) = -\frac{1}{2} \log(1-x^{-2}) \leq \frac{1}{2} \log \frac{4}{3} < 0.15$ for even χ (both for $x \geq 2$); and $r_0(x, \chi) = O_q(\log x)$ in either parity — for even χ the floor $O_q(\log x)$ is genuinely attained, by the $-\log x$ of Step 3. Combining Steps 1–5,

$$\psi(x, \chi) = - \sum_{|\gamma| \leq T'} \frac{x^\rho}{\rho} + O_q \left(\frac{x \log^2(xT)}{T} + \log x \right).$$

Step 7 ($T' \rightarrow T$). The zeros with $T < |\gamma| \leq T' \leq T+1$ number $\ll \log(T+2)$ by (F3), and each satisfies $|x^\rho/\rho| \leq x^\beta/|\gamma| \leq x/T$; restoring the summation range to $|\gamma| \leq T$ therefore costs $\ll x \log(T+2)/T \ll x \log^2(xT)/T$. This proves (1). Every estimate above is uniform in $T \geq 2$, and no step compared T with x . $\square$

Lemma 2 (shifted truncated explicit formula). *Assume $\operatorname{GRH}(\chi)$ and $\operatorname{NCZ}(\chi)$, and let $m \in (-\frac{1}{2}, \infty)$ be fixed. For $\psi_m(x) := \sum_{n \leq x} \Lambda(n) \chi(n) n^m$ and all $x \geq 2$, $T \geq 2$ (no inequality between T and x is assumed; Lemma 1),*

$$\psi_m(x) = - \sum_{|\gamma| \leq T} \frac{x^{\rho+m}}{\rho+m} + O_{m,\chi} \left(\frac{x^{m+1} \log^2(xT)}{T} + (1+x^m) \log x \right). \quad (2)$$

For $m \leq 0$ the floor $(1+x^m) \log x \leq 2 \log x$ recovers the earlier form of this lemma.

Proof. Since $\psi(t, \chi) = 0$ for $t < 2$ (when $2 \nmid q$ the term $n = 2$ is present, and the Stieltjes lower limit 2^- captures its jump), Riemann–Stieltjes partial summation gives exactly

$$\psi_m(x) = \int_{2^-}^x t^m d\psi(t, \chi) = x^m \psi(x, \chi) - m \int_2^x t^{m-1} \psi(t, \chi) dt.$$

Insert (1) (Lemma 1, valid for every $T \geq 2$), with the same T , at x and at every $t \in [2, x]$. The zero terms produce, for each ρ with $|\gamma| \leq T$,

$$x^m \frac{x^\rho}{\rho} - m \int_2^x t^{m-1} \frac{t^\rho}{\rho} dt = \frac{x^{\rho+m}}{\rho+m} + \frac{m 2^{\rho+m}}{\rho(\rho+m)},$$

an identity (integrate $t^{\rho+m-1}$; note $\operatorname{Re}(\rho+m) = m + \frac{1}{2} > 0$ for every $m > -\frac{1}{2}$, and $|\rho+m| \geq |\gamma| \geq \gamma_1(\chi) > 0$ — every ordinate is nonzero because under GRH a zero with $\gamma = 0$ would sit

at $s = \frac{1}{2}$, excluded by $\text{NCZ}(\chi)$ — so no denominator degenerates at any m in the range; the constants below depend on χ through $\gamma_1(\chi)$ and $\sum_\gamma \gamma^{-2}$. The endpoint terms are *summable*: since $|\rho| \geq |\gamma|$ and $|\rho + m| \geq |\gamma|$,

$$\sum_{|\gamma| \leq T} \left| \frac{m 2^{\rho+m}}{\rho(\rho+m)} \right| \leq |m| 2^{m+1/2} \sum_{|\gamma| \leq T} \frac{1}{\gamma^2} \ll_{m,\chi} 1,$$

uniformly in T . (Bounding each term by $O_\gamma(1)$ alone would not suffice, as there are $\asymp T \log T$ of them.) The error terms of (1) contribute

$$x^m \cdot O\left(\frac{x \log^2(xT)}{T} + \log x\right) + |m| \int_2^x t^{m-1} O\left(\frac{t \log^2(tT)}{T} + \log t\right) dt.$$

The truncation parts are $\ll_m x^{m+1} \log^2(xT)/T$, using $\log^2(tT) \leq \log^2(xT)$ and $\int_2^x t^m dt \leq x^{m+1}/(m+1)$ for $m > -1$. The log-floor parts split into three cases:

$$|m| \int_2^x t^{m-1} \log t dt \ll_m \begin{cases} x^m \log x, & m > 0 \quad (\leq \log x \cdot \int_2^x m t^{m-1} dt \leq x^m \log x), \\ 0, & m = 0 \text{ (the factor } m \text{ vanishes)}, \\ 1, & m < 0 \quad (\int_2^\infty t^{m-1} \log t dt = O_m(1)). \end{cases}$$

Since $\log x \geq \log 2$, every $O_m(1)$ above (including the endpoint sum) is $\ll_m \log x$, so all non-zero-sum contributions are $O_m(x^{m+1} \log^2(xT)/T + (1 + x^m) \log x)$, proving (2). Neither half of the floor can be dropped on the full range: for $m < 0$ the quantity $x^m \log x$ tends to 0 while genuinely $O_m(1)$ terms (the endpoint sum, the constants inside (1)) persist, so the summand 1 is needed; for $m > 0$ the term $x^m \log x$ arises from $m \int_2^x t^{m-1} O(\log t) dt$, and removing it would require sign cancellation in the $O(\log t)$ errors of (1) that we do not know. After the normalization by $(2m+1)x^{-(m+1/2)}$, the floor becomes, with $x = e^u$,

$$(1 + e^{mu}) u e^{-(m+1/2)u} = u(e^{-(m+1/2)u} + e^{-u/2}) \leq 2u e^{-\kappa_m u}, \quad \kappa_m := \min(m + \tfrac{1}{2}, \tfrac{1}{2}) > 0,$$

an $L^2(0, \infty)$ function of u for every $m > -\frac{1}{2}$; the mean-square hypothesis (3) used in the proof of (i) is therefore unaffected by the extension. $\square$

Lemma 3 (prime-power separation). *Fix $m \in (-\frac{1}{2}, \infty)$. For $x \geq 4$,*

$$-R_m(x) = \psi_m(x) - \sum_{p \leq \sqrt{x}} p^{2m} \log p + O_{q,m}(x^{m+1/3} \log x + \log^2 x),$$

and

$$\sum_{p \leq \sqrt{x}} p^{2m} \log p = \frac{x^{m+1/2}}{2m+1} + O_m(x^{m+1/2} e^{-c\sqrt{\log x}})$$

with $c > 0$ absolute and the implied constants depending on m . Both error terms are $o(x^{m+1/2})$. The $\log^2 x$ term originates solely in the prime powers p^k , $k \geq 3$, with $km \leq -1$; such k exist only for $m < 0$, and for $m > -\frac{1}{3}$ the term is absorbed by $O_m(x^{m+1/3} \log x)$, so it is genuinely present only for $m \in (-\frac{1}{2}, -\frac{1}{3}]$.

Proof. ψ_m splits over prime powers $p^k \leq x$. The $k = 1$ part is $\sum_{p \leq x} \chi(p) p^m \log p = -R_m(x)$. The $k = 2$ part is

$$\sum_{p^2 \leq x} \chi(p)^2 p^{2m} \log p = \sum_{\substack{p \leq \sqrt{x} \\ p \nmid q}} p^{2m} \log p = \sum_{p \leq \sqrt{x}} p^{2m} \log p + O_{q,m}(1),$$

since $\chi(p)^2 = 1$ for $p \nmid q$ and $\chi(p) = 0$ for $p \mid q$: the subtracted terms are the at most $\omega(q)$ primes dividing q , each contributing $p^{2m} \log p \leq (1 + q^{2m}) \log q$, a constant depending only on q and m (for $q = 4$ this is the single term $p = 2$, which is absent from both sides).

The $k \geq 3$ terms. Here $3 \leq k \leq \log x / \log 2$ and $y := x^{1/k}$. Let A be Chebyshev's absolute constant in $\theta(y) \leq Ay$.

If $m \geq 0$ then $km \geq 0 > -1$ for every k — no case split occurs — and monotonicity gives, uniformly in k ,

$$\sum_{p \leq y} p^{km} \log p \leq y^{km} \theta(y) \leq Ay^{km+1} = Ax^{(km+1)/k} = Ax^{m+1/k} \leq Ax^{m+1/3};$$

summing over the $\leq \log x / \log 2$ values of k yields $O(x^{m+1/3} \log x)$ with an absolute constant.

If $m < 0$, split on the sign of $km + 1$. For the (fewer than $1/|m|$, hence $O_m(1)$ many) k with $-1 < km < 0$, partial summation with $\theta(t) \leq At$ gives

$$\sum_{p \leq y} p^{km} \log p = y^{km} \theta(y) + |km| \int_2^y t^{km-1} \theta(t) dt \leq Ay^{km+1} \left(1 + \frac{|km|}{km+1}\right) \ll_m x^{m+1/k} \leq x^{m+1/3},$$

the constant depending on m through the least positive value of $km + 1$ over the finitely many admissible k ; the total over these k is $O_m(x^{m+1/3})$. For the k with $km \leq -1$ (these exist iff $m < 0$, namely $k \geq 1/|m|$), we have $p^{km} \log p \leq p^{-1} \log p$, so by Mertens

$$\sum_{p \leq y} p^{km} \log p \leq \sum_{p \leq y} \frac{\log p}{p} = \log y + O(1) \ll \log x,$$

uniformly in k ; over $\leq \log x / \log 2$ values of k the total is $O(\log^2 x)$. Combining the cases, the $k \geq 3$ contribution is $O_m(x^{m+1/3} \log x + \log^2 x)$ for every $m \in (-\frac{1}{2}, \infty)$, and the provenance stated in the lemma is exactly the case analysis just made; for $m > -\frac{1}{3}$ one has $\log x \ll_m x^{m+1/3}$, so there the $\log^2 x$ term may be dropped — we keep it so that a single statement covers the whole range. Both error terms are $o(x^{m+1/2})$ uniformly: $x^{m+1/3} \log x / x^{m+1/2} = x^{-1/6} \log x \rightarrow 0$ and $\log^2 x \cdot x^{-(m+1/2)} \rightarrow 0$ since $m + \frac{1}{2} > 0$. (The crude bound “ $O(x^{1/3})$ prime powers times $\log x$ ” is *not* sufficient here: $x^{1/3} \geq x^{m+1/2}$ for $m \leq -\frac{1}{6}$.)

The squares sum. The argument is unchanged from the $(-\frac{1}{2}, 0]$ version, and we record that its only structural input is $2m + 1 > 0$, so it holds on the whole range. Write $\theta(y) = y + E(y)$, $E(y) \ll ye^{-2c\sqrt{\log y}}$ (prime number theorem with the classical de la Vallée Poussin error). Partial summation gives, with $y = \sqrt{x}$,

$$\sum_{p \leq y} p^{2m} \log p = y^{2m} \theta(y) - 2m \int_2^y t^{2m-1} \theta(t) dt = \frac{y^{2m+1}}{2m+1} + O_m(1) + y^{2m} E(y) - 2m \int_2^y t^{2m-1} E(t) dt,$$

the $O_m(1)$ being the endpoint term $2m 2^{2m+1}/(2m+1)$. Here $y^{2m}E(y) \ll y^{2m+1}e^{-2c\sqrt{\log y}}$, and splitting the integral at $\sqrt{y}$,

$$\int_2^y t^{2m} e^{-2c\sqrt{\log t}} dt \leq \frac{y^{(2m+1)/2}}{2m+1} + e^{-2c\sqrt{(\log y)/2}} \frac{y^{2m+1}}{2m+1} \ll_m y^{2m+1} e^{-c\sqrt{\log y}},$$

where both steps use only $2m+1 > 0$: the first integral was extended using $\int_2^{\sqrt{y}} t^{2m} dt \leq y^{(2m+1)/2}/(2m+1)$, and $y^{(2m+1)/2} \ll_m y^{2m+1} e^{-c\sqrt{\log y}}$ because $e^{c\sqrt{\log y}}$ grows slower than any positive power of y . As $y^{2m+1} = x^{m+1/2} \rightarrow \infty$, the $O_m(1)$ is absorbed. All constants depend on m : they blow up like $(2m+1)^{-1}$ as $m \rightarrow -\frac{1}{2}^+$ and (through 2^{2m+1} and the factor $2m+1$) grow with m as $m \rightarrow \infty$; the theorem is for fixed m , and no uniformity in m is used anywhere below — the continuity in (ii) is proved at the level of the limit random variables, not of the explicit formula. $\square$

Proof of (i). We verify the hypotheses of [7, Thm. 1.2 / Cor. 1.3(a)] literally. Their setting: $\phi(y) = c + \operatorname{Re} \sum_{\lambda_n \leq X} r_n e^{i\lambda_n y} + E(y, X)$ for all $y \geq y_0$ and all $X \geq X_0$ (*their* (1.7)), with

$$\lim_{Y \rightarrow \infty} \frac{1}{Y} \int_{y_0}^Y |E(y, e^Y)|^2 dy = 0 \quad (3)$$

(*their* (1.8); note the truncation point is e^Y , tied to the averaging window — this is why Lemma 2 must allow $T > x$). Corollary 1.3(a) then requires $r_n \ll \lambda_n^{-\beta}$ for some $\beta > \frac{1}{2}$ and the counting bound $\sum_{T < \lambda_n \leq T+1} 1 \ll \log T$.

Take $\phi(u) = E_m(e^u)$, $y_0 = 2$, $c = 1$, $\lambda_n = \gamma_n$ the positive ordinates in increasing order (with multiplicity), and

$$r_\gamma(m) = \frac{2(2m+1)}{m + \frac{1}{2} + i\gamma}, \quad \text{so } |r_\gamma(m)| = 2a_\gamma(m).$$

Pairing ρ and $\bar{\rho}$ in Lemma 2 (the zeros of $L(s, \chi)$ are symmetric under conjugation since χ is real) and combining with Lemma 3, for $u \geq 2$ and any $T \geq 2$,

$$E_m(e^u) = 1 + \operatorname{Re} \sum_{0 < \gamma \leq T} r_\gamma(m) e^{i\gamma u} + \mathcal{E}(u, T),$$

$$\mathcal{E}(u, T) \ll_m \frac{e^{u/2} \log^2(e^u T)}{T} + u^2 e^{-\kappa_m u} + u e^{-u/6} + e^{-c\sqrt{u}}, \quad \kappa_m := \min\left(m + \frac{1}{2}, \frac{1}{2}\right) > 0.$$

(The four terms come from, in order: the truncation error of (2) times $(2m+1)x^{-(m+1/2)}$; the $(1+x^m) \log x$ floor of (2) together with the $\log^2 x$ prime-power floor of Lemma 3, which after normalization are $(1+e^{mu})u e^{-(m+1/2)u} + u^2 e^{-(m+1/2)u} \leq 3u^2 e^{-\kappa_m u}$, since $e^{-(m+1/2)u} \leq e^{-\kappa_m u}$ and $e^{mu} e^{-(m+1/2)u} = e^{-u/2} \leq e^{-\kappa_m u}$; the $k \geq 3$ term $x^{m+1/3} \log x \cdot x^{-(m+1/2)} = x^{-1/6} \log x = u e^{-u/6}$; and the PNT error. All exponents are strictly negative for every $m > -\frac{1}{2}$, which is the only property used.) This establishes their (1.7) with our $\mathcal{E}$ defined by the display, for every $T \geq 2$; square-integrability of ϕ on $[0, y_0]$ is trivial since E_m is bounded there.

For (3), put $T = e^Y$ and $u \in [2, Y]$, so $e^u \leq e^Y = T$: the first term is $\ll e^{u/2} (2Y)^2 e^{-Y}$, whose square integrates to $\frac{1}{Y} \int_2^Y 16Y^4 e^{u-2Y} du \ll Y^3 e^{-Y} \rightarrow 0$; each of the remaining terms is

a fixed $L^2(0, \infty)$ function of u (here $\kappa_m > 0$ is used), so its contribution to the Cesàro mean square is $O_m(1/Y) \rightarrow 0$.

The coefficient conditions: $|r_\gamma(m)| \leq 2(2m+1)/\gamma$, so $\beta = 1 > \frac{1}{2}$; and $\sum_{T < \gamma \leq T+1} 1 \ll_q \log T$ is the Riemann–von Mangoldt formula for $L(s, \chi)$, $N(T, \chi) = \frac{T}{2\pi} \log \frac{qT}{2\pi e} + O_q(\log T)$ ([17, §16]), which holds unconditionally. Hence [7, Cor. 1.3(a)] applies verbatim and ν_m exists. (The only structural difference from their unshifted applications is that $\operatorname{Re}(\rho + m) = m + \frac{1}{2}$ replaces $\frac{1}{2}$; it is constant along the zeros, which is all their hypotheses see. Shifted denominators $\rho - \alpha$ of exactly this type already occur in their weighted Möbius and Liouville applications [7, Lem. 4.3, Cor. 1.6].)

Under $\text{LI}(\chi)$ the multiset $\{\gamma\}$ is linearly independent over $\mathbb{Q}$, and [7, Thm. 1.9] (proved in their §5) gives

$$\hat{\nu}_m(\xi) = e^{-i\xi} \prod_{\gamma > 0} J_0(2a_\gamma(m)\xi),$$

which is precisely the characteristic function (in their Fourier normalization) of $X_m = 1 + \sum 2a_\gamma \cos \theta_\gamma$; by uniqueness of Fourier transforms of probability measures, ν_m is the law of X_m . $\square$

Proof of (ii). Fix $m \in (-\frac{1}{2}, \infty)$ and write $V_m = X_m - 1 = \sum_{\gamma > 0} 2a_\gamma \cos \theta_\gamma$ (convergent a.s. and in L^2).

Absolute continuity. Each summand $2a_\gamma \cos \theta_\gamma$ has the arcsine law with density $\pi^{-1}(4a_\gamma^2 - x^2)^{-1/2}$ on $(-2a_\gamma, 2a_\gamma)$, which is absolutely continuous ($a_\gamma > 0$ because $2m+1 > 0$). Split off the first zero: $V_m = 2a_{\gamma_1} \cos \theta_{\gamma_1} + W$ with W independent of the first summand. If $\mu \ll \text{Leb}$ and ν is *any* probability law, then for Lebesgue-null A , $(\mu * \nu)(A) = \int \mu(A - w) \nu(dw) = 0$; hence $\text{law}(V_m) = \text{arcsine} * \text{law}(W)$ is absolutely continuous. One absolutely continuous independent component suffices; in particular $\nu_m(\{x\}) = 0$ for every x .

Existence of the density $\delta(m)$. Since $\nu_m(\partial(0, \infty)) = \nu_m(\{0\}) = 0$, sandwiching $\mathbf{1}_{(0, \infty)}$ between bounded continuous functions in the defining property of ν_m shows that the logarithmic density of $\{x : E_m(x) > 0\} = \{x : R_m(x) > 0\}$ exists and equals $\nu_m((0, \infty))$; this is the same passage as [7, eq. (1.22)] and [1, §2].

Full support. We show that $\mathbb{P}(|V_m - v| < \varepsilon) > 0$ for every $v \in \mathbb{R}$ and $\varepsilon > 0$; this gives $\text{supp}(V_m) = \mathbb{R}$. Let $\gamma_1 < \gamma_2 < \dots$ be the positive ordinates (distinct under LI) and $c_n := 2a_{\gamma_n}(m) > 0$. Three facts: (α) $c_n \rightarrow 0$, since $a_\gamma(m) \leq (2m+1)/\gamma$; (β) $\sum_n c_n = \infty$: for every $\gamma \geq m + \frac{1}{2}$ — all but finitely many ordinates — one has $(m + \frac{1}{2})^2 \leq \gamma^2$, hence $a_\gamma(m) \geq (2m+1)/(\gamma\sqrt{2})$, and $\sum_{\gamma \leq T} \gamma^{-1} \gg_q (\log T)^2$ by partial summation against $N(T, \chi) \sim \frac{T}{2\pi} \log \frac{qT}{2\pi e}$; (γ) $\sum_n c_n^2 < \infty$. By symmetry of V_m we may take $v \geq 0$ (for $v < 0$ replace $\cos \theta \approx 1$ by $\cos \theta \approx -1$ in Step 3).

Step 1 (a finite block reaching v to within $\varepsilon/4$). By (α) fix N_0 with $c_n < \varepsilon/4$ for all $n > N_0$. By (β), $\sum_{n > N_0} c_n = \infty$, so there is a minimal $N_1 \geq N_0$ with $S := \sum_{n=N_0+1}^{N_1} c_n \geq v$ (if $v = 0$ take $N_1 = N_0$, $S = 0$). Minimality and $c_{N_1} < \varepsilon/4$ give $v \leq S < v + \varepsilon/4$. Put $B = \{N_0 + 1, \dots, N_1\}$.

Step 2 (tail). By (γ) fix $N \geq N_1$ with $\sum_{n > N} 2a_{\gamma_n}^2 \leq \varepsilon^2/64$. The tail $W := \sum_{n > N} 2a_{\gamma_n} \cos \theta_{\gamma_n}$ (convergent a.s. and in L^2) has mean 0 and $\text{Var}(W) = \sum_{n > N} 2a_{\gamma_n}^2 \leq \varepsilon^2/64$ (each summand has variance $2a_{\gamma_n}^2$), so Chebyshev gives $\mathbb{P}(|W| \geq \varepsilon/4) \leq (\varepsilon^2/64)(16/\varepsilon^2) = \frac{1}{4}$, i.e. $\mathbb{P}(|W| < \varepsilon/4) \geq \frac{3}{4}$.

Step 3 (head). Put $\delta := \varepsilon / (4(1 + \sum_{n \leq N} c_n)) > 0$ and

$$A := \bigcap_{n \in B} \{|\cos \theta_{\gamma_n} - 1| < \delta\} \cap \bigcap_{\substack{n \leq N \\ n \notin B}} \{|\cos \theta_{\gamma_n}| < \delta\}.$$

Each event has positive probability (θ_{γ_n} is uniform on $[0, 2\pi)$, and $\cos \theta$ lies within δ of 1, resp. of 0, on a set of positive measure), the events concern distinct phases, and the phases are independent, so $\mathbb{P}(A) > 0$. On A the head $H := \sum_{n \leq N} c_n \cos \theta_{\gamma_n}$ satisfies $|H - S| \leq \delta \sum_{n \leq N} c_n < \varepsilon/4$.

Step 4 (combine). A and $\{|W| < \varepsilon/4\}$ involve disjoint sets of independent phases, hence are independent, and $V_m = H + W$ a.s. On their intersection, $|V_m - v| \leq |H - S| + |S - v| + |W| < \frac{3}{4}\varepsilon < \varepsilon$, and

$$\mathbb{P}(|V_m - v| < \varepsilon) \geq \frac{3}{4} \mathbb{P}(A) > 0.$$

Every open interval therefore carries positive mass: $\text{supp}(V_m) = \mathbb{R}$ (cf. the positivity-of-densities argument of [1], of which this is the quantitative form).

Strict inequalities. V_m is symmetric and atomless, so $\mathbb{P}(V_m \geq 0) = \frac{1}{2}$ and $\delta(m) = \mathbb{P}(V_m > -1) = \frac{1}{2} + \mathbb{P}(-1 < V_m < 0) > \frac{1}{2}$, while $1 - \delta(m) = \mathbb{P}(V_m \leq -1) > 0$; both strict by full support. Hence $\frac{1}{2} < \delta(m) < 1$.

Continuity. The characteristic function of V_m is $\Phi_m(t) = \prod_{\gamma} J_0(2a_{\gamma}(m)t)$ (dominated convergence from the finite products). Fix $t_0 \geq 1$ and a compact $K \subset (-\frac{1}{2}, \infty)$, and put

$$A_K := 2 \max K + 1 \in (0, \infty), \quad \text{so} \quad a_{\gamma}(m') \leq \frac{2m' + 1}{\gamma} \leq \frac{A_K}{\gamma} \quad \text{for all } m' \in K, \gamma > 0.$$

For $\Gamma \geq 2A_K t_0$ and $\gamma > \Gamma$ we have $|2a_{\gamma}(m')t| \leq 2A_K t_0 / \gamma \leq 1$, and for $|z| \leq 1$ the alternating series gives $J_0(z) \geq 1 - z^2/4 \geq \frac{3}{4}$, whence $|\log J_0(z)| \leq z^2/3$. Therefore

$$\left| \sum_{\gamma > \Gamma} \log J_0(2a_{\gamma}(m')t) \right| \leq \frac{4A_K^2 t_0^2}{3} \sum_{\gamma > \Gamma} \gamma^{-2} =: \eta_K(\Gamma) \rightarrow 0 \quad (\Gamma \rightarrow \infty),$$

uniformly in $m' \in K$ and $|t| \leq t_0$. Writing $\Phi_{m'}$ as (finite product over $\gamma \leq \Gamma$, jointly continuous in (m', t) since each $a_{\gamma}(\cdot)$ is continuous on $(-\frac{1}{2}, \infty) \times \exp(\sum_{\gamma > \Gamma} \log J_0)$, and noting all factors have modulus ≤ 1 , a standard 3ε argument gives $\Phi_{m'}(t) \rightarrow \Phi_m(t)$ for every t as $m' \rightarrow m$ within $(-\frac{1}{2}, \infty)$. Each $\Phi_{m'}$ is a characteristic function and the limit Φ_m is continuous at 0, so by Lévy's continuity theorem $X_{m'} \Rightarrow X_m$ weakly. Since $\nu_m(\{0\}) = 0$, the portmanteau theorem for the set $(0, \infty)$ gives $\delta(m') \rightarrow \delta(m)$: the map $m \mapsto \delta(m)$ is continuous on $(-\frac{1}{2}, \infty)$. $\square$

Proof of (iii). Let $V_m = X_m - 1$ as above; $\mathbb{E}V_m = 0$ and, by independence and L^2 convergence,

$$\sigma_m^2 := \text{Var}(V_m) = \sum_{\gamma > 0} 2a_{\gamma}(m)^2 \leq (2m + 1)^2 \sum_{\gamma > 0} \frac{2}{\gamma^2} = C(\chi)(2m + 1)^2,$$

since $a_{\gamma}(m)^2 \leq (2m + 1)^2 / \gamma^2$. Chebyshev's inequality gives

$$1 - \delta(m) = \mathbb{P}(V_m \leq -1) \leq \mathbb{P}(|V_m| \geq 1) \leq \sigma_m^2 \leq C(\chi)(2m + 1)^2.$$

(Cantelli's one-sided inequality would give the marginally better $\sigma_m^2 / (1 + \sigma_m^2)$; we keep the clean form.)

For the exponential bound, set $A := \sum_{\gamma>0} a_\gamma(m)^2 = \sigma_m^2/2 \leq \frac{C(\chi)}{2}(2m+1)^2$ and let $S_N = \sum_{n \leq N} 2a_{\gamma_n} \cos \theta_{\gamma_n}$. For $s > 0$, independence gives

$$\mathbb{E} e^{-sS_N} = \prod_{n \leq N} I_0(2a_{\gamma_n}s) \leq \exp\left(s^2 \sum_{n \leq N} a_{\gamma_n}^2\right) \leq e^{s^2 A},$$

where I_0 is the modified Bessel function, $\mathbb{E} e^{x \cos \theta} = I_0(x)$, and $I_0(x) = \sum_k (x/2)^{2k}/(k!)^2 \leq \sum_k (x^2/4)^k/k! = e^{x^2/4}$ term by term (Montgomery's inequality; it is inequality (33) of [5], from [18, §3, eq. (2)]). Chernoff's bound with the optimal $s = (1 - \varepsilon)/(2A)$ yields, for $0 < \varepsilon < 1$,

$$\mathbb{P}(S_N \leq -(1 - \varepsilon)) \leq \exp\left(-\frac{(1 - \varepsilon)^2}{4A}\right).$$

Since $S_N \rightarrow V_m$ in L^2 , $\mathbb{P}(V_m \leq -1) \leq \mathbb{P}(S_N \leq -(1 - \varepsilon)) + \mathbb{P}(|V_m - S_N| \geq \varepsilon)$, and letting $N \rightarrow \infty$ then $\varepsilon \rightarrow 0^+$,

$$1 - \delta(m) = \mathbb{P}(V_m \leq -1) \leq \exp\left(-\frac{1}{4A}\right) \leq \exp\left(-\frac{1}{2C(\chi)(2m+1)^2}\right).$$

Both stated bounds follow, for every $m \in (-\frac{1}{2}, \infty)$ — no step above used $m \leq 0$ — and $e^{-1/(2z)} < z$ for all $z > 0$ (equivalent to $\log w < w/2$ at $w = 1/z$), so the exponential bound is always the stronger one. $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$ is immediate.

Scope. Although valid on the whole range, the bounds carry information only near the left endpoint, and we state this openly. The Chebyshev bound $C(\chi)(2m+1)^2$ is below the trivial estimate $\frac{1}{2}$ of part (ii) only for $m < \frac{1}{2}((2C(\chi))^{-1/2} - 1)$ and is ≥ 1 for $m \geq \frac{1}{2}(C(\chi)^{-1/2} - 1)$; the exponential bound is below $\frac{1}{2}$ only for $m < m_0(\chi) = \frac{1}{2}((2C(\chi) \log 2)^{-1/2} - 1)$. At the two computed moduli: for χ_4 ($C = 0.1560296\dots$) these thresholds are $0.395058\dots, 0.76580\dots$ and $m_0(\chi_4) = 0.57507\dots$; for χ_3 ($C = 0.1134021\dots$) they are $0.549891\dots, 0.984770\dots$ and $m_0(\chi_3) = 0.761048\dots$. For $m \geq m_0(\chi)$, part (iii) asserts nothing beyond part (ii). This limitation is consistent with the expected truth: the computed densities approach $\frac{1}{2}$ from above (Table 1), so an upper bound on $1 - \delta(m)$ tending to 0 for large m would contradict the numerics, and none should be sought. The large- m statement, $\delta_q(m) - \frac{1}{2} \sim (2\pi \text{Var}_q(m))^{-1/2}$ with $\text{Var}_q(m)$ as in Results item 3 (asymptotically equal to $\sigma_m^2 = \text{Var}(V_m)$ above), is the dissolution law of the Results section; it is not proved by this theorem, but it *is* proved in this paper: Theorem 2 of Section 4 establishes it, with rate and a second-order term, for the limiting variable of every real primitive χ under $\text{GRH}(\chi)$ and $\text{NCZ}(\chi)$, and, under $\text{LI}(\chi)$, part (ii) of the present theorem transfers it to the logarithmic density of the race itself (Corollary 3 records the case $q = 4$).

Remark on constants. Plain Hoeffding (range form: $\mathbb{P}(S \leq -t) \leq \exp(-2t^2/\sum_i (\text{range}_i)^2)$ with $\text{range}_\gamma = 4a_\gamma$) gives $\exp(-1/(8A)) \leq \exp(-1/(4C(\chi)(2m+1)^2))$, valid for all m as well; the arcsine moment-generating function is exactly I_0 , and $I_0(x) \leq e^{x^2/4}$ beats Hoeffding's lemma ($e^{x^2/2}$ for a $[-x, x]$ -valued variable) by a factor 2 in the exponent; no smallness restriction on m is needed anywhere.

Numerical consistency ($\chi = \chi_4$; deterministic Gil-Pelaez inversion, 511 ordinates plus Gaussianized tail; computational, not part of the proof): at $m = 0$, $1 - \delta = 4.073 \times 10^{-3}$ against the bounds $\min(0.156, e^{-1/(2C(\chi_4))} = 0.0406)$; at $m = -0.1$, 3.12×10^{-4} vs 6.7×10^{-3} ; at $m = -0.2$, 6.28×10^{-7} vs 1.36×10^{-4} ; at $m = -0.3$ the exact-MGF Chernoff value

certifies $1 - \delta \leq 1.2 \times 10^{-15}$ vs the stated bound 2.0×10^{-9} . The bounds are (as they must be) weaker than the truth at every tested point. For comparison, the Hoeffding-form bound crosses below Chebyshev at $(2m+1)^2 \approx 0.744$, i.e. $m \approx -0.069$; the I_0 -form bound is stronger than Chebyshev for all m . $\square$

Proof of (iv). Immediate from (i)–(iii) together with the cited endpoint results: [1] and [9] at $m = 0$ (our deterministic recomputation agrees to six decimals), and [2, 3, 14] at $m = -\frac{1}{2}$ for the unlogged race as quoted in the statement. We emphasize the logical status: $\delta(m)$ is defined for $m \in (-\frac{1}{2}, \infty)$ and $m = -\frac{1}{2}$ is not a value of the family; the connection asserted is $\delta(m) \rightarrow 1$ as $m \rightarrow -\frac{1}{2}^+$ (proved in (iii) under GRH+LI) matching the almost-sure single-sign regime established at $m = -\frac{1}{2}$ by different methods and under different hypotheses (DRH, resp. GRH variants, no LI). For the θ -form race at the threshold itself, the prime-square mechanism of Lemma 3 produces a $\frac{1}{2} \log x$ drift ($\sum_{p \leq \sqrt{x}} p^{-1} \log p \sim \frac{1}{2} \log x$); the $\frac{1}{2} \log \log x$ drift of [2, 3] belongs to the unlogged race $D_{-1/2}$, and the two are compatible by partial summation. We do not claim the $m = -\frac{1}{2}$ results, nor any new statement at $m = -\frac{1}{2}$. At the right end of the family we prove existence, continuity, and $\frac{1}{2} < \delta(m) < 1$ for every m , but no statement about the rate of approach to $\frac{1}{2}$; the decrease of the computed values in Table 1 is numerical observation, not a theorem of this paper. $\square$

Proposition 1 (transfer to the unlogged races). *Assume $GRH(\chi)$ and $NCZ(\chi)$, and fix $m \in (-\frac{1}{2}, \infty)$. Let $D_m(x) = -\sum_{p \leq x} \chi(p) p^m$ (for $\chi = \chi_4$ this is $\sum_{p \leq x, p \equiv 3(4)} p^m - \sum_{p \leq x, p \equiv 1(4)} p^m$) and*

$$E'_m(x) = (2m+1) x^{-(m+1/2)} (\log x) D_m(x).$$

Then $u \mapsto E'_m(e^u)$ has the same limiting distribution ν_m as $E_m(e^u)$; consequently, under $LI(\chi)$, all of (ii)–(iii) hold verbatim for the unlogged race, with the same $\delta_\chi(m)$.

Proof. Partial summation: $D_m(x) = \int_2^x (\log t)^{-1} dR_m(t) = \frac{R_m(x)}{\log x} + \int_2^x \frac{R_m(t)}{t \log^2 t} dt$. Insert the expansion of R_m from Lemmas 2–3 (truncation T , valid uniformly in $t \leq x$),

$$R_m(t) = \frac{t^{m+1/2}}{2m+1} + \sum_{|\gamma| \leq T} \frac{t^{\rho+m}}{\rho+m} + F(t, T),$$

$$F(t, T) \ll_m \frac{t^{m+1} \log^2(tT)}{T} + t^{m+1/3} \log t + t^{m+1/2} e^{-c\sqrt{\log t}} + (1+t^m) \log t + \log^2 t,$$

the fourth term being the full-range floor of (2) (for $m \leq 0$ it is $\leq 2 \log t$ and merges with the last term, giving back the earlier display).

Write $s = \rho + m = m + \frac{1}{2} + i\gamma$, so $|s| \asymp_{m, \chi} |\gamma|$ (here the lower bound $|\gamma| \geq \gamma_1(\chi) > 0$ supplied by NCZ enters) and $\operatorname{Re} s = m + \frac{1}{2} > 0$.

Bias term. With $a = m + \frac{1}{2}$, two integrations by parts give $\int_2^x t^{a-1} \log^{-2} t dt = \frac{x^a}{a \log^2 x} (1 + O_a(1/\log x)) + O_a(1)$, whence

$$\frac{1}{2m+1} \left(\frac{x^{m+1/2}}{\log x} + \int_2^x \frac{t^{m+1/2}}{t \log^2 t} dt \right) = \frac{x^{m+1/2}}{(2m+1) \log x} \left(1 + O_m \left(\frac{1}{\log x} \right) \right).$$

Zero terms. For each zero, one integration by parts gives

$$\frac{1}{s} \int_2^x t^{s-1} \log^{-2} t dt = \frac{x^s}{s^2 \log^2 x} - \frac{2^s}{s^2 \log^2 2} + \frac{2}{s^2} \int_2^x t^{s-1} \log^{-3} t dt,$$

so the ρ -contribution to $D_m(x)$ is $\frac{x^s}{s \log x} (1 + O(\frac{1}{|s| \log x})) + O(|s|^{-2}) + O_m(|s|^{-2} x^{m+1/2} \log^{-3} x)$. The summability in γ is what makes the bookkeeping close: summing over $|\gamma| \leq T$, the three error families contribute, after multiplication by the normalizer $(2m+1)x^{-(m+1/2)} \log x$,

$$\left| \sum_{0 < \gamma \leq T} r_\gamma(m) e^{i\gamma u} O\left(\frac{1}{\gamma u}\right) \right| \ll_m \frac{1}{u} \sum_{\gamma} \frac{1}{\gamma^2} \ll_m \frac{1}{u}, \quad u e^{-(m+1/2)u} \sum_{\gamma} \gamma^{-2} \ll_m u e^{-(m+1/2)u}, \quad \frac{1}{\log^2 x} \sum_{\gamma} \gamma^{-2} \ll$$

all uniform in T (here $\sum_{\gamma} a_\gamma(m)/\gamma \ll_m \sum \gamma^{-2} < \infty$ is the key convergence).

Truncation/PNT term. Each component of F passes through $\frac{R_m(x)}{\log x}$ and $\int_2^x F(t, T) t^{-1} \log^{-2} t dt$ with at most one extra factor $\log$. For the components carrying a power t^a with $a > 0$ ($a = m+1$ and $a = m + \frac{1}{2}$ always; $a = m + \frac{1}{3}$ when $m > -\frac{1}{3}$), splitting the integral at $\sqrt{x}$ gives $\int_2^x t^{a-1} \log^{-2} t dt \ll_a x^a \log^{-2} x$, the PNT component retaining its factor $e^{-c\sqrt{\log t}}$ (decreasing, so bounded on $[\sqrt{x}, x]$ by $e^{-(c/\sqrt{2})\sqrt{\log x}}$). For $m \leq -\frac{1}{3}$ the $k \geq 3$ component has $m + \frac{1}{3} \leq 0$, so $t^{m+1/3} \log t \leq (\log 2)^{-1} \log^2 t$ for $t \geq 2$ and it is absorbed by the $\log^2 t$ component (this case must be separated: for $a \leq 0$ the integral $\int_2^x t^{a-1} \log^{-2} t dt$ is $\asymp_m 1$, not $\ll x^a \log^{-2} x$). The $\log^2 t$ component gives $\int_2^x t^{-1} dt = \log x - \log 2$; and the new floor $(1+t^m) \log t$ contributes $\int_2^x (1+t^m) t^{-1} \log^{-1} t dt \ll_m \log \log x + 1 + x^m / \log x$. After multiplication by the normalizer $(2m+1)x^{-(m+1/2)} \log x$, all of these are covered by the display below (the floor's contribution is $\ll_m u^2 e^{-\kappa_m u}$).

Collecting, for $u \geq 2$, $T \geq 2$,

$$E'_m(e^u) = 1 + \operatorname{Re} \sum_{0 < \gamma \leq T} r_\gamma(m) e^{i\gamma u} + \mathcal{E}'(u, T), \quad \mathcal{E}'(u, T) \ll_m \frac{u e^{u/2} \log^2(e^u T)}{T} + \frac{1}{u} + u^2 e^{-\kappa_m u} + u e^{-u/6} + u e^{-c\sqrt{u}},$$

with $\kappa_m = \min(m + \frac{1}{2}, \frac{1}{2})$ as in the proof of (i), and at $T = e^Y$, $u \in [2, Y]$ each term's Cesàro mean square tends to 0 exactly as in (i) (the first is $\ll Y^5 e^{-Y}$ after integration; $1/u$ contributes $\frac{1}{Y} \int_2^Y u^{-2} du \ll 1/Y$; the remaining three are fixed $L^2(0, \infty)$ functions of u since $\kappa_m > 0$). Thus $E'_m(e^u)$ satisfies the same [7] hypotheses with the *same* constant 1 and the *same* coefficients $r_\gamma(m)$, so its limiting distribution is again ν_m . At $m = 0$ this recovers the classical passage of [1] between θ - and π -races. □

Remark 2. *Two further points of independent interest. (1) Without LI, part (i) still gives existence of ν_m , and [7, Thm. 1.14] (Parseval, literally their eq. (1.24) with $c = 1$) evaluates the mean square of the normalized race: $\lim_Y \frac{1}{Y} \int_0^Y E_m(e^u)^2 du = 1 + \frac{1}{2} \sum_{\gamma} |r_\gamma|^2 = 1 + \sum_{\gamma} 2a_\gamma(m)^2 \leq 1 + C(\chi)(2m+1)^2$, so the spread about the unit bias collapses quadratically in $(2m+1)$ under GRH alone. We do not pursue LI-free density statements (for which see [14], and [8] for the possible-atom phenomenology), because without LI ν_m could in principle carry an atom at 0 and the density interpretation requires care. (2) The bounds of (iii) hold with $\sigma_m^2 = \sum_{\gamma} 2a_\gamma(m)^2$ in place of $C(\chi)(2m+1)^2$ throughout, which is numerically slightly sharper (for χ_4 at $m = 0$: 0.15557 vs 0.15603); we state the $C(\chi)(2m+1)^2$ form because the m -dependence is transparent.*

Monotonicity in the weight

Every computed value in this note is consistent with a strict monotonicity that the dissolution asymptotics alone do not give, and Section 6 now proves it for all sufficiently large weights:

under GRH, $\delta'_4(m) < 0$ for every $m \geq 16.82$ and $\delta'_3(m) < 0$ for every $m \geq 18.62$ (Corollary 7; for $M = m + \frac{1}{2} \geq 300$ the proof is zero-data-free, and on the remaining explicit range it is a verified criterion in the rigor class of Table 5), and eventual strict decrease holds for every real primitive character with $L(\frac{1}{2}, \chi) \neq 0$ (Theorem 4). What remains open is the compact initial range:

Conjecture 1. *For $q \in \{3, 4\}$ the map $m \mapsto \delta_q(m)$ is strictly decreasing on all of $(-\frac{1}{2}, \infty)$; equivalently (given Corollary 7), on $(-\frac{1}{2}, m_1]$ with $m_1 = 16.82$ resp. 18.62.*

Three remarks on its status. (a) *Numerical evidence.* Deterministic Gil–Pelaez inversion with validated enclosures — the unseen-zero tail variance evaluated in closed form from the completed L -function (the Hadamard-product identity behind the evaluation of C), the tail’s departure from Gaussianity bounded by a proved quartic estimate, truncation and quadrature budgeted into the interval radius; every enclosure width below 5×10^{-7} — on the 30-point grid

$$m \in \{-0.2, -0.15, -0.1, -0.05, 0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 15, 20, 25, 30, 40, 50,$$

yields enclosures that are pairwise disjoint and decreasing at all 29 consecutive pairs, for both moduli; the smallest gap between adjacent enclosures is 2.9×10^{-5} for modulus 4 and 8.6×10^{-7} for modulus 3 (both at the leftmost pair). In particular the grid covers the conjectural range $(-0.2 \leq m \leq m_1)$ densely, and its last points overlap the proved range. The enclosures are rigorous modulo the stated computational budget (IEEE-double quadrature with step-doubling control; 40-digit floating-point evaluation of the closed-form variance; no interval arithmetic is claimed). Left of the grid the inversion loses resolution: at $m = -0.2$ itself the enclosures give $1 - \delta_4 \in [4.8, 7.7] \times 10^{-7}$ and $1 - \delta_3 \leq 2.5 \times 10^{-7}$, and on $(-\frac{1}{2}, -0.2)$ the only control we have is the Chernoff bound of Theorem 1(iii) (applied with $\chi = \chi_4$, resp. χ_3 ; cf. Corollary 1), a monotone upper envelope for $1 - \delta$ which is consistent with — but does not imply — monotonicity of δ itself.

(b) *Why the conjecture is plausible — now a proof mechanism for large m .* With $M = m + \frac{1}{2}$, every amplitude is strictly increasing in the weight:

$$\frac{\partial a_\gamma}{\partial M} = \frac{2\gamma^2}{(M^2 + \gamma^2)^{3/2}} > 0,$$

so as m grows each noise component of $X_m = 1 + \sum_{\gamma > 0} 2a_\gamma \cos \theta_\gamma$ strictly grows while the mean stays fixed at 1. Section 6 turns exactly this into a theorem for $m \geq m_1$, by differentiating the Gil–Pelaez integral: the logarithmic derivative of each Bessel factor of the characteristic function is negative at every frequency $t < j_{0,1}/4$ (all Taylor coefficients of $-\log J_0$ are positive — a Rayleigh-sum fact that replaces peakedness), and the frequencies beyond are superexponentially damped once enough zeros lie below $2M$.

(c) *Why the remaining range is genuinely open.* If the family were a dilation, $V_m = c(m)V$ with c increasing, monotonicity would be immediate; it is not — the ratio $a_\gamma(m')/a_\gamma(m)$ depends on γ (low zeros saturate near 2, high zeros still grow linearly in M). The natural tool is a peakedness ordering: for the symmetric variable $V_m = X_m - 1$ one has $\delta(m) = 1 - \frac{1}{2} \mathbb{P}(|V_m| \geq 1)$, so it would suffice that $V_{m'}$ is more peaked than V_m for $m' < m$ in the sense of Birnbaum [24]. Scaling up a single component always makes that component less peaked, but the classical results that push such comparisons through convolution — Birnbaum’s lemma [24] and Proschan’s theorem on convex combinations [25] — require symmetric *unimodal* summands

(Proschan's in the stronger log-concave form), and the arcsine components $2a_\gamma \cos \theta_\gamma$ are symmetric but not unimodal (their density is U-shaped and diverges at the endpoints). The large- m proof does not help here: its tail control collapses when fewer than a dozen zeros lie below $2M$, precisely the regime $m < m_1$; and neither the Chernoff envelope near $m = -\frac{1}{2}$ nor the dissolution asymptotics controls $\delta(m) - \delta(m')$ at nearby small weights.

3 The variance in closed form

The variance $\sigma_m^2 = \sum_{\gamma>0} 2a_\gamma(m)^2$ of the limiting variable X_m has so far entered this paper through a truncation: 511 computed ordinates plus a tail estimated from the zero-counting density. This section removes the estimate. The variance is the value of an explicitly computable function of the weight, real-analytic on $(-\frac{1}{2}, \infty)$, with three consequences: the constant $C = C(\chi_4)$ of Theorem 1 (this section computes at the modulus 4; the general identity is (8)) acquires a closed form (Corollary 2); the variance half of the dissolution law of the introduction becomes a theorem with an explicit constant term; and the Gaussianized tails of the numerical work become exactly computable quantities.

Throughout, $\Lambda(s) := (4/\pi)^{(s+1)/2} \Gamma(\frac{s+1}{2}) L(s, \chi_4)$ denotes the completed L -function and $\Xi(z) := \Lambda(\frac{1}{2} + z)$.

Proposition 2 (variance identity). *Assume $GRH(\chi_4)$. For every real $m > -\frac{1}{2}$, with $M = m + \frac{1}{2}$,*

$$\sigma_m^2 = \sum_{\gamma>0} 2a_\gamma(m)^2 = 8M^2 \sum_{\gamma>0} \frac{1}{M^2 + \gamma^2} = 4M \frac{\Lambda'}{\Lambda}(m+1). \quad (4)$$

GRH enters only by placing the zeros on the critical line, so that the ordinates γ are real; no linear-independence hypothesis is involved.

Proof. The first equality in (4) is the definition together with $a_\gamma(m)^2 = (2M)^2/(M^2 + \gamma^2)$. The content is the last equality.

Step 1: Ξ is entire of order 1 and even. Since χ_4 is primitive and nonprincipal, Λ is entire of order 1 [17, Chs. 9, 12]. For a primitive character $\chi \bmod q$ with $\chi(-1) = (-1)^a$, the functional equation reads $\Lambda(1-s, \bar{\chi}) = (i^a \sqrt{q}/\tau(\chi)) \Lambda(s, \chi)$ with $\tau(\chi) = \sum_n \chi(n) e^{2\pi i n/q}$ [17, Ch. 9]. Here $q = 4$, $a = 1$ (as $\chi_4(-1) = \chi_4(3) = -1$), $\bar{\chi}_4 = \chi_4$, and the Gauss sum is a two-term computation:

$$\tau(\chi_4) = e^{2\pi i/4} - e^{2\pi i \cdot 3/4} = i - (-i) = 2i, \quad \text{so} \quad \frac{i^a \sqrt{q}}{\tau(\chi_4)} = \frac{2i}{2i} = 1,$$

i.e. the root number is $+1$ and $\Lambda(s) = \Lambda(1-s)$. Hence $\Xi(z) = \Xi(-z)$. (The script verifies the evenness numerically to 4×10^{-51} relative at several complex points.)

Step 2: no central zero. $L(\frac{1}{2}, \chi_4) = \sum_{k \geq 0} (-1)^k (2k+1)^{-1/2} > 0$ by the alternating series test (terms strictly decreasing to 0); numerically $L(\frac{1}{2}, \chi_4) = 0.667691457 \dots$. So $\Xi(0) = \Lambda(\frac{1}{2}) > 0$ (each factor of $\Lambda(\frac{1}{2})$ is positive), and no zero of Ξ sits at the origin.

Step 3: Hadamard product. The zeros of Ξ are exactly $z_\rho = \rho - \frac{1}{2}$ over the nontrivial zeros ρ of $L(s, \chi_4)$; under GRH and Step 2 they are $\pm i\gamma$ with $\gamma > 0$ real, and the multiset of ordinates is symmetric under $\gamma \mapsto -\gamma$ because Λ is real on the real axis. By the Riemann–von

Mangoldt formula $N(T, \chi_4) \ll T \log T$ [17, §16], $\sum_{\gamma > 0} \gamma^{-2} < \infty$. Hadamard's factorization for the order-1 function Ξ gives

$$\Xi(z) = e^{A+Bz} \prod_{\rho} \left(1 - \frac{z}{z_{\rho}}\right) e^{z/z_{\rho}},$$

the product converging absolutely and locally uniformly (indeed $\sum |z/z_{\rho}|^2 = |z|^2 \sum \gamma^{-2} < \infty$ locally uniformly, and $|\log\{(1-w)e^w\}| \leq |w|^2$ for $|w| \leq \frac{1}{2}$), so its factors may be grouped at will. Grouping $\pm i\gamma$ cancels the exponential factors pairwise:

$$\Xi(z) = e^{A+Bz} \prod_{\gamma > 0} \left(1 + \frac{z^2}{\gamma^2}\right).$$

The paired product is even in z ; evenness of Ξ then forces $e^{Bz} = e^{-Bz}$, i.e. $B = 0$ (alternatively, $\Xi'(0) = 0$ since Ξ is even and analytic, and the paired product has vanishing derivative at 0). Setting $z = 0$ identifies $e^A = \Xi(0) \neq 0$:

$$\frac{\Xi(z)}{\Xi(0)} = \prod_{\gamma > 0} \left(1 + \frac{z^2}{\gamma^2}\right). \quad (5)$$

Step 4: logarithmic differentiation on the real axis. Under GRH all zeros of Ξ are purely imaginary and nonzero, so $\Xi > 0$ on $\mathbb{R}$ (it is real there, positive at 0, and nonvanishing). Taking logarithms in (5) at real $z = x$,

$$\log \frac{\Xi(x)}{\Xi(0)} = \sum_{\gamma > 0} \log \left(1 + \frac{x^2}{\gamma^2}\right),$$

a series of nonnegative terms dominated on $|x| \leq A$ by A^2/γ^2 ; the termwise-differentiated series $\sum_{\gamma} 2x/(x^2 + \gamma^2)$ is dominated on $|x| \leq A$ by $2A/\gamma^2$. Both therefore converge uniformly on compact subsets of $\mathbb{R}$, and the standard theorem on differentiation of series (pointwise convergence plus locally uniform convergence of the derivative series) yields, for every real x ,

$$\frac{\Xi'}{\Xi}(x) = \sum_{\gamma > 0} \frac{2x}{x^2 + \gamma^2}.$$

At $x = M > 0$, since $\Xi'/\Xi(M) = (\Lambda'/\Lambda)(M + \frac{1}{2}) = (\Lambda'/\Lambda)(m + 1)$,

$$4M \frac{\Lambda'}{\Lambda}(m + 1) = 8M^2 \sum_{\gamma > 0} \frac{1}{M^2 + \gamma^2} = \sigma_m^2. \quad \square$$

Remark 3. *The right-hand side of (4) is computable to arbitrary precision with no zero data: $(\Lambda'/\Lambda)(s) = \frac{1}{2} \log \frac{4}{\pi} + \frac{1}{2} \psi\left(\frac{s+1}{2}\right) + (L'/L)(s, \chi_4)$, with $L(s, \chi_4) = 4^{-s}(\zeta(s, \frac{1}{4}) - \zeta(s, \frac{3}{4}))$. As a consistency check on both the identity and our ordinate list, the truncated zero sum $\sum_{\gamma \leq \gamma_{511}} 2a_{\gamma}^2$ plus the Riemann–von Mangoldt tail integral reproduces (4) at $m \in \{0, 1, 2, 3, 8, 20, 100\}$ within the expected fluctuation of the zero-counting remainder (3.2×10^{-6} at $m = 0$, growing to 0.13 absolute — 1.5×10^{-4} relative — at $m = 100$), and $\log(\Xi(z)/\Xi(0)) - \sum_{\gamma \leq \gamma_{511}} \log(1 + z^2/\gamma^2)$ matches the corresponding tail integral $\int_{\gamma_{511}}^{\infty} \log(1 + z^2/t^2) dN(t)$ to 1.6×10^{-6} at $z = 1$.*

Corollary 2 (the constant C in closed form). *Assume $GRH(\chi_4)$. With G Catalan's constant,*

$$C = \sum_{\gamma>0} \frac{2}{\gamma^2} = (\log \Lambda)''(\tfrac{1}{2}) = \tfrac{1}{4}\psi'(\tfrac{3}{4}) + (\log L)''(\tfrac{1}{2}, \chi_4), \quad \psi'(\tfrac{3}{4}) = \pi^2 - 8G,$$

and numerically

$$C = 0.156029649889644884607682\dots, \\ \tfrac{1}{4}\psi'(\tfrac{3}{4}) = 0.635469911917901\dots, \quad (\log L)''(\tfrac{1}{2}, \chi_4) = -0.479440262028256\dots$$

The cruder zero-sum estimate 0.156033 appeared in earlier versions of this manuscript; its excess $+3.4 \times 10^{-6}$ over the closed form is the neglected fluctuation of the zero-counting remainder (the unrounded zero-sum-plus-tail estimate is 0.1560328, an excess of 3.2×10^{-6} ; rounding to six printed decimals accounts for the rest). All bounds stated with C only improve.

Proof. Differentiating $\sum_{\gamma} 2x/(x^2 + \gamma^2)$ termwise once more is legitimate on $|x| \leq 1$: the differentiated terms $2(\gamma^2 - x^2)/(\gamma^2 + x^2)^2$ are bounded there by $2/\gamma^2$ (numerator at most γ^2 , denominator at least γ^4 ; recall $\gamma_1 > 6$), giving uniform convergence. Evaluating at $x = 0$ yields $(\log \Xi)''(0) = \sum_{\gamma>0} 2/\gamma^2 = C$, and $(\log \Xi)''(0) = (\log \Lambda)''(\tfrac{1}{2})$. Since the linear factor of $\log \Lambda(s) = \frac{s+1}{2} \log \frac{4}{\pi} + \log \Gamma(\frac{s+1}{2}) + \log L(s, \chi_4)$ has vanishing second derivative, $(\log \Lambda)''(\tfrac{1}{2}) = \tfrac{1}{4}\psi'(\tfrac{3}{4}) + (\log L)''(\tfrac{1}{2}, \chi_4)$; the argument $\tfrac{3}{4}$ is $(s+1)/2$ at $s = \tfrac{1}{2}$, and $\log L$ is well defined near $s = \tfrac{1}{2}$ by Step 2 above. Finally $\psi'(\tfrac{1}{4}) + \psi'(\tfrac{3}{4}) = \pi^2/\sin^2(\pi/4) = 2\pi^2$ (reflection) and, writing $\psi'(x) = \sum_{n \geq 0} (n+x)^{-2}$, $\psi'(\tfrac{1}{4}) - \psi'(\tfrac{3}{4}) = 16 \sum_{n \geq 0} [(4n+1)^{-2} - (4n+3)^{-2}] = 16\beta(2) = 16G$, whence $\psi'(\tfrac{3}{4}) = \pi^2 - 8G$.

On the numerical claim, an honesty note: the closed form is exact, and its decimal evaluation is a routine high-precision computation (mpmath at 50 digits with guard digits; the routes $\tfrac{1}{4}\psi'(\tfrac{3}{4}) + (\log L)''(\tfrac{1}{2})$ and direct numerical $(\log \Lambda)''(\tfrac{1}{2})$ agree to 38 digits, stably under doubling of the working precision). We have not run certified interval arithmetic; the quantities are derivatives of entire functions evaluated far from any singularity. $\square$

Proposition 3 (rigorous variance asymptotic). *Assume $GRH(\chi_4)$. As $m \rightarrow \infty$,*

$$\sigma_m^2 = 2M \log \frac{4M}{2\pi} + 1 + \frac{1}{12M} + O\left(\frac{1}{M^2}\right). \quad (6)$$

More generally, let χ be a real primitive character mod q with $\chi(-1) = (-1)^a$, $a \in \{0, 1\}$, assume $GRH(\chi)$ and $L(\tfrac{1}{2}, \chi) \neq 0$, and set $\text{Var}_q(m) = \sum_{\gamma_{\chi}>0} 2a_{\gamma}(m)^2$ over the ordinates of $L(s, \chi)$. Then

$$\text{Var}_q(m) = 2M \log \frac{qM}{2\pi} + (2a-1) + \frac{1}{12M} + O\left(\frac{1}{M^2}\right), \quad (7)$$

with an absolute implied constant. In particular the constant term of the dissolution law of the introduction is $+1$ for odd real characters ($q = 3, 4$ included) and -1 for even ones.

Proof. Proposition 2 and its proof apply verbatim to χ : the completed function $\xi(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$ is entire of order 1; a real primitive character has root number $+1$ — Gauss's evaluation of the sign of the quadratic Gauss sum, $\tau(\chi) = \sqrt{q}$ for $a = 0$ and $i\sqrt{q}$ for $a = 1$ (see [17, Ch. 2] for prime modulus; the script evaluates τ directly for the moduli

3, 4, 5 used in this paper) — so $\xi(s, \chi) = \xi(1-s, \chi)$; the hypothesis $L(\frac{1}{2}, \chi) \neq 0$ replaces Step 2 (for $q = 3, 4$ it holds unconditionally by grouping the Dirichlet series at $s = \frac{1}{2}$ into positive blocks); and

$$\text{Var}_q(m) = 4M \frac{\xi'}{\xi}(m+1, \chi) = 4M \left[\frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi\left(\frac{m+1+a}{2}\right) + \frac{L'}{L}(m+1, \chi) \right]. \quad (8)$$

Digamma term. Stirling's series $\psi(x) = \log x - \frac{1}{2x} - \frac{1}{12x^2} + O(x^{-4})$, $x \rightarrow \infty$, at $x = \frac{m+1+a}{2} = \frac{M}{2} + \frac{2a+1}{4}$. For $a = 1$, $x = \frac{M}{2} \left(1 + \frac{3}{2M}\right)$ and, expanding each ingredient to two orders,

$$\log x = \log \frac{M}{2} + \frac{3}{2M} - \frac{9}{8M^2} + O(M^{-3}), \quad \frac{1}{2x} = \frac{1}{M} - \frac{3}{2M^2} + O(M^{-3}), \quad \frac{1}{12x^2} = \frac{1}{3M^2} + O(M^{-3}),$$

so that, with $-\frac{9}{8} + \frac{3}{2} - \frac{1}{3} = \frac{1}{24}$,

$$\psi\left(\frac{m+2}{2}\right) = \log \frac{M}{2} + \frac{1}{2M} + \frac{1}{24M^2} + O(M^{-3}).$$

For $a = 0$, $x = \frac{M}{2} \left(1 + \frac{1}{2M}\right)$ and the same computation ($-\frac{1}{8} + \frac{1}{2} - \frac{1}{3} = \frac{1}{24}$ again) gives $\psi\left(\frac{m+1}{2}\right) = \log \frac{M}{2} - \frac{1}{2M} + \frac{1}{24M^2} + O(M^{-3})$. Multiplying by $4M \cdot \frac{1}{2}$ and adding $2M \log \frac{q}{\pi}$,

$$4M \left[\frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi\left(\frac{m+1+a}{2}\right) \right] = 2M \log \frac{qM}{2\pi} + (2a-1) + \frac{1}{12M} + O(M^{-2}),$$

using $\frac{q}{\pi} \cdot \frac{M}{2} = \frac{qM}{2\pi}$; the sign of the $\pm \frac{1}{2M}$ term is the sole source of the parity-dependent constant $2a-1$.

The L'/L term is exponentially small. For $m \geq 1$,

$$\left| \frac{L'}{L}(m+1, \chi) \right| = \left| \sum_{n \geq 2} \Lambda(n) \chi(n) n^{-(m+1)} \right| \leq 2^{-(m-1)} \sum_{n \geq 2} \Lambda(n) n^{-2} \leq 0.57 \cdot 2^{-(m-1)}$$

(using $n^{-(m+1)} \leq n^{-2} \cdot 2^{-(m-1)}$ for $n \geq 2$ and $\sum_n \Lambda(n) n^{-2} = -\frac{\zeta'}{\zeta}(2) = 0.56996\dots$), so $4M|L'/L(m+1, \chi)| \ll M2^{-m}$, absorbed into $O(M^{-2})$ (with an absolute constant, uniformly in q). For χ_4 the decay is in fact $3^{-(m+1)} \log 3 (1+o(1))$, since $\chi_4(2^k) = 0$ and the leading surviving term is $n = 3$ with $\chi_4(3) = -1$; the script confirms $|L'/L(21, \chi_4)| = 1.050 \times 10^{-10} = 3^{-21} \log 3$ to three digits. Combining the three displays proves (7), and $q = 4$, $a = 1$ gives (6). $\square$

Remark 4. (1) Numerically, $M(\sigma_m^2 - 2M \log \frac{4M}{2\pi} - 1)$ equals 0.080845, 0.082709, 0.083083 at $m = 100, 400, 1000$, converging to $\frac{1}{12} = 0.08333\dots$ exactly as (6) requires; for the even real character mod 5 (where $L(\frac{1}{2}, \chi_5) = 0.23175\dots \neq 0$ numerically), the same computation gives $\text{Var}_5(m) - 2M \log \frac{5M}{2\pi} \rightarrow -1$, confirming $c_0(q, 0) = -1$. (2) The law is asymptotic, not a bound: at $m = 1$ the main term $2M \log \frac{4M}{2\pi} = -0.138$ is negative and useless (Table 2). (3) What is not proved in this section: the density half of the dissolution law, $\delta_q(m) - \frac{1}{2} \sim (2\pi \text{Var}_q(m))^{-1/2}$. Proposition 3 certifies only its variance input.

Remark 5 (exact tail variances for the numerical densities). The Gil-Pelaez and Monte Carlo computations of Sections 1 and 7 replace the unseen zeros $\gamma > \Gamma := \gamma_{511} = 639.9157\dots$ by a Gaussian whose variance was previously estimated from the zero-counting density. By Proposition 2 that variance is now exact:

$$\sigma_{\text{tail}}^2(\Gamma) = \sum_{\gamma > \Gamma} 2a_\gamma(m)^2 = 4M \frac{\Lambda'}{\Lambda}(m+1) - \sum_{\gamma \leq \Gamma} 2a_\gamma(m)^2,$$

the subtracted sum being exact up to the 25-digit accuracy of the computed ordinates (the resulting uncertainty is below 10^{-20}). Table 2 gives the exact variances against the law of the introduction. The smoothed tail estimate used previously deviates from the exact tail by at most 1.5×10^{-4} relative to σ_m^2 over the computed range (worst at $m = 100$, where the tail carries 16.7% of the variance). Re-running the Gil–Pelaez inversion with the exact tail variances moves the densities by a few units in their sixth decimal — well within the 10^{-4} Monte Carlo cross-check, but not below the printed resolution: the six-decimal values from the earlier Gaussianized-tail computation deviate from the enclosure midpoints of Section 7 by up to 3.5×10^{-6} (at $(q, m) = (3, 1)$; last column of Table 5), of which the estimated tail variance accounts for -3.2×10^{-6} at that worst point and the coarser quadrature of the original inversion for the small remainder, so ten of the fourteen sixth decimals change by up to three units. Table 1 prints the corrected values (the enclosure midpoints rounded to six decimals), and the tail-variance estimation error in them is exactly zero. For the record, the identity also gives $\sigma_0^2 = 0.1555679799235859 \dots$ (the rounded 0.15557 quoted in Section 2 is unaffected).

m	σ_m^2 (exact, (4))	$2M \log \frac{4M}{2\pi}$	$\sigma_m^2 - 2M \log \frac{4M}{2\pi} - 1$	$\sigma_{\text{tail}}^2(\gamma_{511})$	tail share
1	1.368555	-0.138353	0.506908	0.031353	2.3%
3	6.776846	5.608262	0.168585	0.170697	2.5%
8	29.712468	28.704219	0.008249	1.006718	3.4%
20	106.326000	105.322529	0.003470	5.854189	5.5%
100	836.874384	835.873579	0.000804	139.712453	16.7%

Table 2: Exact variances from the identity (4) against the paper’s dissolution-law main term, modulus 4. The residual column converges to 0 like $\frac{1}{12M}$ once the constant $c_0 = 1$ of Proposition 3 is subtracted; at small m it is dominated by the (exponentially small in m , but not yet small) L'/L term and higher Stirling corrections. The last columns show the exact variance of the Gaussianized tail beyond the 511th ordinate and its share of σ_m^2 .

4 The dissolution law: from heuristic to theorem

The introduction stated the dissolution law $\delta_q(m) - \frac{1}{2} \sim (2\pi \text{Var}_q(m))^{-1/2}$ as a Gaussian-comparison heuristic with numerical confirmation, and Proposition 3 made its variance input $\text{Var}_q(m) = 2M \log \frac{qM}{2\pi} + O(1)$ rigorous. This section proves the density half, with a rate, and in fact with a proved second-order term. The method is direct Fourier inversion: the limiting variable is $1 + V$ with V a sum of independent scaled cosines, its characteristic function is a product of Bessel factors $J_0(2a_\gamma t)$, and

$$\delta_q(m) - \frac{1}{2} = \frac{1}{\pi} \int_0^\infty \Phi_m(t) \frac{\sin t}{t} dt, \quad \Phi_m(t) = \prod_{\gamma>0} J_0(2a_\gamma(m)t).$$

Near $t = 0$ the product is Gaussian to high accuracy, with variance σ_m^2 and a fourth-cumulant correction governed by $S_4(m) := \sum_{\gamma>0} (2a_\gamma(m))^4$; away from $t = 0$ the $\gtrsim M \log M$ zeros in the window $(M, 2M]$, each carrying an amplitude $a_\gamma \in [2/\sqrt{5}, \sqrt{2}]$ bounded away from 0 and ∞ , force superexponential decay. Both σ_m^2 and $S_4(m)$ turn out to be exact values of the completed L -function (Proposition 2 and Lemma 6), so every coefficient below is a closed form.

Setting and statement

Let χ be a real primitive character mod $q \geq 3$, $\chi(-1) = (-1)^a$ with $a \in \{0, 1\}$, and let $\xi(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$ be its completed L -function. We assume $\text{GRH}(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$ (for $q = 3, 4$ the nonvanishing is unconditional, as noted in the proof of Proposition 3); γ runs over the positive ordinates of $L(s, \chi)$, listed with multiplicity. For $m > 0$ put $M = m + \frac{1}{2}$,

$$a_\gamma(m) = \frac{2M}{\sqrt{M^2 + \gamma^2}}, \quad X_{m, \chi} = 1 + V_m, \quad V_m = \sum_{\gamma > 0} 2a_\gamma(m) \cos \theta_\gamma,$$

with θ_γ i.i.d. uniform on $[0, 2\pi)$ (the series converges a.s. and in L^2 , as in Section 2), and

$$\delta_q(m) := \mathbb{P}(X_{m, \chi} > 0), \quad \sigma_m^2 := \sum_{\gamma > 0} 2a_\gamma(m)^2, \quad S_4(m) := \sum_{\gamma > 0} (2a_\gamma(m))^4.$$

For every real primitive χ under the standing hypotheses, Theorem 1(ii) identifies $\delta_q(m)$, under $\text{LI}(\chi)$, with the logarithmic density $\delta_\chi(m)$ of the set where the weighted race $R_{m, \chi}$ is positive ($q = 4$ and $q = 3$ are the computed cases). Note that the theorem below does *not* assume LI : independence of the phases is built into the definition of $X_{m, \chi}$, and LI is what transfers the conclusion to the race.

Theorem 2 (dissolution law). *Let χ be a real primitive character mod q ; assume $\text{GRH}(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$. Then, with $\sigma_m^2 = 4M \xi'_\xi(m+1, \chi)$ as in (8) and $S_4(m)$ as in Lemma 6, as $m \rightarrow \infty$:*

$$(i) \quad \delta_q(m) - \frac{1}{2} = \frac{1}{\sqrt{2\pi\sigma_m^2}} \left(1 + O\left(\frac{1}{M \log M}\right) \right);$$

$$(ii) \quad \text{more precisely, } \delta_q(m) - \frac{1}{2} = \frac{1}{\sqrt{2\pi\sigma_m^2}} \left(1 - \frac{1}{6\sigma_m^2} - \frac{3S_4(m)}{64\sigma_m^4} + O\left(\frac{1}{(M \log M)^2}\right) \right);$$

(iii) consequently, by Proposition 3,

$$\delta_q(m) - \frac{1}{2} = \left(4\pi M \log \frac{qM}{2\pi} \right)^{-1/2} \left(1 + O\left(\frac{1}{M \log M}\right) \right).$$

All implied constants depend on q alone; they are effectively computable in principle and we have not optimized them. Nothing is claimed uniformly in q .

Corollary 3 (the race itself, $q = 4$). *Assume $\text{GRH}(\chi_4)$ and $\text{LI}(\chi_4)$. Then the logarithmic density $\delta(m)$ of $\{x \geq 2 : R_m(x) > 0\}$ of Theorem 1 satisfies (i)–(iii) of Theorem 2 verbatim. In particular $\delta(m) \rightarrow \frac{1}{2}$ as $m \rightarrow \infty$ at the rate $(4\pi M \log \frac{2M}{\pi})^{-1/2}$, closing the gap left open in Theorem 1(iii).*

Corollary 4 (closed second order). *Under the hypotheses of Theorem 2, with $L := \log \frac{qM}{2\pi}$,*

$$\delta_q(m) - \frac{1}{2} = \frac{1}{\sqrt{2\pi\sigma_m^2}} \left(1 - \frac{11}{24ML} + O\left(\frac{1}{ML^2}\right) \right) \quad (m \rightarrow \infty).$$

The relative error in (i) is thus not just $O(1/(M \log M))$ but asymptotically equal to $-\frac{11}{24}(ML)^{-1}$: the density approaches the Gaussian prediction from *below*, as every entry of Table 3 shows.

We assemble the proof from four lemmas: elementary Bessel estimates (Lemma 4); a count of zeros in the dyadic window $(M, 2M]$ (Lemma 5); the closed form and asymptotics of S_4 (Lemma 6); and the inversion formula (Lemma 7) together with the tail bound that justifies it (Lemma 8).

Lemmas

Lemma 4 (Bessel facts). *Let $J_0(z) = \sum_{k \geq 0} (-1)^k (z^2/4)^k / (k!)^2$. For real z :*

(J1) $|J_0(z)| \leq 1$, with strict inequality for $z \neq 0$; moreover for every $z_0 > 0$,

$$\rho(z_0) := \sup_{z \geq z_0} |J_0(z)| < 1.$$

(J2) $|J_0(z)| \leq \frac{8}{\pi} z^{-1/2}$ for all $z > 0$.

(J3) For $0 \leq z \leq 1$: $J_0(z) \geq \frac{3}{4}$ and $|\log J_0(z) + \frac{z^2}{4}| \leq \frac{z^4}{17}$.

(J4) For $0 \leq z \leq 1$: $|\log J_0(z) + \frac{z^2}{4} + \frac{z^4}{64}| \leq \frac{z^6}{80}$.

Proof. Throughout we use the integral representation

$$J_0(z) = \frac{2}{\pi} \int_0^{\pi/2} \cos(z \sin \varphi) d\varphi = \frac{2}{\pi} \int_0^1 \frac{\cos(zu)}{\sqrt{1-u^2}} du, \quad (9)$$

obtained by expanding $\cos(z \sin \varphi)$ into its (uniformly convergent) power series and integrating termwise with Wallis's formula $\int_0^{\pi/2} \sin^{2k} \varphi d\varphi = \frac{\pi}{2} \binom{2k}{k} 4^{-k}$, which reproduces the defining series; the second form is the substitution $u = \sin \varphi$.

(J1) The first form of (9) gives $|J_0(z)| \leq 1$. If $z \neq 0$, then $\varphi \mapsto z \sin \varphi$ is injective on $[0, \pi/2]$, so $\cos(z \sin \varphi) = \pm 1$ for at most finitely many φ , and by continuity $|J_0(z)| < 1$ strictly. For the supremum: by (J2), $|J_0(z)| \leq \frac{1}{2}$ for $z \geq 26 > (16/\pi)^2$; on the compact interval $[z_0, 26]$ the continuous function $|J_0|$ attains a maximum, which is < 1 by the strict inequality just proved. Hence $\rho(z_0) = \max(\max_{[z_0, 26]} |J_0|, \frac{1}{2}) < 1$ (for $z_0 \geq 26$ the compact piece is vacuous and $\rho(z_0) \leq \frac{1}{2}$ directly). (No numerical value of ρ is used in the proofs; for orientation, $\rho(1/(2\sqrt{5})) = J_0(1/(2\sqrt{5})) = 0.98754$, computed in the verification script.)

(J2) For $0 < z < 1$ the bound exceeds $8/\pi > 1$ and is trivial. For $z \geq 1$ set $\delta = 1/z \in (0, 1]$ and split (9) at $u = 1 - \delta$. The singular piece:

$$\int_{1-\delta}^1 \frac{du}{\sqrt{1-u^2}} \leq \int_{1-\delta}^1 \frac{du}{\sqrt{1-u}} = 2\sqrt{\delta}.$$

The oscillatory piece, integrating by parts ($\cos(zu) = \frac{1}{z} \frac{d}{du} \sin(zu)$) and using $1-u^2 \geq \delta(2-\delta) \geq \delta$ at $u = 1 - \delta$ together with $\int_0^{1-\delta} u(1-u^2)^{-3/2} du = (1-u^2)^{-1/2} \Big|_0^{1-\delta} \leq \delta^{-1/2}$:

$$\left| \int_0^{1-\delta} \frac{\cos(zu)}{\sqrt{1-u^2}} du \right| \leq \frac{1}{z} \cdot \frac{1}{\sqrt{\delta}} + \frac{1}{z} \int_0^{1-\delta} \frac{u du}{(1-u^2)^{3/2}} \leq \frac{2}{z\sqrt{\delta}}.$$

With $\delta = 1/z$ both pieces are $2z^{-1/2}$, so $|J_0(z)| \leq \frac{2}{\pi} \cdot 4z^{-1/2} = \frac{8}{\pi} z^{-1/2}$.

(J3) For $|z| \leq 1$ the series terms $c_k = (z^2/4)^k / (k!)^2$ decrease strictly ($c_{k+1}/c_k = z^2/(4(k+1)^2) \leq \frac{1}{4}$), so the alternating series gives

$$1 - \frac{z^2}{4} \leq J_0(z) \leq 1 - \frac{z^2}{4} + \frac{z^4}{64}, \quad \text{hence } J_0(z) \geq \frac{3}{4},$$

and, writing $J_0 = 1 - w$ with $w = \frac{z^2}{4} - R$,

$$R := \sum_{k \geq 2} (-1)^k c_k \in [0, \frac{z^4}{64}], \quad w \in [\frac{z^2}{4} - \frac{z^4}{64}, \frac{z^2}{4}] \subset [0, \frac{1}{4}].$$

Then $\log J_0 + \frac{z^2}{4} = R - \sum_{j \geq 2} \frac{w^j}{j}$, and $\sum_{j \geq 2} \frac{w^j}{j} \leq \frac{w^2}{2(1-w)} \leq \frac{2}{3} w^2 \leq \frac{2}{3} (\frac{z^2}{4})^2 = \frac{z^4}{24}$, so the total is at most $\frac{z^4}{64} + \frac{z^4}{24} = \frac{11z^4}{192} \leq \frac{z^4}{17}$.

(J4) Refine each piece by one order. First, $R - \frac{z^4}{64} = \sum_{k \geq 3} (-1)^k c_k$ has absolute value at most $c_3 = \frac{z^6}{2304}$. Next, $w^2 = \frac{z^4}{16} - \frac{z^2 R}{2} + R^2$, so

$$\log J_0 + \frac{z^2}{4} + \frac{z^4}{64} = \left(R - \frac{z^4}{64}\right) + \frac{z^2 R}{4} - \frac{R^2}{2} - \sum_{j \geq 3} \frac{w^j}{j},$$

and, for $|z| \leq 1$: $\frac{z^2 R}{4} \leq \frac{z^6}{256}$, $\frac{R^2}{2} \leq \frac{z^8}{8192} \leq \frac{z^6}{8192}$, and $\sum_{j \geq 3} \frac{w^j}{j} \leq \frac{w^3}{3(1-w)} \leq \frac{4}{9} (\frac{z^2}{4})^3 = \frac{z^6}{144}$. The sum of the four bounds is $z^6 (\frac{1}{2304} + \frac{1}{256} + \frac{1}{8192} + \frac{1}{144}) < 0.0115 z^6 < \frac{z^6}{80}$. (The true limit ratio as $z \rightarrow 0$ is $\frac{1}{576}$; the script checks the claimed bounds on a grid in multiprecision, float64 being insufficient near $z = 0$.) $\square$

Lemma 5 (zeros in the dyadic window). *Unconditionally, with $N_\chi(T) := \#\{\gamma \in (0, T]\}$, there is $M_0(q)$ such that for all $M \geq M_0(q)$,*

$$N_0 := N_\chi(2M) - N_\chi(M) \geq \frac{M}{4\pi} \log M \quad (\geq 5).$$

Proof. For a primitive $\chi \bmod q$ the Riemann–von Mangoldt formula gives $N_\chi(T) = \frac{T}{2\pi} \log \frac{qT}{2\pi e} + O(\log qT)$ for $T \geq 2$ [17, §16] (Davenport counts zeros with $|\gamma| \leq T$; for real χ the multiset of ordinates is symmetric under $\gamma \mapsto -\gamma$, as used in Proposition 2, so the one-sided count is half). Hence

$$N_\chi(2M) - N_\chi(M) = \frac{M}{2\pi} \left(\log \frac{qM}{2\pi e} + 2 \log 2 \right) + O_q(\log M) \geq \frac{M}{2\pi} \log \frac{qM}{2\pi e} - C_q \log M,$$

which exceeds $\frac{M}{4\pi} \log M$ once $M \geq M_0(q)$. $\square$

Lemma 6 (fourth moment of the amplitudes). *Assume $GRH(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$. Then for every $m > -\frac{1}{2}$,*

$$S_4(m) = \sum_{\gamma > 0} \frac{256 M^4}{(M^2 + \gamma^2)^2} = 16 \sigma_m^2 - 64 M^2 (\log \xi)''(m+1, \chi), \quad (10)$$

and, as $m \rightarrow \infty$,

$$S_4(m) = 32M \left(\log \frac{qM}{2\pi} - 1 \right) + 32(2a-1) + O_q\left(\frac{1}{M}\right). \quad (11)$$

Moreover, unconditionally on the size of m , $S_4(m) \leq 32 \sigma_m^2$ and $S_6(m) := \sum_\gamma (2a_\gamma)^6 \leq 16 S_4(m)$.

Proof. The trivial bounds are immediate from $(2a_\gamma)^2 \leq 16$ and $\sum_\gamma (2a_\gamma)^2 = 2\sigma_m^2$.

For (10): exactly as in the proof of the closed form for C (Corollary 2, applied to χ in place of χ_4 as in Proposition 3), the series $(\log \Xi)'(x) = \sum_{\gamma>0} 2x/(x^2 + \gamma^2)$, $\Xi(z) := \xi(\frac{1}{2} + z, \chi)$, may be differentiated termwise on a neighbourhood of $x = M$: the differentiated terms $2(\gamma^2 - x^2)/(x^2 + \gamma^2)^2$ are dominated by $2/(x^2 + \gamma^2) \leq 2/\gamma^2$, uniformly on compacts. Hence

$$(\log \Xi)''(M) = \sum_{\gamma>0} \frac{2(\gamma^2 - M^2)}{(M^2 + \gamma^2)^2},$$

and the identity is termwise algebra:

$$16 \cdot \frac{8M^2}{M^2 + \gamma^2} - 64M^2 \cdot \frac{2(\gamma^2 - M^2)}{(M^2 + \gamma^2)^2} = \frac{128M^2(M^2 + \gamma^2) - 128M^2\gamma^2 + 128M^4}{(M^2 + \gamma^2)^2} = \frac{256M^4}{(M^2 + \gamma^2)^2},$$

summed over $\gamma > 0$, using $\sigma_m^2 = \sum_\gamma 8M^2/(M^2 + \gamma^2)$ (Proposition 2, (8)) and $(\log \Xi)''(M) = (\log \xi)''(m+1, \chi)$.

For (11): $(\log \xi)''(s) = \frac{1}{4}\psi'(\frac{s+a}{2}) + (\frac{L'}{L})'(s, \chi)$. At $s = m+1$, $x := \frac{m+1+a}{2} = \frac{M}{2} + \frac{2a+1}{4}$, and Euler–Maclaurin for $\psi'(x) = \sum_{n>0} (n+x)^{-2}$ (comparison with $\int_0^\infty (t+x)^{-2} dt = 1/x$; correction $f(0)/2 = 1/(2x^2)$; remainder at most $\frac{1}{12}|f'(0)| + \frac{1}{12}\int_0^\infty |f''| \leq \frac{1}{2x^3}$) gives $\psi'(x) = \frac{1}{x} + \frac{1}{2x^2} + O(x^{-3})$, whence

$$\frac{1}{4}\psi'(x) = \frac{1}{2M} \cdot \frac{1}{1 + \frac{2a+1}{2M}} + \frac{1}{2M^2} + O\left(\frac{1}{M^3}\right) = \frac{1}{2M} - \frac{2a+1}{4M^2} + \frac{1}{2M^2} + O\left(\frac{1}{M^3}\right) = \frac{1}{2M} + \frac{1-2a}{4M^2} + O\left(\frac{1}{M^3}\right),$$

so $64M^2 \cdot \frac{1}{4}\psi'(x) = 32M + 16(1-2a) + O(1/M)$. The Dirichlet series $(\frac{L'}{L})'(s, \chi) = \sum_{n \geq 2} \Lambda(n)\chi(n)(\log n)n^{-s}$ is bounded at $s = m+1$ by $2^{-(m-1)} \sum_n \Lambda(n)(\log n)n^{-2} \ll 2^{-m}$, so $64M^2(\frac{L'}{L})'(m+1) \ll M^2 2^{-m} = O(1/M)$ eventually. Combining with $16\sigma_m^2 = 32M \log \frac{qM}{2\pi} + 16(2a-1) + O(1/M)$ (Proposition 3) yields (11):

$$S_4(m) = 32M \log \frac{qM}{2\pi} + 16(2a-1) - 32M - 16(1-2a) + O(1/M). \quad \square$$

Numerically (10) and (11) agree to the expected order: at $(q, m) = (4, 100)$, the closed form gives $S_4 = 10190.02$, the 511 ordinates plus Riemann–von Mangoldt tail give 10190.11, and (11) without its $O(1/M)$ gives 10189.98.

Lemma 7 (characteristic function and inversion). *Let $V_m = \sum_{\gamma>0} 2a_\gamma(m) \cos \theta_\gamma$ as above. Then:*

(a) *the law of V_m is symmetric, and its characteristic function is $\Phi_m(t) = \prod_{\gamma>0} J_0(2a_\gamma(m)t)$, the limit of the partial products, which is real and even; moreover for every finite set F of positive ordinates, $|\Phi_m(t)| \leq \prod_{\gamma \in F} |J_0(2a_\gamma(m)t)|$;*

(b) *if $\Phi_m \in L^1(\mathbb{R})$, then V_m has a bounded continuous even density and*

$$\delta_q(m) - \frac{1}{2} = \mathbb{P}(V_m > -1) - \frac{1}{2} = \frac{1}{\pi} \int_0^\infty \Phi_m(t) \frac{\sin t}{t} dt,$$

the integral converging absolutely.

Proof. (a) For a single summand, $\mathbb{E} e^{it \cdot 2a \cos \theta} = \frac{1}{2\pi} \int_0^{2\pi} e^{2iat \cos \theta} d\theta = J_0(2at)$: expand the exponential and integrate termwise (odd powers vanish; $\frac{1}{2\pi} \int_0^{2\pi} \cos^{2k} \theta d\theta = \binom{2k}{k} 4^{-k}$), recovering the series of J_0 ; in particular each summand is symmetric ($\theta \mapsto \theta + \pi$ preserves the uniform law and flips the sign), hence so are the partial sums $S_N = \sum_{n \leq N} 2a_{\gamma_n} \cos \theta_{\gamma_n}$ and their a.s. limit V_m . Independence gives the characteristic function of S_N as the N -th partial product; $S_N \rightarrow V_m$ a.s. implies convergence in distribution, so the characteristic functions converge pointwise to that of V_m , which is therefore the limit of the partial products, real (symmetry) and even. For the subset bound, take N large enough that the first N ordinates contain F ; all factors have modulus ≤ 1 by (J1), so the modulus of the N -th partial product is at most $\prod_{\gamma \in F} |J_0|$, and let $N \rightarrow \infty$.

(b) By the Fourier inversion theorem for integrable characteristic functions [26, Ch. XV.3], the law of V_m has the bounded continuous density $f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \Phi_m(t) dt$, which is even since Φ_m is real and even. Hence, using symmetry and then Fubini (justified by $\int_{\mathbb{R}} \int_{-1}^1 |\Phi_m(t)| dx dt = 2\|\Phi_m\|_1 < \infty$),

$$\mathbb{P}(V_m > -1) = \frac{1}{2} + \frac{1}{2} \mathbb{P}(|V_m| < 1) = \frac{1}{2} + \frac{1}{2} \int_{-1}^1 f = \frac{1}{2} + \frac{1}{4\pi} \int_{\mathbb{R}} \Phi_m(t) \left(\int_{-1}^1 e^{-itx} dx \right) dt,$$

and $\int_{-1}^1 e^{-itx} dx = 2 \sin t/t$, so $\mathbb{P}(V_m > -1) - \frac{1}{2} = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi_m(t) \frac{\sin t}{t} dt = \frac{1}{\pi} \int_0^\infty \Phi_m(t) \frac{\sin t}{t} dt$ by evenness. Finally $\delta_q(m) = \mathbb{P}(1 + V_m > 0) = \mathbb{P}(V_m > -1)$. $\square$

Lemma 8 (superexponential decay for $t \geq \frac{1}{8}$). *Assume $GRH(\chi)$. Let $\rho := \rho(1/(2\sqrt{5})) < 1$ and $\tilde{\rho} := \max(\rho, \frac{1}{2}) < 1$ (absolute constants, by (J1)). There is $M_0(q)$ such that for all m with $M \geq M_0(q)$,*

$$\int_{1/8}^\infty |\Phi_m(t)| \frac{|\sin t|}{t} dt \leq 17 \tilde{\rho}^{N_0} \leq \exp\left(-\frac{\log(1/\tilde{\rho})}{8\pi} M \log M\right),$$

with $N_0 = N_\chi(2M) - N_\chi(M)$ as in Lemma 5; in particular $\Phi_m \in L^1(\mathbb{R})$.

Proof. For $\gamma \in (M, 2M]$ we have $M^2 + \gamma^2 \in (2M^2, 5M^2]$, so

$$a_\gamma(m) = \frac{2M}{\sqrt{M^2 + \gamma^2}} \in \left[\frac{2}{\sqrt{5}}, \sqrt{2} \right), \quad 2a_\gamma(m)t \geq \frac{4}{\sqrt{5}} t.$$

By Lemma 7(a) with $F = \{\gamma \in (M, 2M]\}$ (a set of N_0 ordinates, nonempty for $M \geq M_0(q)$): (i) for $t \geq \frac{1}{8}$, every factor has argument $\geq \frac{4}{\sqrt{5}} \cdot \frac{1}{8} = \frac{1}{2\sqrt{5}}$, so by (J1) $|\Phi_m(t)| \leq \rho^{N_0}$; (ii) for any $t > 0$, by (J2),

$$|\Phi_m(t)| \leq \left(\frac{8}{\pi} \left(\frac{4t}{\sqrt{5}} \right)^{-1/2} \right)^{N_0} = \left(\frac{c_3}{\sqrt{t}} \right)^{N_0}, \quad c_3 := \frac{4 \cdot 5^{1/4}}{\pi} = 1.9040 \dots < 2,$$

so that $c_3/\sqrt{t} \leq \frac{1}{2}$ for $t \geq 16$. Using $|\sin t|/t \leq 1$ on $[\frac{1}{8}, 16]$ and $|\sin t| \leq 1$ beyond, and (for $M \geq M_0(q)$, by Lemma 5) $N_0 \geq 5$:

$$\int_{1/8}^{16} |\Phi_m| \frac{|\sin t|}{t} dt \leq 16 \rho^{N_0}, \quad \int_{16}^\infty |\Phi_m| \frac{dt}{t} \leq 2^{5-N_0} c_3^5 \int_{16}^\infty t^{-7/2} dt = 2^{5-N_0} c_3^5 \cdot \frac{2}{5 \cdot 1024} \leq \frac{2^{-N_0}}{2},$$

where the middle step keeps five factors $(c_3/\sqrt{t})^5$ and bounds the remaining $N_0 - 5$ by $2^{-(N_0-5)}$. The sum is at most $17\tilde{\rho}^{N_0}$, and with $N_0 \geq \frac{M}{4\pi} \log M$ the final bound follows for $M \geq M_0(q)$ enlarged if necessary (to absorb the factor 17 into half the exponent). The same estimates with $|\sin t|/t$ replaced by 1 on $[\frac{1}{8}, 16]$ and by nothing beyond show $\int_{1/8}^\infty |\Phi_m| < \infty$; on $[0, \frac{1}{8}]$, $|\Phi_m| \leq 1$. Hence $\Phi_m \in L^1$. $\square$

Proof of Theorem 2

Proof. Throughout, $M \geq M_0(q)$ so that Lemmas 5 and 8 apply, and $G := (2\pi\sigma_m^2)^{-1/2}$. By Lemma 8, $\Phi_m \in L^1$, so Lemma 7(b) gives

$$\delta_q(m) - \frac{1}{2} = \frac{1}{\pi} \int_0^{1/8} \Phi_m(t) \frac{\sin t}{t} dt + O(e^{-c_q M \log M}), \quad c_q := \frac{\log(1/\tilde{\rho})}{8\pi}. \quad (12)$$

The Gaussian factorization on $[0, \frac{1}{8}]$. For $0 \leq t \leq \frac{1}{8}$ every argument satisfies $2a_\gamma(m)t \leq 4t \leq \frac{1}{2}$, since $a_\gamma \leq 2$. By (J3) each factor is $\geq \frac{3}{4} > 0$ and $\sum_\gamma |\log J_0(2a_\gamma t)| \leq \sum_\gamma (a_\gamma^2 t^2 + (2a_\gamma t)^4/17) < \infty$, so the limit of the partial products is

$$\Phi_m(t) = \exp\left(-\frac{\sigma_m^2 t^2}{2} + E(t)\right), \quad E(t) := \sum_{\gamma > 0} \left(\log J_0(2a_\gamma t) + \frac{(2a_\gamma t)^2}{4}\right), \quad (13)$$

using $\sum_\gamma (2a_\gamma)^2 t^2/4 = \sigma_m^2 t^2/2$. By (J3), $|E(t)| \leq \frac{t^4}{17} S_4$, and since $S_4 \leq 32\sigma_m^2$ (Lemma 6) and $t \leq \frac{1}{8}$,

$$|E(t)| \leq \frac{t^2}{17} \cdot \frac{1}{64} \cdot 32\sigma_m^2 = \frac{\sigma_m^2 t^2}{34} \quad (0 \leq t \leq \frac{1}{8}), \quad (14)$$

so that $|\Phi_m(t)| \leq e^{-(8/17)\sigma_m^2 t^2}$ there: the Gaussian decay survives the correction uniformly.

Decomposition. Write, on $[0, \frac{1}{8}]$, $\Phi_m \frac{\sin t}{t} = e^{-\sigma^2 t^2/2} - e^{-\sigma^2 t^2/2} \left(1 - \frac{\sin t}{t}\right) + e^{-\sigma^2 t^2/2} (e^E - 1) \frac{\sin t}{t}$ (abbreviating $\sigma = \sigma_m$), so that

$$\frac{1}{\pi} \int_0^{1/8} \Phi_m \frac{\sin t}{t} dt = G - T_1 - T_2 + T_3, \quad (15)$$

with, using $\frac{1}{\pi} \int_0^\infty e^{-\sigma^2 t^2/2} dt = G$ and the moments $\frac{1}{\pi} \int_0^\infty t^{2k} e^{-\sigma^2 t^2/2} dt = (2k-1)!! G \sigma^{-2k}$:

T_1 : the completed Gaussian tail, $0 \leq T_1 = \frac{1}{\pi} \int_{1/8}^\infty e^{-\sigma^2 t^2/2} dt \leq \frac{8}{\pi\sigma^2} e^{-\sigma^2/128}$;

T_2 : the sin correction: since $0 \leq 1 - \frac{\sin t}{t} \leq \frac{t^2}{6}$ for all $t > 0$ (from $t - \frac{t^3}{6} \leq \sin t \leq t$),

$$0 \leq T_2 = \frac{1}{\pi} \int_0^{1/8} e^{-\sigma^2 t^2/2} \left(1 - \frac{\sin t}{t}\right) dt \leq \frac{1}{6\pi} \int_0^\infty t^2 e^{-\sigma^2 t^2/2} dt = \frac{G}{6\sigma^2};$$

T_3 : the Bessel correction: by (14) and $|e^w - 1| \leq |w|e^{|w|}$,

$$|T_3| \leq \frac{1}{\pi} \int_0^{1/8} e^{-\sigma^2 t^2/2} \cdot \frac{t^4 S_4}{17} e^{\sigma^2 t^2/34} dt \leq \frac{S_4}{17\pi} \int_0^\infty t^4 e^{-\frac{8}{17}\sigma^2 t^2/2} dt = \frac{3S_4}{8 \cdot 17\sqrt{\pi}} \left(\frac{17}{8}\right)^{5/2} \sigma^{-5} \leq \frac{S_4}{12\sigma^5} \leq \frac{G S_4}{4\sigma^4},$$

the last step by $\sigma^{-5} = \sqrt{2\pi} G \sigma^{-4}$ and $\sqrt{2\pi}/12 < \frac{1}{4}$.

Part (i). By Proposition 3 there is $M_1(q) \geq M_0(q)$ with $ML \leq \sigma_m^2 \leq 3ML$ for $M \geq M_1(q)$, where $L = \log \frac{qM}{2\pi} \asymp_q \log M$; and $S_4 \leq 33ML$ by (11). Hence

$$\frac{T_2}{G} \leq \frac{1}{6ML}, \quad \frac{|T_3|}{G} \leq \frac{33ML}{4M^2L^2} \leq \frac{9}{ML}, \quad \frac{T_1 + O(e^{-c_q M \log M})}{G} \ll \sigma e^{-\sigma^2/128} + \sigma e^{-c_q M \log M} \leq \frac{1}{ML}$$

for M large, and (12)–(15) give $\delta_q(m) - \frac{1}{2} = G(1 + O_q(1/(ML)))$, which is (i).

Part (ii). Refine T_2 and T_3 by one order each. For T_2 : on $[0, \frac{1}{8}]$ the alternating series of $\frac{\sin t}{t}$ has decreasing terms, so $|1 - \frac{\sin t}{t} - \frac{t^2}{6}| \leq \frac{t^4}{120}$, and extending the integral to ∞ costs $O(e^{-\sigma^2/129})$:

$$T_2 = \frac{1}{6\pi} \int_0^\infty t^2 e^{-\sigma^2 t^2/2} dt + O\left(\frac{1}{120\pi} \int_0^\infty t^4 e^{-\sigma^2 t^2/2} dt\right) + O(e^{-\sigma^2/129}) = \frac{G}{6\sigma^2} + O(G\sigma^{-4}).$$

For T_3 : by (J4), $E(t) = -\frac{t^4}{64}S_4 + E_2(t)$ with $|E_2(t)| \leq \frac{t^6}{80}S_6 \leq \frac{t^6}{5}S_4$ (Lemma 6); and $e^E - 1 = E + \omega$ with $|\omega| \leq \frac{E^2}{2}e^{|E|} \leq \frac{t^8 S_4^2}{578} e^{\sigma^2 t^2/34}$ by (14). Hence

$$T_3 = -\frac{S_4}{64\pi} \int_0^{1/8} t^4 e^{-\sigma^2 t^2/2} \frac{\sin t}{t} dt + O\left(\frac{S_4}{5\pi} \int_0^\infty t^6 e^{-\sigma^2 t^2/2} dt\right) + O\left(\frac{S_4^2}{578\pi} \int_0^\infty t^8 e^{-\frac{8}{17}\sigma^2 t^2} dt\right),$$

and in the first integral $\frac{\sin t}{t} = 1 + O(t^2)$, so, with the moment values and extension to ∞ as before,

$$T_3 = -\frac{3S_4}{64} G \sigma^{-4} + O(G S_4 \sigma^{-6}) + O(G S_4^2 \sigma^{-8}).$$

Since $\sigma^2 \asymp_q ML$ and $S_4 \asymp_q ML$, the error terms $G \cdot O(\sigma^{-4} + S_4 \sigma^{-6} + S_4^2 \sigma^{-8})$ are all $G \cdot O_q((ML)^{-2})$, and the exponentially small terms T_1 , (12) are absorbed. Collecting,

$$\delta_q(m) - \frac{1}{2} = G\left(1 - \frac{1}{6\sigma_m^2} - \frac{3S_4}{64\sigma_m^4} + O_q((M \log M)^{-2})\right).$$

Part (iii). By Proposition 3, $\sigma_m^2 = 2ML + O(1)$, so $(2\pi\sigma_m^2)^{-1/2} = (4\pi ML)^{-1/2}(1 + O(1/(ML)))$; combine with (i). $\square$

Proof of Corollary 3. $L(\frac{1}{2}, \chi_4) > 0$ unconditionally, so Theorem 2 applies to χ_4 ; and under $\text{GRH}(\chi_4) + \text{LI}(\chi_4)$, Theorem 1(ii) gives $\delta(m) = \nu_m((0, \infty)) = \mathbb{P}(X_{m, \chi_4} > 0) = \delta_4(m)$. $\square$

Proof of Corollary 4. Insert $\sigma_m^2 = 2ML + O(1)$ and $S_4 = 32M(L - 1) + O(1)$ (Proposition 3, Lemma 6) into (ii):

$$\frac{1}{6\sigma_m^2} = \frac{1}{12ML} + O\left(\frac{1}{M^2L^2}\right), \quad \frac{3S_4}{64\sigma_m^4} = \frac{96M(L - 1)}{256M^2L^2} + O\left(\frac{1}{M^2L^2}\right) = \frac{3}{8ML} + O\left(\frac{1}{ML^2}\right),$$

and $\frac{1}{12} + \frac{3}{8} = \frac{11}{24}$; the error $O((ML)^{-2})$ of (ii) is also $O(1/(ML^2))$. $\square$

Remark 6 (numerical consistency). *Table 3 compares the theorem with deterministic Gil-Pelaez reference densities (computational, not part of any proof): 511 resp. 537 ordinates, unseen zeros Gaussianized with the exact tail variance of Remark 5, trapezoidal inversion with grid-halving and window-doubling stable to 10^{-9} . Both σ_m^2 and S_4 are evaluated by their closed forms (8), (10) — no zero data enters the predictions. Gaussianizing the unseen zeros biases*

the reference values themselves: on $t \leq \frac{1}{8}$ the T_3 estimate, applied to the tail factors alone, rigorously bounds the relative bias by $0.21 S_4^{\text{tail}}/\sigma_m^4$; on $t > \frac{1}{8}$ that estimate does not apply, and we instead bound the remaining bias by the same integrand with the computed $|\Phi_m|$ inserted, which is below 9×10^{-7} at every table point. The total — 9.9×10^{-6} at $(q, m) = (4, 100)$, less elsewhere in the table — is the accuracy floor of the comparison; the $t > \frac{1}{8}$ piece is a numerical evaluation, not a theorem, on the same footing as the reference densities themselves. The validated enclosures of Table 5, computed by an independent route, contain the (unrounded) reference values used here at all four table points (e.g. $0.51377836 \in [0.5137782134, 0.5137783837]$ at $(4, 100)$; the table entry 0.5137784 is that reference rounded to seven decimals and lies just above the enclosure’s upper end, consistent with the one-sided Gaussianization bias of the reference noted above).

(q, m)	σ_m^2	S_4	$\delta_q(m)$ (ref.)	$\frac{\delta_q(m) - \frac{1}{2}}{(2\pi\sigma_m^2)^{-1/2}} - 1$	$-\frac{1}{6\sigma_m^2} - \frac{3S_4}{64\sigma_m^4}$	residual
(4, 8)	29.7125	219.83	0.5718701	-1.80×10^{-2}	-1.73×10^{-2}	-7.4×10^{-4}
(4, 20)	106.3260	1061.32	0.5384553	-6.05×10^{-3}	-5.97×10^{-3}	-7.9×10^{-5}
(4, 100)	836.8744	10190.02	0.5137784	-8.80×10^{-4}	-8.81×10^{-4}	$+7.0 \times 10^{-7}$
(3, 60)	407.9701	4607.56	0.5197175	-1.71×10^{-3}	-1.71×10^{-3}	-4.9×10^{-6}

Table 3: The dissolution law against deterministic reference densities. Column 5 is the observed relative deviation from the leading term of Theorem 2(i); column 6 is the second-order term of (ii); the residual — the relative error of the two-term prediction, equal to (col. 5 – col. 6)/(1 + col. 6) — shrinks like $(M \log M)^{-2}$, as (ii) requires — indeed at $(4, 100)$ it is below the 9.9×10^{-6} modelling floor of the reference value, i.e. (ii) is confirmed there to the full accuracy of the computation. The $m = 100$ row is the consistency check promised in the introduction: $(2\pi \cdot 836.8744)^{-1/2} = 0.0137905$ against $\delta_4(100) - \frac{1}{2} = 0.0137784$, a relative gap of 8.8×10^{-4} , against the theorem’s error scale $1/(M \log \frac{qM}{2\pi}) = 1/418 = 2.4 \times 10^{-3}$.

Corollary 4 at $(4, 100)$: $-\frac{11}{24}(ML)^{-1} = -1.10 \times 10^{-3}$ against the observed -8.80×10^{-4} ; the difference, 2.2×10^{-4} , is consistent with the corollary’s $O(1/(ML^2)) = O(5.8 \times 10^{-4})$ — at these heights $L \approx 4.2$ is still small, which is why the table works with (ii) rather than the corollary. The reference values $\delta_4(100) = 0.5137784$ and $\delta_3(60) = 0.5197175$ are seven-decimal refinements of the corresponding entries of Table 1, 0.513778 and $0.519717^\dagger$ (the dagger marking the sixth decimal left unresolved by the enclosure).

Remark 7 (scope, and what is not proved). (1) Race interpretation. *Theorem 1 is stated and proved for every real primitive χ (under $\text{GRH}(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$), so under the additional hypothesis $\text{LI}(\chi)$ its part (ii) transfers Theorem 2 to the weighted prime race itself for any such χ ; Corollary 3 records the case $q = 4$, and the mod-3 densities of Table 1 instantiate $q = 3$. Without $\text{LI}(\chi)$, Theorem 2 is, as before, a theorem about the limiting random variable $X_{m, \chi}$ only. (2) Hypotheses. LI is not among the hypotheses of Theorem 2; GRH enters through the reality of the ordinates and the closed forms (8), (10), and $L(\frac{1}{2}, \chi) \neq 0$ (automatic for $q = 3, 4$) through the Hadamard product. (3) Constants. The constants are effective in principle (ρ is a computable supremum, numerically 0.98754 ; the Riemann–von Mangoldt constants for fixed q are classical) but the large- t bound uses only the zeros in $(M, 2M]$ and is extremely wasteful; no attempt has been made to make $M_0(q)$ small, and at the heights of Table 3 the*

proved bounds on the tail terms are not yet numerically dominant — the table is a consistency check on the asymptotics, not a validated enclosure. (4) Uniformity. A version uniform in q requires the window count and Proposition 3 with explicit q -dependence; this is carried out in Section 5 (cf. the q -aspect asymptotics of Fiorilli–Martin [9], whose densities-near- $\frac{1}{2}$ regime is the natural companion statement). (5) The reference densities of Remark 6 remain conditional computations under GRH+LI, as throughout this literature.

5 A joint (q, m) -uniform dissolution law

Theorem 2 dissolves the bias in the weight aspect: q fixed, $m \rightarrow \infty$, with all constants depending on q . The classical Chebyshev literature dissolves it in the modulus aspect: the weight fixed, $q \rightarrow \infty$, as in the asymptotic series of Fiorilli and Martin [9] for the unweighted race. This section proves one estimate containing both aspects and their joint interior: the Gaussian prediction $(2\pi\sigma_{m,\chi}^2)^{-1/2}$ approximates $\delta_\chi(m) - \frac{1}{2}$ with *relative error* $O(1/\sigma_{m,\chi}^2)$, *uniformly over all real primitive characters and all weights m bounded below*, with no lower bound on $\sigma_{m,\chi}^2$ required for the inequality itself. The mechanism is the observation that the Lindeberg-type ratio S_4/σ^4 obeys the *absolute* bound $32/\sigma^2$ — the amplitudes $2a_\gamma \leq 4$ are bounded by a constant free of (q, m) — so the entire small- t analysis of Section 4 is already uniform; what has to be reworked is the large- t decay, whose zero count must come from the q -uniform Riemann–von Mangoldt formula and whose window must be widened to survive small M .

Setting and statement

Throughout this section, hypothesis (H) means: χ is a real primitive character mod $q \geq 3$, GRH(χ) holds, and $L(\frac{1}{2}, \chi) \neq 0$. For $m > -\frac{1}{2}$ put $M = m + \frac{1}{2}$ and, as in Section 4,

$$a_\gamma(m) = \frac{2M}{\sqrt{M^2 + \gamma^2}}, \quad X_{m,\chi} = 1 + \sum_{\gamma>0} 2a_\gamma(m) \cos \theta_\gamma, \quad \delta_\chi(m) := \mathbb{P}(X_{m,\chi} > 0),$$

$$\sigma^2 = \sigma_{m,\chi}^2 := \sum_{\gamma>0} 2a_\gamma(m)^2 = 4M \frac{\xi'}{\xi}(m+1, \chi), \quad S_4 = S_4(m, \chi) := \sum_{\gamma>0} (2a_\gamma(m))^4,$$

the closed form for σ^2 being (8). We write δ_χ rather than δ_q because a modulus can carry two real primitive characters (e.g. $q = 8$); all bounds below are uniform over the choice, and for the moduli 3, 4 of the rest of the paper $\delta_\chi(m) = \delta_q(m)$. Set

$$\overline{M} := \max(M, 1), \quad \mathcal{L} := \log(q\overline{M}) \quad (\geq \log 3).$$

Theorem 3 (joint uniform dissolution law). *For every $\delta_0 \in (0, 1]$ there is an effectively computable constant $C(\delta_0) < \infty$ such that the following holds for every pair (χ, m) with (H) and $M = m + \frac{1}{2} \geq \delta_0$. With $G := (2\pi\sigma^2)^{-1/2}$:*

- (i) $\left| \delta_\chi(m) - \frac{1}{2} - G \right| \leq C(\delta_0) \frac{G}{\sigma^2};$
- (ii) more precisely, $\left| \delta_\chi(m) - \frac{1}{2} - G \left(1 - \frac{1}{6\sigma^2} - \frac{3S_4}{64\sigma^4} \right) \right| \leq C(\delta_0) \frac{G}{\sigma^4}.$

No lower bound on σ^2 is assumed: the inequalities hold for all $\sigma^2 > 0$ and are informative once σ^2 is large, which by Proposition 4 happens exactly when $\overline{M}\mathcal{L}$ is large. In particular, taking $\delta_0 = 1$: (i) and (ii) hold with absolute constants uniformly over all real primitive χ satisfying (H) and all $m \geq \frac{1}{2}$. An inspection of the proof shows one may take $C(\delta_0) \leq C\delta_0^{-6}$ with C absolute; we have not optimized this.

The scale of the error is calibrated by the following uniform comparability, proved after the theorem.

Proposition 4 (uniform size of σ^2 and S_4). *There is an absolute constant C_9 such that under (H), for all $M > 0$,*

$$\sigma^2 \leq C_9 \overline{M}\mathcal{L}, \quad S_4 \leq C_9 \overline{M}\mathcal{L}, \quad (16)$$

and there is an absolute constant $c_9 > 0$ such that for all $\delta_0 \in (0, 1]$ and all $M \geq \delta_0$,

$$\sigma^2 \geq c_9 \delta_0^2 \overline{M}\mathcal{L}. \quad (17)$$

Consequently, for $M \geq \delta_0$,

$$\sigma^2 \asymp_{\delta_0} \overline{M}\mathcal{L}, \quad \text{and} \quad \frac{S_4}{\sigma^4} \leq \frac{C_9}{c_9^2 \delta_0^4} \cdot \frac{1}{\overline{M}\mathcal{L}}. \quad (18)$$

Corollary 5 (the joint asymptotic). *Fix $\delta_0 \in (0, 1]$. Uniformly over all (χ, m) with (H) and $M \geq \delta_0$,*

$$\delta_\chi(m) - \frac{1}{2} = \frac{1}{\sqrt{2\pi\sigma^2}} \left(1 + O_{\delta_0} \left(\frac{1}{\overline{M} \log(q\overline{M})} \right) \right).$$

In particular:

- (a) weight aspect (q fixed, $m \rightarrow \infty$): the error is $O_q(1/(M \log M))$, recovering Theorem 2(i) — now with no threshold $M \geq M_0(q)$, since the inequality holds for all $M \geq \delta_0$;
- (b) modulus aspect ($m > -\frac{1}{2}$ fixed, $q \rightarrow \infty$ through moduli admitting a real primitive character with (H)): taking $\delta_0 = \min(M, 1)$, the error is $O_m(1/\log q)$, so $\delta_\chi(m) \rightarrow \frac{1}{2}$ at the rate $(2\pi\sigma^2)^{-1/2} \asymp_m (\log q)^{-1/2}$;
- (c) any joint passage: the right-hand error tends to 0 exactly when $q\overline{M} \rightarrow \infty$, i.e. when $q + m \rightarrow \infty$.

Corollary 6 (closed second order, uniform). *There is an absolute constant C such that under (H), for all $M \geq 16$ and $qM \geq 2\pi e^2$, with $L := \log \frac{qM}{2\pi} (\geq 2)$,*

$$\left| \delta_\chi(m) - \frac{1}{2} - \frac{1}{\sqrt{2\pi\sigma^2}} \left(1 - \frac{11}{24ML} \right) \right| \leq \frac{C}{\sqrt{2\pi\sigma^2}} \cdot \frac{1}{ML^2}.$$

Thus the constant $-\frac{11}{24}$ of Corollary 4 is not a fixed- q artifact: the approach to the Gaussian prediction is from below, by the same universal second-order coefficient, in every aspect at once.

Lemmas

The Bessel facts (J1)–(J4) of Lemma 4, the inversion formula of Lemma 7, and the bounds $S_4 \leq 32\sigma^2$, $S_6 := \sum_{\gamma} (2a_{\gamma})^6 \leq 16S_4$ of Lemma 6 involve no parameter and are used as they stand; recall also $\sup_{\gamma} 2a_{\gamma} \leq 4$. The new input is q -uniformity in the zero counts, a widened window, and a rate for the constant $\rho(z_0)$ of (J1).

Lemma 9 (uniform Riemann–von Mangoldt). *Let $N_f(T, \chi)$ denote the number of nontrivial zeros $\rho = \beta + i\gamma$ of $L(s, \chi)$ with $|\gamma| \leq T$, and under (H) let $N_{\chi}(T) := \#\{\gamma \in (0, T]\}$, so that $N_{\chi}(T) = \frac{1}{2}N_f(T, \chi)$ (the ordinates are symmetric under $\gamma \mapsto -\gamma$ as in Proposition 2, and $\gamma = 0$ does not occur since $L(\frac{1}{2}, \chi) \neq 0$). There are absolute constants C_R, C'_R such that for every primitive $\chi \bmod q \geq 3$:*

- (a) for $T \geq 2$, $\left| N_f(T, \chi) - \frac{T}{\pi} \log \frac{qT}{2\pi e} \right| \leq C_R \log(qT)$;
- (b) under (H), for all $T \geq 1$, $N_{\chi}(T) \leq C'_R T \log(qT)$.

Proof. (a) is [27, Theorem 14.5], whose implied constant is absolute — this uniformity in q is the point of the modulus-aspect literature and the reason we cite [27] rather than the fixed- q formulation used in Lemma 5 (cf. [17, §16]). It is moreover available in fully explicit form: for every character of conductor $q > 1$ and every $T \geq \frac{5}{7}$, $\left| N_f(T, \chi) - \frac{T}{\pi} \log \frac{qT}{2\pi e} \right| \leq 0.298 \log(qT) + 4.358$ [28, Corollary 1.2], so for $T \geq 2$ and $q \geq 3$ — where $\log(qT) \geq \log 6$, and $0.298 + 4.358/\log 6 < 2.74$ — one may take $C_R = 2.8$; no estimate below depends on the numerical value. (b) For $T \geq 2$, (a) gives $N_{\chi}(T) \leq \frac{T}{2\pi} \log(qT) + \frac{C_R}{2} \log(qT) \leq (\frac{1}{2\pi} + C_R) T \log(qT)$. For $1 \leq T < 2$, $N_{\chi}(T) \leq N_{\chi}(2) \leq (1 + C_R) \log(2q) \leq 2(1 + C_R) \log q \leq 2(1 + C_R) T \log(qT)$, using $\log(2q) \leq 2 \log q$ for $q \geq 2$. Take $C'_R = 2 + 2C_R$. $\square$

Lemma 10 (uniform window). *There is an absolute constant $A_1 \geq 2$ with the following property. Under (H), for every $M > 0$, set $A := A_1 \overline{M}$ and*

$$N_0 := N_{\chi}(2A) - N_{\chi}(A) = \#\{\gamma \in (A, 2A]\}.$$

Then

$$N_0 \geq \frac{A_1}{8\pi} \overline{M} \mathcal{L} \quad (\geq 7), \quad \text{and} \quad a_{\gamma}(m) \geq \frac{2}{\sqrt{5} A_1} \min(M, 1) \quad \text{for all } \gamma \in (A, 2A]. \quad (19)$$

Proof. By Lemma 9(a) at $T = A$ and $T = 2A$ (both $\geq A_1 \geq 2$),

$$2N_0 = N_f(2A) - N_f(A) \geq \frac{A}{\pi} \left(\log \frac{qA}{2\pi e} + 2 \log 2 \right) - C_R \log(2qA) - C_R \log(qA) \geq \frac{A}{\pi} \log \frac{qA}{2\pi e} - 3C_R \log(qA),$$

using $\log(2qA) \leq \frac{3}{2} \log(qA)$ (valid since $qA \geq 3A_1 \geq 4$). If $A_1 \geq (2\pi e)^2/3$ then $qA \geq 3A_1 \geq (2\pi e)^2$, so $\log \frac{qA}{2\pi e} \geq \frac{1}{2} \log(qA)$; if moreover $A \geq 12\pi C_R$ — guaranteed by $A_1 \geq 12\pi C_R$ — then $\frac{A}{2\pi} - 3C_R \geq \frac{A}{4\pi}$ and

$$N_0 \geq \frac{A}{8\pi} \log(qA) \geq \frac{A_1}{8\pi} \overline{M} \log(q\overline{M}),$$

since $A = A_1 \overline{M} \geq \overline{M}$ and $A_1 \geq 1$. Choosing

$$A_1 := \max\left(2, \frac{(2\pi e)^2}{3}, 12\pi C_R, \frac{56\pi}{\log 3}\right)$$

also forces $N_0 \geq \frac{A_1}{8\pi} \log 3 \geq 7$. For the amplitudes: $\gamma \in (A, 2A]$ and $M \leq \overline{M} \leq A$ give $M^2 + \gamma^2 \leq A^2 + 4A^2 = 5A_1^2 \overline{M}^2$, so $a_{\gamma} = 2M/\sqrt{M^2 + \gamma^2} \geq 2M/(\sqrt{5} A_1 \overline{M}) = \frac{2}{\sqrt{5} A_1} \min(M, 1)$. $\square$

Lemma 11 (rate in (J1)). *Let $\rho(z_0) = \sup_{z \geq z_0} |J_0(z)|$ as in (J1). There is an absolute constant $\kappa > 0$ such that $\log(1/\rho(z_0)) \geq \kappa z_0^2$ for all $z_0 \in (0, 1]$.*

Proof. Split the supremum at $z = 1$. On $[z_0, 1]$, (J3) gives $J_0(z) \geq \frac{3}{4} > 0$ and the alternating series bound of its proof gives $J_0(z) \leq 1 - \frac{z^2}{4} + \frac{z^4}{64} \leq 1 - \frac{z^2}{8}$ (the last step since $z^2 \leq 8$), which is decreasing in z , so $\sup_{[z_0, 1]} |J_0| \leq 1 - \frac{z_0^2}{8} \leq e^{-z_0^2/8}$. On $[1, \infty)$, $|J_0| \leq \rho(1) < 1$ by (J1). Hence $\log(1/\rho(z_0)) \geq \min(\frac{z_0^2}{8}, \log(1/\rho(1))) \geq \kappa z_0^2$ with $\kappa := \min(\frac{1}{8}, \log(1/\rho(1)))$. $\square$

Lemma 12 (uniform superexponential tail). *Under (H), $\Phi_m(t) = \prod_{\gamma > 0} J_0(2a_\gamma(m)t)$ is in $L^1(\mathbb{R})$ for every $M > 0$. Moreover, for every $\delta_0 \in (0, 1]$ there are constants $C_{11}(\delta_0)$ and $c_{10} > 0$ (c_{10} absolute) such that for all (χ, m) with (H) and $M \geq \delta_0$,*

$$\int_{1/8}^{\infty} |\Phi_m(t)| \frac{|\sin t|}{t} dt \leq C_{11}(\delta_0) \exp(-c_{10} \delta_0^2 \sigma^2), \quad C_{11}(\delta_0) \leq \frac{C}{\delta_0} \quad (C \text{ absolute}). \quad (20)$$

Proof. Let $F = \{\gamma \in (A, 2A]\}$ be the window of Lemma 10, $\#F = N_0 \geq 7$, and $\alpha := \frac{2}{\sqrt{5}A_1} \min(M, 1) > 0$, so $a_\gamma \geq \alpha$ on F . By Lemma 7(a), $|\Phi_m(t)| \leq \prod_{\gamma \in F} |J_0(2a_\gamma t)|$.

Moderate t . For $t \geq \frac{1}{8}$ every window argument satisfies $2a_\gamma t \geq \frac{\alpha}{4} =: z_0 \in (0, 1]$, so by (J1) $|\Phi_m(t)| \leq \rho(z_0)^{N_0}$ for all $t \geq \frac{1}{8}$.

Large t . By (J2) each window factor is at most $c_6 t^{-1/2}$ with $c_6 := \frac{8}{\pi} (2\alpha)^{-1/2}$; setting $T_1 := \frac{128}{\pi^2 \alpha}$ one checks $c_6 T_1^{-1/2} = \frac{1}{2}$, so for $t \geq T_1$ each factor is $\leq \frac{1}{2}$. Keeping seven factors and bounding the remaining $N_0 - 7$ by $2^{-(N_0-7)}$,

$$\int_{T_1}^{\infty} |\Phi_m| \frac{|\sin t|}{t} dt \leq 2^{-(N_0-7)} c_6^7 \int_{T_1}^{\infty} t^{-9/2} dt = 2^{-(N_0-7)} c_6^7 \cdot \frac{2}{7} T_1^{-7/2} = \frac{2}{7} \cdot 2^{-N_0} \leq 2^{-N_0},$$

using $c_6^7 T_1^{-7/2} = 2^{-7}$. On $[\frac{1}{8}, T_1]$, using $|\sin t|/t \leq 1$, $\int_{1/8}^{T_1} |\Phi_m| \frac{|\sin t|}{t} dt \leq T_1 \rho(z_0)^{N_0}$. With $\tilde{\rho} := \max(\rho(z_0), \frac{1}{2}) < 1$ the total is at most $(T_1 + 1) \tilde{\rho}^{N_0}$; this is finite for each fixed (χ, m) with $M > 0$, and the same bounds with $|\sin t|/t$ replaced by 1 on $[\frac{1}{8}, T_1]$ (and $|\Phi_m| \leq 1$ on $[0, \frac{1}{8}]$) give $\Phi_m \in L^1$.

Calibration. Now let $M \geq \delta_0$, so $\alpha \geq \alpha_0 := \frac{2}{\sqrt{5}A_1} \delta_0$ and $z_0 \geq \frac{\alpha_0}{4}$, whence $T_1 \leq C/\delta_0$ and, by Lemma 11, $\log(1/\tilde{\rho}) \geq \min(\kappa \frac{\alpha_0^2}{16}, \log 2) \geq \kappa_1 \delta_0^2$ with κ_1 absolute. By (19) and the upper bound (16) (proved below, independently of this lemma), $N_0 \geq \frac{A_1}{8\pi} \overline{M} \mathcal{L} \geq \frac{A_1}{8\pi C_9} \sigma^2$, so

$$\tilde{\rho}^{N_0} = \exp(-N_0 \log(1/\tilde{\rho})) \leq \exp\left(-\frac{A_1 \kappa_1}{8\pi C_9} \delta_0^2 \sigma^2\right),$$

which is (20) with $c_{10} := A_1 \kappa_1 / (8\pi C_9)$. $\square$

Proof of Proposition 4

Proof. Upper bounds (16). Write $\sigma^2 = \sum_{\gamma > 0} 8M^2/(M^2 + \gamma^2)$ (Proposition 2, applied to χ as in (8)) and split at $\overline{M}$. Each term is ≤ 8 , so by Lemma 9(b) the ordinates $\gamma \leq \overline{M}$ contribute at most $8N_\chi(\overline{M}) \leq 8C'_R \overline{M} \mathcal{L}$. For $\gamma > \overline{M}$ use $8M^2/(M^2 + \gamma^2) \leq 8\overline{M}^2/\gamma^2$ and partial summation: with $N^*(t) := N_\chi(t) - N_\chi(\overline{M}) \leq N_\chi(t) \leq C'_R t \log(qt)$,

$$\sum_{\gamma > \overline{M}} \gamma^{-2} = 2 \int_{\overline{M}}^{\infty} N^*(t) t^{-3} dt \leq 2C'_R \int_{\overline{M}}^{\infty} t^{-2} \log(qt) dt = 2C'_R \frac{\mathcal{L} + 1}{\overline{M}} \leq \frac{4C'_R \mathcal{L}}{\overline{M}}$$

(the boundary term vanishes since $N^*(t) \ll t \log qt$, and $\mathcal{L} \geq \log 3 \geq 1$), contributing at most $32C'_R \overline{M} \mathcal{L}$. For $S_4 = \sum_{\gamma} 256M^4/(M^2 + \gamma^2)^2$ the same split gives $256N_{\chi}(\overline{M})$ from $\gamma \leq \overline{M}$ and, from $\gamma > \overline{M}$, using $\int_{\overline{M}}^{\infty} t^{-4} \log(qt) dt = \frac{\mathcal{L}}{3\overline{M}^3} + \frac{1}{9\overline{M}^3} \leq \frac{\mathcal{L}}{2\overline{M}^3}$,

$$256 \overline{M}^4 \sum_{\gamma > \overline{M}} \gamma^{-4} = 256 \overline{M}^4 \cdot 4 \int_{\overline{M}}^{\infty} N^*(t) t^{-5} dt \leq 512 C'_R \overline{M} \mathcal{L}.$$

Take $C_9 := 768 C'_R + 40C'_R$, say.

Lower bound (17). Keep only the window of Lemma 10: each $\gamma \in (A, 2A]$ has $M^2 + \gamma^2 \leq 5A_1^2 \overline{M}^2$, so

$$\sigma^2 \geq N_0 \cdot \frac{8M^2}{5A_1^2 \overline{M}^2} = \frac{8 \min(M, 1)^2}{5A_1^2} N_0 \geq \frac{8\delta_0^2}{5A_1^2} \cdot \frac{A_1}{8\pi} \overline{M} \mathcal{L} = \frac{\delta_0^2}{5\pi A_1} \overline{M} \mathcal{L},$$

using $\min(M, 1) \geq \delta_0$ for $M \geq \delta_0$, $\delta_0 \leq 1$. Take $c_9 := 1/(5\pi A_1)$. The two displays give (18) at once. $\square$

Proof of Theorem 3

Proof. Fix $\delta_0 \in (0, 1]$ and (χ, m) with (H), $M \geq \delta_0$. All constants denoted C below are absolute unless the dependence (δ_0) is written; we freely use $\sup_{x>0} x^{\beta} e^{-cx} = (\beta/(ec))^{\beta}$ for $\beta, c > 0$.

By Lemma 12, $\Phi_m \in L^1$, so Lemma 7(b) applies:

$$\delta_{\chi}(m) - \frac{1}{2} = \frac{1}{\pi} \int_0^{1/8} \Phi_m(t) \frac{\sin t}{t} dt + \Theta_{\text{tail}}, \quad |\Theta_{\text{tail}}| \leq C_{11}(\delta_0) e^{-c_{10}\delta_0^2\sigma^2}. \quad (21)$$

Small t : the uniform Gaussian factorization. For $0 \leq t \leq \frac{1}{8}$ every argument satisfies $2a_{\gamma}t \leq 4t \leq \frac{1}{2}$, so by (J3), exactly as in (13),

$$\Phi_m(t) = \exp\left(-\frac{\sigma^2 t^2}{2} + E(t)\right), \quad |E(t)| \leq \frac{t^4}{17} S_4 \leq \frac{t^2}{17} \cdot \frac{1}{64} \cdot 32\sigma^2 = \frac{\sigma^2 t^2}{34},$$

where the middle bound used $t^2 \leq \frac{1}{64}$ and the *absolute* inequality $S_4 \leq 32\sigma^2$ of Lemma 6. Hence $|\Phi_m(t)| \leq e^{-(8/17)\sigma^2 t^2}$ on $[0, \frac{1}{8}]$: every constant so far is free of (q, m) .

Decomposition. As in (15),

$$\frac{1}{\pi} \int_0^{1/8} \Phi_m \frac{\sin t}{t} dt = G - T_1 - T_2 + T_3,$$

with $T_1 = \frac{1}{\pi} \int_{1/8}^{\infty} e^{-\sigma^2 t^2/2} dt$, $T_2 = \frac{1}{\pi} \int_0^{1/8} e^{-\sigma^2 t^2/2} (1 - \frac{\sin t}{t}) dt$, $T_3 = \frac{1}{\pi} \int_0^{1/8} e^{-\sigma^2 t^2/2} (e^E - 1) \frac{\sin t}{t} dt$, and, verbatim as in the proof of Theorem 2 (whose estimates for these three terms use no property of (q, m) beyond $S_4 \leq 32\sigma^2$):

$$0 \leq T_1 \leq \frac{8}{\pi\sigma^2} e^{-\sigma^2/128}, \quad 0 \leq T_2 \leq \frac{G}{6\sigma^2}, \quad |T_3| \leq \frac{G S_4}{4\sigma^4} \leq \frac{8G}{\sigma^2}. \quad (22)$$

Part (i). Divide by $G = (2\pi\sigma^2)^{-1/2}$. From (22), $T_2/G \leq \frac{1}{6\sigma^2}$ and $|T_3|/G \leq \frac{8}{\sigma^2}$;

$$\frac{T_1}{G} \leq \sqrt{2\pi} \frac{8}{\pi\sigma} e^{-\sigma^2/128} \leq \frac{C}{\sigma^2} \quad (\text{as } \sigma e^{-\sigma^2/128} \leq C), \quad \frac{|\Theta_{\text{tail}}|}{G} \leq \sqrt{2\pi} \sigma C_{11}(\delta_0) e^{-c_{10}\delta_0^2\sigma^2} \leq \frac{C(\delta_0)}{\sigma^2},$$

the last step since $\sigma^3 e^{-c_{10}\delta_0^2\sigma^2} \leq C(c_{10}\delta_0^2)^{-3/2}$. Summing proves (i), for *every* $\sigma^2 > 0$.

Part (ii). Refine T_2 and T_3 by one order, as in the proof of Theorem 2(ii); we record why each constant is absolute. For T_2 : $|1 - \frac{\sin t}{t} - \frac{t^2}{6}| \leq \frac{t^4}{120}$ on $[0, \frac{1}{8}]$, and each extension of a moment integral from $[0, \frac{1}{8}]$ to $[0, \infty)$ costs at most $Ce^{-\sigma^2/300} \cdot \sigma^{-k}$ -type terms, all $\leq CG\sigma^{-4}$ by the supremum bound; hence $T_2 = \frac{G}{6\sigma^2} + O(G\sigma^{-4})$. For T_3 : by (J4), $E(t) = -\frac{t^4}{64}S_4 + E_2(t)$ with $|E_2| \leq \frac{t^6}{80}S_6 \leq \frac{t^6}{5}S_4$ (Lemma 6), and $e^E - 1 = E + \omega$ with $|\omega| \leq \frac{E^2}{2}e^{|E|} \leq \frac{t^8 S_4^2}{578}e^{\sigma^2 t^2/34}$; the moment integrals give

$$T_3 = -\frac{3S_4}{64}G\sigma^{-4} + O(GS_4\sigma^{-6}) + O(GS_4^2\sigma^{-8}) = -\frac{3S_4}{64}G\sigma^{-4} + O(G\sigma^{-4}),$$

the final step by $S_4 \leq 32\sigma^2$ (so $S_4\sigma^{-6} \leq 32\sigma^{-4}$ and $S_4^2\sigma^{-8} \leq 1024\sigma^{-4}$) — this is where the joint statement diverges from the fixed- q proof, which at this point invoked $\sigma^2 \asymp_q M \log M$. The exponentially small terms T_1 and Θ_{tail} are $\leq C(\delta_0)G\sigma^{-4}$ by the supremum bound with $\beta = \frac{5}{2}$. Collecting proves (ii).

The constant. Tracing the δ_0 -dependence: $C_{11}(\delta_0) \leq C\delta_0^{-1}$ and $(c_{10}\delta_0^2)^{-5/2} \leq C\delta_0^{-5}$, all other constants absolute; hence $C(\delta_0) \leq C\delta_0^{-6}$. $\square$

Proof of Corollary 5. Combine Theorem 3(i) with the lower bound (17): the relative error is at most $C(\delta_0)/\sigma^2 \leq C(\delta_0)/(c_9\delta_0^2 \overline{M}\mathcal{L})$. For (a), $\mathcal{L} \asymp_q \log M$ as $M \rightarrow \infty$; the absence of a threshold is because Theorem 3 holds for all $M \geq \delta_0$. For (b), $\overline{M}\mathcal{L} \geq \log q$. For (c): $\overline{M}\mathcal{L} \rightarrow \infty$ iff $q\overline{M} \rightarrow \infty$, since $\overline{M}\mathcal{L} \geq \log(q\overline{M})$, while if $q\overline{M}$ stays bounded then so does $\overline{M}\mathcal{L}$. $\square$

Proof of Corollary 6. Let $M \geq 16$ and $qM \geq 2\pi e^2$, so $L = \log \frac{qM}{2\pi} \geq 2$; take $\delta_0 = 1$ in Theorem 3(ii), so all constants below are absolute. The proofs of Proposition 3 and of (11) in Lemma 6 carry absolute constants in the stated ranges: Stirling's series for ψ, ψ' at $x = \frac{M}{2} + \frac{2a+1}{4} \geq \frac{33}{4}$ has absolute remainders, and the Dirichlet-series terms obey $4M|\frac{L'}{L}(m+1, \chi)| \leq 2.28M2^{-(M-3/2)} \leq CM^{-1}$ and $64M^2|(\frac{L'}{L})'(m+1, \chi)| \leq CM^22^{-M} \leq CM^{-1}$ for $M \geq 16$, uniformly in q . Hence, with absolute O 's,

$$\sigma^2 = 2ML + (2a-1) + O(1/M) = 2ML(1 + O((ML)^{-1})), \quad S_4 = 32M(L-1) + 32(2a-1) + O(1/M).$$

Substituting into Theorem 3(ii): $\frac{1}{6\sigma^2} = \frac{1}{12ML} + O((ML)^{-2})$;

$$\frac{3S_4}{64\sigma^4} = \frac{96M(L-1) + O(1)}{64 \cdot 4M^2L^2(1 + O((ML)^{-1}))} = \frac{3}{8ML} - \frac{3}{8ML^2} + O\left(\frac{1}{M^2L^2}\right);$$

and the error $CG\sigma^{-4} \leq CG(ML)^{-2}$. Since $\frac{1}{12} + \frac{3}{8} = \frac{11}{24}$ and both $(ML)^{-2}$ and $(ML^2)^{-1} \cdot M^{-1}$ -type terms are $O((ML^2)^{-1})$ (as $M \geq 16, L \geq 2$), the corollary follows. $\square$

The two classical aspects as boundary cases

Remark 8 (boundary cases; relation to Fiorilli–Martin). (1) Weight aspect (q fixed). *Corollary 5(a) and Theorem 3(ii) reproduce Theorem 2(i)–(ii) with the same error scales ($1/\sigma^2 \asymp_q 1/(M \log M)$ and $1/\sigma^4 \asymp_q (M \log M)^{-2}$), so the fixed- q theorem is literally the q -constant slice of Theorem 3; in addition the joint form dispenses with the threshold $M_0(q)$ and makes the second-order constant $-\frac{11}{24}$ aspect-independent (Corollary 6).*

(2) Modulus aspect (m fixed). Fiorilli and Martin [9] prove, under GRH and LI, an asymptotic series for the classical race density $\delta(q; a, b)$ as $q \rightarrow \infty$, in descending powers of the variance of the associated limiting distribution, whose leading correction to $\frac{1}{2}$ is of order $(\text{variance})^{-1/2}$. Corollary 5(b) is the exact structural analogue for the single-character variable $X_{m,\chi}$: leading term $(2\pi\sigma^2)^{-1/2} \asymp_m (\log q)^{-1/2}$, relative error $O_m(1/\log q)$, and a proved second-order term (Theorem 3(ii)). We stress, honestly, that this is an analogue, not a specialization: the Fiorilli–Martin density aggregates the zeros of all non-principal characters mod q and its bias term is driven by the count of square roots of the residues raced, whereas $X_{m,\chi}$ carries the single real character χ , with the constant 1 (the squares’ contribution in the m -weighted race of Theorem 1) playing the role of the mean. Neither statement implies the other; Theorem 3 is the common roof for the one-character object only. In the modulus aspect the hypothesis (H) rides on $L(\frac{1}{2}, \chi) \neq 0$ for every χ in the family — conjecturally always true (Chowla), known for a positive proportion of quadratic characters [29], and unconditional at $q = 3, 4$.

(3) Race interpretation. As in Remark 7(1), for $q = 4$ (and, with the verbatim extension recorded there, $q = 3$) the conclusion transfers, under $LI(\chi)$, to the logarithmic density of the weighted race itself via Theorem 1(ii); for all other moduli Theorem 3 is, in this paper, a statement about the limiting random variable $X_{m,\chi}$ only.

Remark 9 (the excluded corner $M \rightarrow 0$, $q \rightarrow \infty$, and other honest limits). Theorem 3 is uniform on $\{M \geq \delta_0\}$ for each fixed $\delta_0 > 0$, with $C(\delta_0) \leq C\delta_0^{-6}$, but it does not cover the corner $M \rightarrow 0$ simultaneously with $q \rightarrow \infty$, and we make no claim there. The corner is genuinely different, not an artifact of the method’s bookkeeping alone. For fixed χ , as $M \rightarrow 0$ one has $\sigma^2 \leq 8M^2 \sum_{\gamma} \gamma^{-2} \rightarrow 0$ and $X_{m,\chi} \rightarrow 1$, so $\delta_{\chi}(m) \rightarrow 1$: the bias saturates rather than dissolves, and the Gaussian prediction $\frac{1}{2} + (2\pi\sigma^2)^{-1/2}$ is not even bounded by 1. In the corner the two behaviours compete: σ^2 can remain of moderate size with M small, because a large modulus can place many ordinates below 1 ($N_{\chi}(1)$ can be as large as $\asymp \log q$, and $\gamma_1 \gg 1/\log q$ is possible), making $\sigma^2 \asymp M^2 \sum_{\gamma \leq 1} \gamma^{-2}$ as large as $\asymp M^2(\log q)^2$; whether $\delta_{\chi}(m) - \frac{1}{2} \sim (2\pi\sigma^2)^{-1/2}$ persists as $M \asymp 1/\log q$, $q \rightarrow \infty$, is open. Two concrete obstructions in our proof: the window amplitudes degenerate ($a_{\gamma} \asymp M$ on any window at height $\gg 1$), so the per-factor decay of $|\Phi_m|$ at $t \asymp 1$ weakens to $\exp(-cM^2)$ and the tail bound (20) decays only like $\exp(-c\sigma^2/\log q)$, which for bounded σ^2 does not decay at all; and the low-lying zeros that then dominate σ^2 are exactly the ones about which the q -uniform count of Lemma 9 says least. A uniform treatment of the corner would need control of $\sum_{\gamma \leq 1/\log q}$ -type sums (low-lying-zero statistics), which we have not carried out.

Remark 10 (numerical sanity check). Table 4 tests the uniform error scales of Theorem 3 at the three points of Table 3 that vary both coordinates; every input (σ_m^2 and S_4 from their closed forms (8), (10); deterministic Gil–Pelaez reference densities) is reused unchanged from that table, and only the ratio columns are new arithmetic. The observed relative deviation from G times σ^2 sits at 0.64–0.74, and the residual after the second-order term times σ^4 sits below 1, at three points with $(q, \overline{M}\mathcal{L})$ ranging over $(4, 90)$, $(3, 315)$, $(4, 603)$: both error scales of the theorem are confirmed with constants of order one — far below the unoptimized proved constants — and the comparability $\sigma^2/\overline{M}\mathcal{L} \in [1.17, 1.39]$ illustrates (18). (At $(4, 100)$ the printed residual 7.0×10^{-7} is below the 9.9×10^{-6} modelling floor of the reference density discussed in Remark 6, so its ratio 0.49 is a bound consistent with zero rather than a measurement.)

(q, m)	σ_m^2	$\frac{\sigma_m^2}{M\mathcal{L}}$	rel. dev. from G	$\sigma^2 \cdot \text{rel. dev.} $	residual after 2nd order	$\sigma^4 \cdot \text{residual} $
(4, 20)	106.3260	1.18	-6.05×10^{-3}	0.64	-7.9×10^{-5}	0.89
(3, 60)	407.9701	1.30	-1.71×10^{-3}	0.70	-4.9×10^{-6}	0.82
(4, 100)	836.8744	1.39	-8.80×10^{-4}	0.74	$+7.0 \times 10^{-7}$	0.49

Table 4: Theorem 3 against the deterministic reference densities of Table 3 (columns 4 and 6 are that table’s columns 5 and 7; $\delta_\chi(m)$ references 0.5384553, 0.5197175, 0.5137784). Column 5 confirms the C/σ^2 scale of (i), column 7 the C/σ^4 scale of (ii), with $O(1)$ constants across a range of (q, m) .

Remark 11 (scope, and what is not proved). (1) The corner $M \rightarrow 0$ with $q \rightarrow \infty$ is excluded and open (Remark 9); on $\{M \geq \delta_0\}$ the constant degrades like δ_0^{-6} , which we have not optimized. (2) The q -uniformity rests on the absolute constant in the Riemann–von Mangoldt formula: we cite [27, Theorem 14.5], and, independently of any textbook formulation, the explicit bound of [28, Corollary 1.2] pins the constant ($C_R = 2.8$ suffices in Lemma 9); the fixed- q formulation of [17, §16] would not suffice as cited. (3) Hypothesis (H) includes $L(\frac{1}{2}, \chi) \neq 0$ for every character in whatever family is considered; in q -aspect corollaries this is an unproved input beyond GRH (cf. [29]). (4) For moduli other than 3, 4 the conclusion concerns the limiting variable $X_{m, \chi}$, not a prime race, no LI-transfer having been stated (Remark 8(3)). (5) Constants are effective in principle but wasteful; no numerical value of $C(\delta_0)$ is claimed, and Table 4 is a consistency check, not a validated enclosure. (6) Theorem 3 makes no claim of an asymptotic series beyond the second order proved in (ii); the Fiorilli–Martin series in the modulus aspect suggests one exists jointly, which we leave open.

6 Monotonicity in the weight for large m

Conjecture 1 asserts that $m \mapsto \delta_q(m)$ is strictly decreasing on all of $(-\frac{1}{2}, \infty)$. This section proves the large- m half. The mechanism is the one the conjecture’s heuristic suggests: every amplitude $a_\gamma(m)$ grows with the weight, and through the Gil–Pelaez integral of Lemma 7 this pointwise growth of the noise can be differentiated. The key structural fact — which replaces the peakedness comparisons that fail for arcsine components — is that the logarithmic derivative of the Bessel factor,

$$\varphi(z) := -\frac{J'_0(z)}{J_0(z)} = \frac{J_1(z)}{J_0(z)},$$

has all Taylor coefficients positive on $(0, j_{0,1})$, where $j_{0,1} = 2.4048\dots$ is the first zero of J_0 (Lemma 13). Consequently the differentiated integrand is *pointwise negative* on the entire initial segment $0 < t \leq \frac{3}{5}$ of the inversion integral — no cancellation has to be tracked there — and only the superexponentially damped range $t > \frac{3}{5}$ has to be beaten. Throughout, notation is that of Section 4: χ is a real primitive character mod q , $\text{GRH}(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$ are assumed, $M = m + \frac{1}{2}$, $a_\gamma = 2M/\sqrt{M^2 + \gamma^2}$, $\Phi_m(t) = \prod_{\gamma>0} J_0(2a_\gamma t)$, $\delta_q(m) = \mathbb{P}(X_{m, \chi} > 0)$, and σ_m^2 , $S_4(m)$ are as in Theorem 2. Derivatives in the weight are taken with respect to M ;

since $M = m + \frac{1}{2}$, $d/dm = d/dM$ and we write $'$ for either. We abbreviate

$$L_M := \log \frac{qM}{2\pi}, \quad D(M) := \sum_{\gamma > 0} \frac{16M\gamma^2}{(M^2 + \gamma^2)^2},$$

the latter being, as Lemma 18 shows, the derivative of the variance: $D = (\sigma_m^2)'$.

Statements

Theorem 4 (partial monotonicity). *Let χ be a real primitive character mod q ; assume $GRH(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$. Then there is an effectively computable $M_1 = M_1(\chi)$ such that for every m with $M = m + \frac{1}{2} \geq M_1$ the function δ_q is differentiable at m with*

$$\delta'_q(m) \leq -0.18 \frac{D(M)}{\sigma_m^3} < 0.$$

In particular $m \mapsto \delta_q(m)$ is strictly decreasing on $[M_1 - \frac{1}{2}, \infty)$. Moreover there is $M_2(\chi) \geq M_1(\chi)$ such that

$$-\delta'_q(m) = \theta(m) \frac{D(M)}{2\sqrt{2\pi} \sigma_m^3}, \quad \theta(m) \in [0.90, 1.14] \quad (M \geq M_2(\chi)),$$

and by Lemma 18(iii) (for $q \in \{3, 4\}$; the general real primitive case is identical with $2a - 1$ in place of $+1$ in the variance constant),

$$-\delta'_q(m) \asymp \frac{2L_M + 2}{2\sqrt{2\pi} (2ML_M)^{3/2}} \asymp_q \frac{1}{M^{3/2} (\log M)^{1/2}} \quad (m \rightarrow \infty).$$

Corollary 7 (explicit thresholds for $q = 4, 3$). *Assume $GRH(\chi_4)$ resp. $GRH(\chi_3)$. Then, in the computational rigor class described in Remark 12,*

$$\delta'_4(m) < 0 \text{ for all } m \geq 16.82, \quad \delta'_3(m) < 0 \text{ for all } m \geq 18.62;$$

i.e. δ_4 is strictly decreasing on $[16.82, \infty)$ and δ_3 on $[18.62, \infty)$. For $M = m + \frac{1}{2} \geq 300$ the conclusion is a theorem in the full sense of Theorem 2 (no zero data enters: Lemma 22); on $17.32 \leq M \leq 320$ (resp. $19.12 \leq M \leq 350$) it rests additionally on the certified ordinate lists and the evaluated criterion of Proposition 5. Under $LI(\chi_4)$ the same statements hold for the logarithmic density $\delta(m)$ of the weighted race itself, by Theorem 1(ii) as in Corollary 3. Combined with Conjecture 1, monotonicity is now proved on $[m_1, \infty)$ and open only on the bounded interval $(-\frac{1}{2}, m_1)$, $m_1 = 16.82$ resp. 18.62 , where the certified grid of Section 2 supports it numerically.

The proof occupies the rest of the section: Bessel structure (Lemmas 13, 14), digamma and zero-counting inputs (Lemmas 15–17), the variance derivative (Lemma 18), differentiation under the integral (Lemma 19), the sign structure and the main term (Lemma 20), the tail (Lemma 21), and the assembly (proof of Theorem 4, Proposition 5, Lemma 22).

Bessel structure

Lemma 13 (positivity of the Bessel log-derivative). *Let $j_{0,1} < j_{0,2} < \dots$ be the positive zeros of J_0 (all real). Then:*

(i) $J_0(z) = \prod_{k \geq 1} (1 - z^2/j_{0,k}^2)$, locally uniformly on $\mathbb{C}$, and for $|z| < j_{0,1}$,

$$-\log J_0(z) = \sum_{n \geq 1} \frac{\lambda_n}{n} z^{2n}, \quad \lambda_n := \sum_{k \geq 1} j_{0,k}^{-2n} > 0,$$

with $\lambda_1 = \frac{1}{4}$, $\lambda_2 = \frac{1}{32}$, $\lambda_3 = \frac{1}{192}$. In particular $\varphi(z) = -J'_0(z)/J_0(z) = \sum_{n \geq 1} 2\lambda_n z^{2n-1}$ has all coefficients positive, so $\varphi > 0$ on $(0, j_{0,1})$ and $\varphi(z) \geq \frac{z}{2}$ there.

(ii) For $0 \leq z \leq \frac{1}{2}$:

$$\frac{z}{2} + \frac{z^3}{16} \leq \varphi(z) \leq \frac{z}{2} + \frac{z^3}{16} + \frac{z^5}{90} \leq 0.517z.$$

(iii) $j_{0,1} > 2.4$.

Proof. (i) J_0 is entire, even, $J_0(0) = 1$, and of order 1 ($|J_0(z)| \leq \sum |z|^{2k}/(4^k(k!)^2) \leq e^{|z|}$ from the series, and J_0 is unbounded on the imaginary axis). All its zeros are real — classical [30, Ch. XV] — hence they are the $\pm j_{0,k}$, and $\sum j_{0,k}^{-2} < \infty$ since an order-1 entire function has convergence exponent ≤ 1 for its zeros. Exactly as in Step 3 of the proof of Proposition 2, Hadamard's factorization plus pairing $\pm j_{0,k}$ cancels the exponential factors, evenness kills the linear exponent, and $J_0(0) = 1$ fixes the constant, giving the product (also stated directly in [30, Ch. XV]). For $|z| < j_{0,1}$ each factor lies in the disc $|w - 1| < 1$, so $-\log J_0(z) = \sum_k -\log(1 - z^2/j_{0,k}^2) = \sum_k \sum_n (z^2/j_{0,k}^2)^n/n$, absolutely convergent, and rearrangement gives the double-series form with the Rayleigh sums $\lambda_n > 0$. Their first three values follow by matching with the directly computed expansion of $-\log J_0$: writing $J_0 = 1 - w$, $w = \frac{z^2}{4} - \frac{z^4}{64} + \frac{z^6}{2304} - \dots$, one gets $-\log J_0 = w + \frac{w^2}{2} + \frac{w^3}{3} + \dots = \frac{z^2}{4} + \frac{z^4}{64} + (\frac{1}{2304} - \frac{1}{256} + \frac{1}{192})z^6 + \dots$ and $\frac{1}{2304} - \frac{1}{256} + \frac{1}{192} = \frac{4}{2304} = \frac{1}{576}$; hence $\lambda_1 = \frac{1}{4}$, $\lambda_2 = 2 \cdot \frac{1}{64} = \frac{1}{32}$, $\lambda_3 = 3 \cdot \frac{1}{576} = \frac{1}{192}$ (the z^2 and z^4 coefficients are those of Lemma 4(J4)). Term-by-term differentiation of the power series gives φ .

(ii) The lower bound is positivity of the coefficients with $n \geq 3$ dropped. For the upper: since $J_0(z) \geq 1 - \frac{z^2}{4} > 0$ for $z < 2$, we have $j_{0,1} > 2$, so $\lambda_n \leq j_{0,1}^{-2(n-3)} \lambda_3 \leq 4^{-(n-3)} \lambda_3$ for $n \geq 3$, and for $z \leq \frac{1}{2}$

$$\sum_{n \geq 3} 2\lambda_n z^{2n-1} \leq 2\lambda_3 z^5 \sum_{j \geq 0} \left(\frac{z^2}{4}\right)^j \leq \frac{2z^5}{192} \cdot \frac{16}{15} = \frac{z^5}{90}.$$

Finally $\frac{1}{2} + \frac{z^2}{16} + \frac{z^4}{90} \leq \frac{1}{2} + \frac{1}{64} + \frac{1}{1440} < 0.517$ at $z \leq \frac{1}{2}$.

(iii) In the series $J_0(2.4) = \sum_k (-1)^k c_k$, $c_k = (1.44)^k/(k!)^2$, the terms decrease strictly from $k = 1$ on ($c_{k+1}/c_k = 1.44/(k+1)^2 \leq 0.36$), so the alternating remainder after c_6 is at most $c_7 < 6 \times 10^{-7}$, and

$$J_0(2.4) \geq 1 - 1.44 + 0.5184 - 0.082945 + 0.0074650 - 0.000430 - 10^{-6} > 0.0024 > 0.$$

Since $J_0 > 0$ on $[0, 2]$ as well (by $1 - \frac{z^2}{4} + \dots$ alternating) and J_0 is continuous, and since J_0 is strictly decreasing on $[0, 2\sqrt{2}]$ by Lemma 14(i) below, $J_0 > 0$ on $[0, 2.4]$, i.e. $j_{0,1} > 2.4$. $\square$

Lemma 14 (envelopes of J_0). *Let $P(z) := J_0(z)^2 + J_1(z)^2$. Then:*

- (i) $J'_0 = -J_1$, $(zJ_1)' = zJ_0$, and $P'(z) = -2J_1(z)^2/z \leq 0$: P is decreasing on $(0, \infty)$, $P \leq P(0) = 1$; in particular $|J_1| \leq 1$ everywhere, and by the alternating series $0 < J_1(z) \leq z/2$ on $(0, 2\sqrt{2}]$, so

$$|J_1(z)| \leq \min(1, \frac{z}{2}) \quad (z \geq 0), \quad J_0 \text{ is strictly decreasing on } [0, 2\sqrt{2}].$$

(ii) $\sup_{z \geq 2\sqrt{2}} |J_0(z)| \leq \sqrt{P(2\sqrt{2})} \leq 0.446$.

(iii) $\sup_{z \geq z_0} |J_0(z)| \leq \max(J_0(z_0), 0.446)$ for $0 < z_0 \leq 2\sqrt{2}$; in particular $\rho_1 := \sup_{z \geq 1.0733} |J_0(z)| \leq 0.733$.

- (iv) On any $[z_{\text{lo}}, z_{\text{hi}}] \subset (0, \infty)$,

$$\sup_{[z_{\text{lo}}, z_{\text{hi}}]} |J_0| = \max(|J_0(z_{\text{lo}})|, |J_0(z_{\text{hi}})|, \max_{j_{1,k} \in (z_{\text{lo}}, z_{\text{hi}})} |J_0(j_{1,k})|),$$

where $j_{1,k}$ are the positive zeros of J_1 (the critical points of J_0), and $|J_0(j_{1,k})| = \sqrt{P(j_{1,k})}$ is decreasing in k , so the inner maximum is attained at the smallest $j_{1,k}$ in the interval.

Proof. (i) $J'_0 = -J_1$ and $(zJ_1)' = zJ_0$ are termwise identities of the defining series (for the first, $\frac{d}{dz} \sum_k (-1)^k \frac{(z/2)^{2k}}{(k!)^2} = -\frac{z}{2} \sum_{k \geq 1} (-1)^{k-1} \frac{(z^2/4)^{k-1}}{(k-1)!k!} = -J_1$; the second is the same computation for zJ_1). Hence $P' = 2J_0J'_0 + 2J_1J'_1 = -2J_0J_1 + 2J_1(J_0 - \frac{J_1}{z}) = -\frac{2J_1^2}{z} \leq 0$. The series $J_1(z) = \sum_k (-1)^k (z/2)^{2k+1}/(k!(k+1)!)$ has terms decreasing in modulus for $z \leq 2\sqrt{2}$ (ratio $z^2/(4(k+1)(k+2)) \leq z^2/8 \leq 1$), so its partial sums bracket it: $0 \leq J_1 \leq \frac{z}{2}$ there, with $J_1 > 0$ on $(0, 2\sqrt{2}]$ (at $z = 2\sqrt{2}$ the bracketing from the third term on gives $J_1 \geq c_2 - c_3 > 0$). So $J'_0 = -J_1 < 0$ on $(0, 2\sqrt{2}]$.

(ii) For $z \geq 2\sqrt{2}$, $|J_0(z)| \leq \sqrt{P(z)} \leq \sqrt{P(2\sqrt{2})}$. At $z = 2\sqrt{2}$, $z^2 = 8$ exactly, and the alternating tails of both series (decreasing from $k = 2$ on) give the enclosures $J_0(2\sqrt{2}) \in [-0.19655, -0.19654]$, $J_1(2\sqrt{2}) \in [0.40019, 0.40020]$ (twelve terms; the script computes the partial sums in exact rational arithmetic), whence $P(2\sqrt{2}) \leq 0.19655^2 + 0.40020^2 < 0.19880$ and $\sqrt{P(2\sqrt{2})} < 0.44587 < 0.446$.

(iii) On $[z_0, 2\sqrt{2}]$, J_0 decreases from $J_0(z_0)$ to $J_0(2\sqrt{2}) \in (-0.197, -0.196)$, so $|J_0| \leq \max(J_0(z_0), 0.197)$ there; beyond, (ii) applies. For $z_0 = 1.0733$ the alternating upper bound gives $J_0(1.0733) \leq 1 - \frac{z_0^2}{4} + \frac{z_0^4}{64} < 0.7328 < 0.733 (> 0.446)$.

(iv) $|J_0|$ can attain an interior maximum only at a critical point of J_0 , i.e. a zero of J_1 ; at such points $J_1 = 0$, so $|J_0(j_{1,k})| = \sqrt{P(j_{1,k})}$, decreasing in k since P is decreasing. $\square$

Digamma and counting inputs

Lemma 15 (Binet bounds). *For $x > 0$,*

$$\log x - \frac{1}{2x} - \frac{1}{12x^2} \leq \psi(x) \leq \log x - \frac{1}{2x}, \quad \frac{1}{x} + \frac{1}{2x^2} \leq \psi'(x) \leq \frac{1}{x} + \frac{1}{2x^2} + \frac{1}{6x^3}.$$

Proof. Binet's first formula [31, Ch. XII]: $\psi(x) = \log x - \frac{1}{2x} - 2 \int_0^\infty \frac{t dt}{(t^2 + x^2)(e^{2\pi t} - 1)}$. The integral is positive and, using $t^2 + x^2 \geq x^2$ and $\int_0^\infty \frac{t dt}{e^{2\pi t} - 1} = \frac{\zeta(2)}{(2\pi)^2} = \frac{1}{24}$, at most $\frac{1}{24x^2}$; doubling gives the

ψ bounds. Differentiating under the integral (dominated convergence; the differentiated integrand is $\leq 2t \cdot 2x x^{-4} / (e^{2\pi t} - 1)$ locally uniformly) yields $\psi'(x) = \frac{1}{x} + \frac{1}{2x^2} + 4x \int_0^\infty \frac{t dt}{(t^2 + x^2)^2 (e^{2\pi t} - 1)}$, and the last integral lies in $[0, \frac{1}{24}x^{-4}]$. $\square$

Lemma 16 (explicit upper zero count). *Assume $GRH(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$, χ real primitive mod q , odd ($a = 1$; this covers $q = 3, 4$). Then for $T \geq \frac{3}{2}$,*

$$N_\chi(T) \leq N^+(T) := \frac{T}{2} \log \frac{q(T+2)}{2\pi} + 0.57T.$$

Proof. Apply the variance identity (8) at weight $M = T$: every zero with $\gamma \leq T$ contributes $\frac{8T^2}{T^2 + \gamma^2} \geq 4$ to the (positive-termed) sum $4T \frac{\xi'}{\xi}(T + \frac{1}{2}, \chi) = \sum_\gamma \frac{8T^2}{T^2 + \gamma^2}$, so $N_\chi(T) \leq T \frac{\xi'}{\xi}(T + \frac{1}{2}, \chi)$. At $s = T + \frac{1}{2} \geq 2$,

$$\frac{\xi'}{\xi}(s, \chi) = \frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi\left(\frac{s+1}{2}\right) + \frac{L'}{L}(s, \chi) \leq \frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \log \frac{T+\frac{3}{2}}{2} + 0.5700,$$

by Lemma 15 and $|L'/L(s, \chi)| \leq \sum_n \Lambda(n)n^{-2} = -\frac{\xi'}{\xi}(2) = 0.56996\dots$ (as in the proof of Proposition 3). Since $\frac{q}{\pi} \cdot \frac{T+3/2}{2} \leq \frac{q(T+2)}{2\pi}$, the claim follows. $\square$

Lemma 17 (self-contained lower zero count). *Under the hypotheses of Lemma 16, for every $Y > 0$,*

$$N_\chi(\sqrt{3}Y) \geq \frac{1}{32} \left(\frac{S_4(Y)}{8} - \sigma^2(Y) \right), \quad (23)$$

where $\sigma^2(Y)$, $S_4(Y)$ denote the closed forms (8), (10) evaluated at weight $M = Y$. Moreover, for $q \in \{3, 4\}$ and $Y \geq 20$, with $\ell_Y := \log \frac{qY}{2\pi}$,

$$N_\chi(\sqrt{3}Y) \geq \frac{Y(\ell_Y - 2)}{16} - \frac{1}{2}. \quad (24)$$

Proof. Write $u_\gamma = Y^2/(Y^2 + \gamma^2) \in (0, 1)$, so $\sigma^2(Y) = \sum_\gamma 8u_\gamma$ and $S_4(Y) = \sum_\gamma 256u_\gamma^2$. Then $\frac{S_4}{8} - \sigma^2 = \sum_\gamma 8u_\gamma(4u_\gamma - 1)$, and $8u(4u - 1)$ is negative for $u < \frac{1}{4}$ (i.e. $\gamma > \sqrt{3}Y$) and at most $24 \leq 32$ for $u \leq 1$. Dropping the negative terms and bounding the rest, $\frac{S_4}{8} - \sigma^2 \leq 32 N_\chi(\sqrt{3}Y)$, which is (23).

For (24) we need explicit bounds at weight Y , from (8), (10) with $x = \frac{Y}{2} + \frac{3}{4}$ and Lemma 15 (all for $q \in \{3, 4\}$, $Y \geq 20$; the script checks each step):

$$\sigma^2(Y) \leq 2Y\ell_Y + 1 + \frac{3}{Y} + 4Y |L'/L| \leq 2Y\ell_Y + 1.4, \quad \sigma^2(Y) \geq 2Y\ell_Y$$

(upper: $2Y\psi(x) \leq 2Y \log x - \frac{Y}{x}$, $2Y \log(1 + \frac{3}{2Y}) \leq 3$, $-\frac{Y}{x} \leq -2 + \frac{3}{Y}$, and $4Y |L'/L(Y + \frac{1}{2})| \leq 2.28 Y 2^{-(Y-\frac{3}{2})} \leq 0.1$; lower: the same expansions with the $-\frac{1}{12x^2}$ and $-\frac{u^2}{2}$ remainders, which total ≥ -1 for $Y \geq 10$); and, using ψ' of Lemma 15 in $S_4 = 16\sigma^2 - 64Y^2(\frac{1}{4}\psi'(x) + (L'/L)')$ together with $16Y^2/x \leq 32Y$, $8Y^2/x^2 \leq 32$, $\frac{8}{3}Y^2/x^3 \leq \frac{64}{3Y}$, and $|(L'/L)'(Y + \frac{1}{2})| \leq 0.89 \cdot 2^{-(Y-\frac{3}{2})}$ (from $\sum_n \Lambda(n)(\log n)n^{-2} \leq 0.89$, verified by the script),

$$S_4(Y) \geq 16 \cdot 2Y\ell_Y - 32Y - 34.$$

Hence $\frac{S_4}{8} - \sigma^2 \geq 4Y\ell_Y - 4Y - 4.3 - 2Y\ell_Y - 1.4 \geq 2Y(\ell_Y - 2) - 6$, and division by 32 gives (24). $\square$

The variance derivative

Lemma 18 (derivative of the variance). *Under $GRH(\chi)$ and $L(\frac{1}{2}, \chi) \neq 0$:*

$$(i) \ a'_\gamma(M) = \frac{2\gamma^2}{(M^2 + \gamma^2)^{3/2}} > 0, \ \sum_\gamma a_\gamma a'_\gamma = \frac{D}{4} < \infty, \text{ and on } (0, \infty)$$

$$(\sigma_m^2)' = D(M) = \sum_{\gamma > 0} \frac{16M\gamma^2}{(M^2 + \gamma^2)^2} = 4 \frac{\xi'}{\xi}(m+1, \chi) + 4M(\log \xi)''(m+1, \chi) > 0.$$

(ii) D is differentiable with $|D'(M)| \leq 3D(M)/M$; hence for $0 < M_a \leq M \leq M_b$,

$$\max(D(M_a), D(M_b)) \left(\frac{M_a}{M_b}\right)^3 \leq D(M) \leq \min(D(M_a), D(M_b)) \left(\frac{M_b}{M_a}\right)^3.$$

(iii) For $q \in \{3, 4\}$ and $M \geq 10$: $|D(M) - 2L_M - 2| \leq 8/M$ and $2ML_M \leq \sigma_m^2 \leq 2ML_M + 3.1$; for $M \geq 20$ also $S_4(m) \leq 32M(L_M - 1) + 100$.

Proof. (i) The formula for a'_γ is a direct computation, and $4a_\gamma a'_\gamma = 16M\gamma^2/(M^2 + \gamma^2)^2$. The series for $\sigma_m^2 = \sum_\gamma 8M^2/(M^2 + \gamma^2)$ may be differentiated termwise: the differentiated terms are dominated on $M \leq A$ by $16A/\gamma^2$, so the derivative series converges uniformly on compacts and the standard theorem applies, exactly as in the proofs of Proposition 2 and Corollary 2. The closed form follows from $\sigma_m^2 = 4M(\log \Xi)'(M)$ and the termwise-differentiated series for $(\log \Xi)'$, $(\log \Xi)''$ established there: $4(\log \Xi)'(M) + 4M(\log \Xi)''(M) = \sum_\gamma \left[\frac{8M}{M^2 + \gamma^2} + \frac{8M(\gamma^2 - M^2)}{(M^2 + \gamma^2)^2} \right] = \sum_\gamma \frac{16M\gamma^2}{(M^2 + \gamma^2)^2}$. Positivity is termwise.

(ii) Termwise again (domination as before), $D'(M) = \sum_\gamma 16\gamma^2 \frac{\gamma^2 - 3M^2}{(M^2 + \gamma^2)^3}$, and $|\gamma^2 - 3M^2| \leq 3(\gamma^2 + M^2)$ gives $|D'| \leq \sum_\gamma 48\gamma^2/(M^2 + \gamma^2)^2 = 3D/M$. Integrating $|(\log D)'| \leq 3/M$ from either endpoint of $[M_a, M_b]$ gives the two-sided comparison.

(iii) With $x = \frac{M}{2} + \frac{3}{4}$, $u = \frac{3}{2M}$,

$$D = 2 \log \frac{q}{\pi} + 2\psi(x) + M\psi'(x) + 4 \frac{L'}{L}(m+1) + 4M \left(\frac{L'}{L}\right)'(m+1).$$

Lemma 15 plus $2 \log x = 2 \log \frac{M}{2} + 2 \log(1 + u)$, $2 \log(1 + u) \in [2u - u^2, 2u]$, $\frac{M}{x} = \frac{2}{1+u}$, and the L -bounds $|4L'/L| + |4M(L'/L)'| \leq (2.28 + 3.56M)2^{-(M-\frac{3}{2})} \leq 1.1/M$ ($M \geq 10$) give, after collecting the $1/M$ terms (which cancel to first order), $|D - 2L_M - 2| \leq 4.3/M + 9/M^2 \leq 8/M$. The σ^2 and S_4 bounds are those derived in the proof of Lemma 17 (with the S_4 upper bound using the lower ψ' estimate: $S_4 \leq 16(2ML_M + 1.4) - 32M + 48 - 27 + 0.1 \leq 32M(L_M - 1) + 100$). $\square$

Differentiating the inversion integral

Lemma 19 (differentiation under the integral). *Let $0 < M_a \leq M_b$ and suppose at least 8 ordinates satisfy $\gamma \leq 2M_a$. Then:*

(a) for every $t \geq 0$ and $M \in [M_a, M_b]$ the function $M \mapsto \Phi_m(t)$ is differentiable, with

$$\partial_M \Phi_m(t) = \sum_{\gamma > 0} J'_0(2a_\gamma t) 2a'_\gamma(M) t \prod_{\gamma' \neq \gamma} J_0(2a_{\gamma'} t), \quad (25)$$

the series converging absolutely, uniformly on $[M_a, M_b] \times [0, T]$ for each T ; for $4t < j_{0,1}$ equivalently

$$\partial_M \Phi_m(t) = -\Phi_m(t) B(t), \quad B(t) := \sum_{\gamma > 0} \varphi(2a_\gamma t) 2a'_\gamma(M) t \geq 0. \quad (26)$$

(b) With $A(t) := \sum_\gamma \min(1, a_\gamma t) 2a'_\gamma(M) t$ one has, for all $t > 0$,

$$|\partial_M \Phi_m(t)| \leq A(t) \cdot \sup_\gamma \prod_{\gamma' \in S, \gamma' \neq \gamma} |J_0(2a_{\gamma'} t)|, \quad A(t) \leq \frac{D(M) t^2}{2} + \frac{4t}{M} N_\chi(2Mt), \quad (27)$$

for any finite set S of ordinates.

(c) δ_q is differentiable on the corresponding m -interval, and

$$\delta'_q(m) = \frac{1}{\pi} \int_0^\infty \partial_M \Phi_m(t) \frac{\sin t}{t} dt.$$

Proof. (a) Fix T and let N_1 be so large that $a_\gamma(M_b) T \leq \frac{1}{4}$ for all but the first N_1 ordinates (possible since $a_\gamma(M_b) \leq 2M_b/\gamma \rightarrow 0$). Split $\Phi_m = P \cdot R$ with P the product over the first N_1 ordinates and R the tail. On the tail every argument is $\leq \frac{1}{2}$, each factor is $\geq \frac{3}{4}$ by (J3), and $R = \exp(\sum_{\text{tail}} \log J_0(2a_\gamma t))$; the termwise M -derivatives $-\varphi(2a_\gamma t) 2a'_\gamma t$ are bounded by $0.517 \cdot 4a_\gamma a'_\gamma t^2$ (Lemma 13(ii)), whose sum converges uniformly on $[M_a, M_b]$ by Lemma 18(i)–(ii) (with $D \leq D_{\max}$ on the interval). Hence R is differentiable with the termwise log-derivative; P is a finite product of differentiable factors; the product rule gives (25), absolutely and locally uniformly (each summand of the tail part carries the convergent majorant above, each of the finitely many head summands is continuous). For $4t < j_{0,1}$ every factor is positive (all arguments are $\leq 4t < j_{0,1}$ since $a_\gamma \leq 2$), so the product form can be rewritten as (26), and $B \geq 0$ because $\varphi \geq 0$ on $[0, j_{0,1})$ and $a'_\gamma > 0$.

(b) In (25) bound the differentiated factor by $|J'_0(2a_\gamma t)| = |J_1(2a_\gamma t)| \leq \min(1, a_\gamma t)$ (Lemma 14(i)), all factors outside $S \setminus \{\gamma\}$ by 1 (J1). For the bound on A : terms with $a_\gamma t \leq 1$ contribute at most $\sum 2a_\gamma a'_\gamma t^2 \leq \frac{D}{2} t^2$; terms with $a_\gamma t > 1$ force $\gamma < 2Mt$ and contribute at most $2a'_\gamma t \leq \frac{4t}{M}$ each, since $a'_\gamma \leq 2/\sqrt{M^2 + \gamma^2} \leq 2/M$.

(c) Integrability of a dominant: on $[0, \frac{1}{8}]$, (26) with $B \leq 0.517 D_{\max} t^2$ and $|\Phi_m| \leq 1$; on $[\frac{1}{8}, 16]$, (27) with the trivial product bound 1 and $A(t) \leq \frac{D_{\max}}{2} t^2 + \frac{4t}{M_a} N^+(2M_b t)$ (Lemma 16), a fixed continuous function; on $[16, \infty)$, take $S = \{\gamma \leq 2M_a\}$ ($|S| = n \geq 8$): each factor obeys (J2) with $2a_\gamma t \geq \frac{4}{\sqrt{5}} t$, so the product over any $n-1$ of them is $\leq (c_3/\sqrt{t})^{n-1} \leq c_3^7 t^{-7/2}$, and the integrand is $\ll t^2 \log t \cdot t^{-1} \cdot t^{-7/2}$, integrable. Every bound is uniform for $M \in [M_a, M_b]$. Since $\Phi_m \in L^1$ (same bounds), Lemma 7(b) gives the inversion formula for each such m , and the standard differentiation-under-the-integral theorem (pointwise differentiability in M plus an integrable uniform dominant for $\partial_M \Phi \cdot \sin t/t$) yields (c). $\square$

Sign structure, main term, tail

Lemma 20 (negativity on $[0, \frac{3}{5}]$, and the main term). *In the setting of Lemma 19, for $0 \leq t \leq \frac{3}{5}$ every factor of $\Phi_m(t)$ is positive and $-\partial_M \Phi_m(t) = \Phi_m(t) B(t) \geq 0$; since also $\sin t/t > 0$ there,*

$$-\delta'_q(m) \geq \underbrace{\frac{1}{\pi} \int_0^{1/8} \Phi_m(t) B(t) \frac{\sin t}{t} dt}_{=: \text{MAIN}} - \underbrace{\frac{1}{\pi} \int_{3/5}^\infty |\partial_M \Phi_m(t)| \frac{|\sin t|}{t} dt}_{=: \text{TAIL}}, \quad (28)$$

and, with $\beta := \frac{\sigma_m^2}{2} + \frac{S_4(m)}{1088}$,

$$\text{MAIN} \geq \frac{0.9973}{\pi} \cdot \frac{D(M)}{2} \left[\frac{\sqrt{\pi}}{4\beta^{3/2}} \operatorname{erf}\left(\frac{\sqrt{\beta}}{8}\right) - \frac{e^{-\beta/64}}{16\beta} \right]. \quad (29)$$

Proof. For $t \leq \frac{3}{5}$ all arguments satisfy $2a_\gamma t \leq 4t \leq \frac{12}{5} < j_{0,1}$ (Lemma 13(iii)), so every factor is positive and (26) applies with $B \geq 0$; the integrand of $-\delta'_q$ is therefore nonnegative on $[0, \frac{3}{5}]$ ($t < \pi$ keeps $\sin t/t > 0$), and discarding the part on $(\frac{1}{8}, \frac{3}{5}]$ and bounding the rest in absolute value gives (28).

On $[0, \frac{1}{8}]$: by (J3), $\log J_0(z) \geq -\frac{z^2}{4} - \frac{z^4}{17}$ for $z \leq \frac{1}{2}$, so

$$\Phi_m(t) \geq \exp\left(-\frac{\sigma_m^2 t^2}{2} - \frac{t^4 S_4}{17}\right) \geq e^{-\beta t^2},$$

using $t^4 \leq t^2/64$ and $\frac{1}{64 \cdot 17} \leq \frac{1}{1088}$. By Lemma 13(ii), $B(t) \geq \sum_\gamma a_\gamma t \cdot 2a'_\gamma t = \frac{D}{2}t^2$; and $\frac{\sin t}{t} \geq 1 - \frac{t^2}{6} \geq 1 - \frac{1}{384} > 0.9973$. Finally $\int_0^{1/8} t^2 e^{-\beta t^2} dt = \frac{\sqrt{\pi}}{4\beta^{3/2}} \operatorname{erf}\left(\frac{\sqrt{\beta}}{8}\right) - \frac{1}{16\beta} e^{-\beta/64}$ (integrate by parts), giving (29). $\square$

Lemma 21 (explicit tail bound). *In the setting of Lemma 19, let $S \subseteq \{\gamma \leq 2M_a\}$ be a set of $n \geq 12$ ordinates, let $\frac{3}{5} = t_0 < t_1 < \dots < t_K = 64$ be any partition, and for a cell $[t_j, t_{j+1}]$ and $\gamma' \in S$ set*

$$b_{\gamma'}^{(j)} := \min\left(1, \sup\left\{|J_0(z)| : 2a_{\gamma'}(M_a)t_j \leq z \leq 2a_{\gamma'}(M_b)t_{j+1}\right\}\right),$$

evaluated via Lemma 14(iv). Then, uniformly for $M \in [M_a, M_b]$,

$$\begin{aligned} \text{TAIL} &\leq \frac{1}{\pi} \sum_{j < K} \left[\frac{D_{\max} t_{j+1}^2}{2} + \frac{4t_{j+1}}{M_a} \nu_j \right] \min\left(1, \frac{1}{t_j}\right) \left(\prod_{\gamma' \in S} b_{\gamma'}^{(j)} \right) / \min_{\gamma' \in S} b_{\gamma'}^{(j)} (t_{j+1} - t_j) \\ &\quad + \frac{1}{\pi} \left(\prod_{i=2}^{n'+1} \kappa_{(i)} \right) \left[\frac{D_{\max}}{2} I_1 + \frac{4M_b}{M_a} (\ell_0 I_1 + I_2) \right], \end{aligned} \quad (30)$$

where $D_{\max}$ is the interval bound of Lemma 18(ii); ν_j is any upper bound for $N_\chi(M_b \sqrt{4t_{j+1}^2 - 1})$ (the certified list count when the threshold is below the list height, N^+ of Lemma 16 otherwise); $\kappa_{(1)} \leq \kappa_{(2)} \leq \dots$ are the sorted values of $\kappa_{\gamma'} = \frac{8}{\pi} (2a_{\gamma'}(M_a))^{-1/2}$, $n' = \min(n-1, 62)$; $\ell_0 = \log \frac{q(2M_b+2)}{2\pi} + 1.14$; and, with $p = \frac{n'}{2} - 1$,

$$I_1 = \frac{64^{1-p}}{p-1}, \quad I_2 = \frac{64^{1-p}((p-1)\log 64 + 1)}{(p-1)^2}.$$

Proof. By (27) with the given S : on a cell, every factor with index in $S \setminus \{\gamma\}$ is bounded by $b_{\gamma'}^{(j)}$ (the argument $2a_{\gamma'}(M)t$ is increasing in both M and t , hence lies in the stated range), and the product over $S \setminus \{\gamma\}$, whatever γ is, is at most the product over S divided by its smallest factor. The count in $A(t)$ uses $a_\gamma t > 1 \iff \gamma < M\sqrt{4t^2 - 1} \leq M_b\sqrt{4t_{j+1}^2 - 1}$, $D(M) \leq D_{\max}$, $t \leq t_{j+1}$, and $|\sin t|/t \leq \min(1, 1/t_j)$; the cell integral is at most the sup of the integrand times the cell length.

For $t \geq 64$: by (J2), $b_{\gamma'}(t) \leq \kappa_{\gamma'} t^{-1/2}$ with $\kappa_{\gamma'} \leq c_3 < 1.905$ (since $a_{\gamma'}(M_a) \geq \frac{2}{\sqrt{5}}$ for $\gamma' \leq 2M_a$), so each such bound is < 1 for $t \geq 64$. For arbitrary γ , the product over $S \setminus \{\gamma\}$ is at most the product over any n' -element subset $U \subseteq S \setminus \{\gamma\}$ of these bounds; choosing U among the $(n' + 1)$ smallest- κ indices and allowing for the worst case that γ is the smallest of them leaves $\prod_{i=2}^{n'+1} \kappa_{(i)} \cdot t^{-n'/2}$. With $A(t)/t \leq \frac{D_{\max}}{2}t + \frac{4}{M_a}N^+(2M_bt)$ and $N^+(2M_bt) \leq M_bt(\ell_0 + \log t)$ for $t \geq 1$ (since $2M_bt + 2 \leq (2M_b + 2)t$), the remainder is at most

$$\frac{1}{\pi} \prod_{i=2}^{n'+1} \kappa_{(i)} \int_{64}^{\infty} \left[\frac{D_{\max}}{2}t + \frac{4M_b}{M_a}t(\ell_0 + \log t) \right] t^{-n'/2} dt,$$

and $\int_{64}^{\infty} t^{-p} dt = I_1$, $\int_{64}^{\infty} t^{-p} \log t dt = I_2$ with $p = \frac{n'}{2} - 1 \geq 2$ evaluate to the stated closed forms. $\square$

Proof of Theorem 4

Proof. Take $M_1(\chi) \geq M_0(q)$ (Lemma 5) large enough that the window $(M, 2M]$ contains $N_0 \geq \frac{M}{4\pi} \log M \geq 14$ zeros; Lemma 19 then applies on a neighbourhood of any $M \geq M_1$ (its hypothesis needs only 8 zeros below $2M$), so δ'_q exists and (28) holds.

Main term. By (11) and Proposition 3, $S_4/\sigma_m^2 \rightarrow 16$, so $S_4 \leq 17\sigma_m^2$ for M large, whence $\beta \leq \frac{\sigma_m^2}{2}(1 + \frac{1}{32})$ and $\beta \rightarrow \infty$; the bracket in (29) is $\geq \frac{\sqrt{\pi}}{4}\beta^{-3/2}(1 - o(1))$ and

$$\text{MAIN} \geq (1 - o(1)) \frac{0.9973}{\pi} \cdot \frac{D}{2} \cdot \frac{\sqrt{\pi}}{4} \left(\frac{2}{\sigma_m^2} \right)^{3/2} \left(\frac{33}{32} \right)^{-3/2} \geq 0.189 (1 - o(1)) \frac{D}{\sigma_m^3},$$

since $\frac{1}{\pi} \cdot \frac{1}{2} \cdot \frac{\sqrt{\pi}}{4} \cdot 2^{3/2} = \frac{1}{2\sqrt{2\pi}} = 0.19947$ and $0.9973 \cdot 0.9548 \cdot 0.19947 > 0.189$.

Tail. For $t \geq \frac{3}{5}$, every window zero $\gamma \in (M, 2M]$ has $2a_\gamma t \geq \frac{4}{\sqrt{5}} \cdot \frac{3}{5} = \frac{12}{5\sqrt{5}} > 1.0733$, so by Lemma 14(iii) its factor is $\leq \rho_1 \leq 0.733$. Take S = the window in Lemma 21 (a one-cell version suffices): on $[\frac{3}{5}, 64]$ the product over S minus one factor is $\leq \rho_1^{N_0-1}$, and with $A(t) \leq \frac{D}{2}t^2 + \frac{4t}{M}N^+(2Mt) \ll_q t^2 \log(qMt)$,

$$\text{TAIL} \ll_q \rho_1^{N_0-1} \log M + c_3^{12} 2^{-(N_0-13)} \ll_q e^{-cM \log M}$$

with $c = \frac{\log(1/0.733)}{4\pi} - \varepsilon$, by $N_0 \geq \frac{M}{4\pi} \log M$ (the $t \geq 64$ remainder keeps 12 decaying factors and bounds the rest by $(c_3/\sqrt{t}) \leq \frac{1}{2}$, exactly as in Lemma 8). Since $\text{MAIN} \gg_q M^{-3/2}(\log M)^{-1/2}$ decays only polynomially, enlarging $M_1(\chi)$ makes $\text{TAIL} \leq 0.005 D\sigma_m^{-3}$ and the $o(1)$ above at most 0.02, and (28) gives $-\delta'_q \geq (0.189 \cdot 0.98 - 0.005) D\sigma_m^{-3} \geq 0.18 D\sigma_m^{-3} > 0$. All steps are effective: $M_1(\chi)$ is determined by $M_0(q)$ of Lemma 5 and the displayed comparisons.

The sandwich. For the upper bound write $-\delta'_q = \frac{1}{\pi}(\int_0^{1/8} + \int_{1/8}^{3/5})(-\partial_M \Phi_m) \frac{\sin t}{t} dt + O(\text{TAIL})$. On $[0, \frac{1}{8}]$ all arguments are $\leq \frac{1}{2}$, so $B(t) \leq 0.517 D t^2$ (Lemma 13(ii)) and, by (13)–(14), $\Phi_m(t) \leq e^{-(8/17)\sigma^2 t^2}$; hence

$$\frac{1}{\pi} \int_0^{1/8} \Phi_m B \frac{\sin t}{t} dt \leq \frac{0.517 D}{\pi} \int_0^{\infty} t^2 e^{-(8/17)\sigma^2 t^2} dt = \frac{0.517 D}{4\sqrt{\pi}} \left(\frac{17}{8\sigma^2} \right)^{3/2} \leq 0.226 \frac{D}{\sigma^3} = 1.133 \cdot \frac{D}{2\sqrt{2\pi} \sigma^3}.$$

On $[\frac{1}{8}, \frac{3}{5}]$ every window argument is $\geq \frac{4}{\sqrt{5}} \cdot \frac{1}{8} = 0.447$, so each window factor is $\leq J_0(0.447) \leq 0.9510$ (alternating series), and by (27) the integrand is $\leq A(t) \cdot 0.951^{N_0-1} \ll_q \log M \cdot 0.951^{N_0}$

— superexponentially small. Hence $-\delta'_q \leq 1.14 \frac{D}{2\sqrt{2\pi}\sigma^3}$ for $M \geq M_2(\chi)$. For the matching lower constant: once $\sigma_m^2 \geq 900$ the bracket of (29) is $\geq 0.996 \frac{\sqrt{\pi}}{4} \beta^{-3/2}$ ($\operatorname{erf}(\sqrt{\beta}/8) \geq \operatorname{erf}(2.65) \geq 0.9998$, and the subtracted term is below $0.003 \frac{\sqrt{\pi}}{4} \beta^{-3/2}$), so the main-term computation above gives $-\delta'_q \geq (0.9973 \cdot 0.9548 \cdot 0.996 - o(1)) \frac{D}{2\sqrt{2\pi}\sigma^3} - \text{TAIL} \geq 0.90 \frac{D}{2\sqrt{2\pi}\sigma^3}$ for $M \geq M_2(\chi)$. Finally the closed form $D = 2L_M + 2 + O(1/M)$ (Lemma 18(iii)) and $\sigma_m^2 = 2ML_M + O(1)$ give the last display of the theorem. $\square$

The explicit criterion and the two moduli

Proposition 5 (evaluated criterion, $q = 4$ and $q = 3$). *Let $q \in \{3, 4\}$ and let $[M_a, M_b]$ be an interval on which the right-hand side of (29) (evaluated with $D_{\min} = \max(D(M_a), D(M_b))(M_a/M_b)^3$, $\beta = \frac{\sigma^2(M_b)}{2} + \frac{S_4(M_b)}{1088}$, all closed forms) strictly exceeds the right-hand side of (30) (evaluated with $S = \{\gamma \leq 2M_a\}$ from the certified ordinate lists, the geometric partition with ratios 1.05 on $[\frac{3}{5}, 16]$ and 1.10 on $[16, 64]$, and per-zero cell bounds $b_{\gamma'}^{(j)}$ from Lemma 14(iv)). Then $\delta'_q(m) < 0$ for every $M \in [M_a, M_b]$. The script `paper/v3/verify_monotonicity.py` evaluates this criterion on a geometric grid of ratio 1.02 and finds that it holds on every subinterval of*

$$[17.317, 319.96] \quad (q = 4), \quad [19.119, 349.89] \quad (q = 3),$$

the right endpoints being $\gamma_{\max}/2$ for the respective lists; the worst relative margin $(\text{MAIN} - \text{TAIL})/\text{MAIN}$ on these ranges is 0.105 ($q = 4$, at the left edge) and 0.035 ($q = 3$, at the left edge); by $M \approx 35$ the tail is below 2×10^{-14} in absolute size, i.e. below 10^{-10} of the main term.

Proof. The claim of the first sentence is Lemmas 20 and 21 combined with (28), the interval monotonicities used being: $a_\gamma(M)$ increasing in M (so the cell argument ranges are correct), σ^2 and S_4 increasing in M (termwise), and the D comparison of Lemma 18(ii). The evaluation is the script's item [4]; its rigor class is discussed in Remark 12. $\square$

Lemma 22 (analytic leg: $M \geq 300$, no zero data). *Let $q \in \{3, 4\}$ and $M \geq 300$. With $\ell = L_M$, $Y_1 = M/\sqrt{3}$, $Y_2 = 2M/\sqrt{3}$,*

$$n_1 = \frac{Y_1(\ell_{Y_1} - 2)}{16} - \frac{1}{2}, \quad n_2 = \frac{Y_2(\ell_{Y_2} - 2)}{16} - \frac{1}{2}, \quad \ell_0 = \log \frac{q(2M+2)}{2\pi} + 1.14,$$

$D_\pm = 2\ell + 2 \pm \frac{8}{M}$, one has $N_\chi(M) \geq n_1$, $N_\chi(2M) \geq n_2$ (Lemma 17) and

$$\begin{aligned} \text{TAIL} \leq \text{TAIL}_{\text{an}}(M) &:= \frac{1}{0.447\pi} (64D_+ + 512\ell_0 + 1165) \left(\frac{0.447}{0.733}\right)^{n_1} (0.733)^{n_2} \\ &\quad + \frac{c_3^9}{\pi} \left(\frac{D_+}{2} I'_1 + 4(\ell_0 I'_1 + I'_2)\right) (0.733)^{n_2-10}, \end{aligned}$$

with $I'_1 = \frac{16^{-2.5}}{2.5}$, $I'_2 = \frac{16^{-2.5}(2.5 \log 16 + 1)}{6.25}$, while

$$\text{MAIN} \geq \text{MAIN}_{\text{an}}(M) := \frac{0.9973}{\pi} \cdot \frac{D_-}{2} \left[\frac{\sqrt{\pi}}{4\beta_+^{3/2}} \operatorname{erf}\left(\frac{\sqrt{\beta_+}}{8}\right) - \frac{e^{-\beta_+/64}}{16\beta_+} \right], \quad \beta_+ = M\ell + 1.55 + \frac{32M(\ell - 1) + 100}{1088}.$$

Then $\text{TAIL}_{\text{an}}(300) < \text{MAIN}_{\text{an}}(300)$ for both moduli (ratios $< 5 \times 10^{-4}$ for $q = 4$ and $< 3 \times 10^{-3}$ for $q = 3$; the script's item [5], which further weakens n_i to $n_i - \frac{9}{2}$, still finds 1.9×10^{-3} resp.

1.2×10^{-2}), and on $[300, \infty)$ the two summands of $\text{TAIL}_{\text{an}}/\text{MAIN}_{\text{an}}$ have logarithmic derivatives ≤ -0.14 and ≤ -0.08 respectively, so the ratio is decreasing there; hence $\delta'_q(m) < 0$ for all $M \geq 300$.

Proof. The tail bound is Lemma 21 degraded to zone form. On $[\frac{3}{5}, 16]$: all $N_\chi(2M)$ zeros below $2M$ have $2a_\gamma t \geq \frac{12}{5\sqrt{5}} > 1.0733$, factor ≤ 0.733 ; those below M have $a_\gamma \geq \sqrt{2}$, hence arguments $\geq 2\sqrt{2}t \geq 2\sqrt{2} \cdot \frac{3}{5} > 1.697$, and $\sup_{z \geq 1.697} |J_0| \leq \max(J_0(1.697), 0.446) \leq 0.447$ by Lemma 14(iii), since the alternating series gives $J_0(1.697) \leq 1 - \frac{(1.697)^2}{4} + \frac{(1.697)^4}{64} \leq 0.410$. The product over all zeros $\leq 2M$ except one is then at most $0.447^{N_\chi(M)} 0.733^{N_\chi(2M) - N_\chi(M)} / 0.447 \leq (0.447/0.733)^{n_1} 0.733^{n_2} / 0.447$, decreasing in the true counts, so the lower bounds n_1, n_2 may be inserted. The A -integral: by (27) and Lemma 16, $\int_{3/5}^{16} A(t) \min(1, \frac{1}{t}) dt \leq \int_{3/5}^{16} [\frac{D_+}{2} t + \frac{4}{M} N^+(2Mt)] dt \leq 64D_+ + 512\ell_0 + 1165$, using $N^+(2Mt) \leq Mt(\ell_0 - 1.14) + 1.14Mt = Mt\ell_0$ for $t \leq 1$ and $N^+(2Mt) \leq Mt(\ell_0 + \log t)$ for $t \geq 1$, plus $\int_{3/5}^{16} t dt \leq 127.9$ and $\int_1^{16} t \log t dt \leq 291.2$. For $t \geq 16$: any 9 of the zeros below $2M$ carry $(c_3/\sqrt{t}) \leq 0.48$, one factor is dropped, the remaining $\geq n_2 - 10$ keep 0.733; the integral closed forms are as in Lemma 21 with $n' = 9$, $p = 3.5$, $T = 16$. The main-term bound is (29) with Lemma 18(iii) ($D \geq D_-$, $\sigma^2 \leq 2M\ell + 3.1$, $S_4 \leq 32M(\ell - 1) + 100$).

The two evaluations at $M = 300$ are the script's item [5] (which further weakens n_i to $n_i - \frac{9}{2}$; the ratios quoted in the statement for the formulas as displayed are smaller). For the log-derivative, $\frac{d}{dM} n_i = \frac{\ell_{Y_i} - 1}{16\sqrt{3}/i}$, and the two summands of TAIL_{an} must be treated separately, since the second — which carries only the factor $(0.733)^{n_2 - 10}$ — dominates the first by three to four orders of magnitude at $M = 300$. For the first summand,

$$\frac{d}{dM} \log(\text{first}) \leq -\log \frac{0.733}{0.447} \cdot \frac{\ell_{Y_1} - 1}{27.8} - \log \frac{1}{0.733} \cdot \frac{\ell_{Y_2} - 1}{13.9} + \frac{640}{M(64D_+ + 512\ell_0 + 1165)} \leq -0.15$$

for $M \geq 300$, $q \geq 3$ (then $\ell_{Y_1} \geq 4.4$, $\ell_{Y_2} \geq 5.1$, and both ℓ_{Y_i} increase in M and q). For the second summand the prefactor grows at rate $\frac{d}{dM} \log(\frac{D_+}{2} I'_1 + 4(\ell_0 I'_1 + I'_2)) \leq \frac{5}{M(D_+/2 + 4\ell_0)} \leq 0.001$ (using $D'_+ \leq 2/M$, $\ell'_0 \leq 1/M$), so

$$\frac{d}{dM} \log(\text{second}) \leq -\log \frac{1}{0.733} \cdot \frac{\ell_{Y_2} - 1}{13.9} + 0.001 \leq -0.09.$$

Meanwhile $\frac{d}{dM} \log \text{MAIN}_{\text{an}} \geq -\frac{3}{2} \cdot \frac{d\beta_+/dM}{\beta_+} \cdot (1 - \frac{0.003}{1})^{-1} \geq -\frac{3}{2} \cdot \frac{1.03\ell+1}{M\ell} \cdot 1.004 \geq -0.007$ (D_- is increasing, the erf factor and the subtracted term of the bracket vary favourably, and the subtracted term is at most 0.003 of the first, as in the proof of Theorem 4). Hence each summand's ratio to MAIN_{an} has logarithmic derivative $\leq -0.15 + 0.007 = -0.143 \leq -0.14$ resp. $\leq -0.09 + 0.007 \leq -0.08$; both ratios decrease on $[300, \infty)$, so their sum $\text{TAIL}_{\text{an}}/\text{MAIN}_{\text{an}}$ stays below its value at $M = 300$, which is below 1. $\square$

Proof of Corollary 7. For $q = 4$: Proposition 5 gives $\delta'_4 < 0$ on $17.317 \leq M \leq 319.96$; Lemma 22 gives it on $M \geq 300$; the union is $M \geq 17.317$, i.e. $m \geq 16.817$, and we round the threshold up to 16.82. For $q = 3$ the same with 19.119 and 349.89, threshold 18.62. Strict negativity of the derivative on an interval implies strict decrease. The transfer to the race under $\text{LI}(\chi_4)$ is Theorem 1(ii) as in Corollary 3: under LI the logarithmic density $\delta(m)$ equals $\delta_4(m)$ for every m , so monotonicity of the latter is monotonicity of the former. $\square$

Remark 12 (rigor classes, and what is computer-verified). *Three classes of statement coexist above, and we distinguish them explicitly. (1) Theorem 4, Lemmas 13–21 and Lemma 22 are proved; their numerical constants (0.446, 0.733, $\frac{1}{90}$, 0.517, $j_{0,1} > 2.4$, the Binet constants, 0.57, 0.89, the closed-form integrals) are established by alternating-series or integral arguments in the proofs, and the script re-verifies each (items [1]–[3], [5]). In particular the range $M \geq 300$ of Corollary 7 uses no zero data and no floating-point evaluation beyond checking finitely many explicit elementary inequalities at $M = 300$ (the two evaluations $\text{TAIL}_{\text{an}}(300) < \text{MAIN}_{\text{an}}(300)$ and the four logarithmic-derivative constants of Lemma 22), which can be done by hand. (2) The range $[17.32, 320]$ resp. $[19.12, 350]$ rests on Proposition 5: the criterion itself is a theorem, but its evaluation uses (a) the certified ordinate lists (25 digits; their assumed completeness below $\gamma_{\max}$ is the same hypothesis underlying the reference densities and Table 5), and (b) float64 evaluation of J_0 , erf, digamma and the von-Mangoldt sums, with an additive safety margin 10^{-9} on every Bessel supremum and a worst-case criterion margin of 3.5%; no interval arithmetic is claimed. This is the rigor class of the paper’s validated enclosures, not of its theorems. (3) The finite differences of item [6] of the script (matching $-D/(2\sqrt{2\pi}\sigma^3)$ to 1–2% at $m \in \{20, 60\}$ for both moduli) are a consistency check only. Two further honesty notes. First, the constant 0.18 of Theorem 4 and the window $[0.90, 1.14]$ are far from optimal; the truth is $-\delta' \sim D/(2\sqrt{2\pi}\sigma^3)$, i.e. $\theta \rightarrow 1$, as the second-order machinery of Theorem 2(ii) would show if pushed through the differentiated integral — we have not done so. Second, nothing here says anything about $(-\frac{1}{2}, m_1)$: the positivity structure of Lemma 20 holds there too, but the tail of the differentiated integral is no longer dominated by our bounds when fewer than a dozen zeros sit below $2M$; on that range the monotonicity remains Conjecture 1, supported by the validated grid (whose points $m = 15$ and $m = 20$ bracket both thresholds, with decreasing, pairwise disjoint enclosures across them).*

Remark 13 (what the theorem adds to the dissolution law). *Theorem 2 says $\delta_q(m) \rightarrow \frac{1}{2}$ at the rate $(2\pi\sigma_m^2)^{-1/2}$; it does not exclude oscillation around that envelope. Theorem 4 and its corollary exclude it: the approach to $\frac{1}{2}$ is eventually monotone — ultimately because each Bessel factor of the characteristic function loses mass at every frequency simultaneously as the weight grows ($\partial_M \log J_0(2a_\gamma t) = -\varphi(2a_\gamma t) 2a'_\gamma t < 0$ for $0 < 4t < j_{0,1}$), a differentiated form of the peakedness heuristic of Section 2 that, unlike Birnbaum–Proschan peakedness, survives the arcsine components’ bimodality.*

7 Methods and verification

Zeros of $L(s, \chi_4)$ and $L(s, \chi_3)$ (511 resp. 537 ordinates to 25 digits) were computed by Hardy Z-scans with derived phases, validated by reality of Z to $\sim 10^{-27}$, Riemann–von Mangoldt counts, and the known first ordinates.

Validated enclosures for the densities

The densities of Table 1 rest on a tail model: the unseen zeros $\gamma > \Gamma$ ($\Gamma = \gamma_{511} = 639.9157\dots$ for χ_4 , $\gamma_{537} = 699.7747\dots$ for χ_3) are replaced by a Gaussian whose variance was originally *estimated* from the zero-counting density. This subsection replaces the estimate by the exact closed form of Proposition 2 and derives validated enclosures for the densities — explicit analytic error bounds throughout, with quadrature error estimated rather than certified —

stating precisely which terms are proved and which remain estimated; Table 1 prints the enclosure midpoints. Throughout, the standing hypotheses are $\text{GRH}(\chi) + \text{LI}(\chi)$ (as for every density in this literature) together with the completeness of the computed ordinate list up to Γ (validated by the Hardy Z -scans and Riemann–von Mangoldt counts described above; the enclosure script repeats a smoothed count check, which matches to 0.6 resp. 1.3 of a zero, within the $S(T)$ fluctuation).

Under LI , $\delta_q(m) = \mathbb{P}(1 + V > 0)$ with $V = \sum_{\gamma > 0} 2a_\gamma(m) \cos \theta_\gamma$, whose characteristic function factors over the computed and unseen ordinates,

$$\Phi(t) = \prod_{\gamma > 0} J_0(2a_\gamma t) = P(t) \Psi(t), \quad P = \prod_{\gamma \leq \Gamma}, \quad \Psi = \prod_{\gamma > \Gamma}.$$

Since $|J_0(x)| \leq \min\{1, \sqrt{2/(\pi x)}\}$ for $x > 0$ (the classical envelope; the extremal value $\sup_{x > 0} \sqrt{x} |J_0(x)| = \sqrt{2/\pi} = 0.79788\dots$ is re-verified numerically in the script, which uses the rounded-up constant 0.8), Φ is integrable, V has a continuous density, and the Gil–Pelaez inversion is an identity:

$$\delta_q(m) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(t) \frac{\sin t}{t} dt. \quad (31)$$

The enclosure controls the three gaps between (31) and what a finite computation can evaluate.

(i) *The tail variance is certified, not estimated.* By Proposition 2,

$$V_T := \sum_{\gamma > \Gamma} 2a_\gamma(m)^2 = 4M \frac{\Lambda'}{\Lambda}(m+1) - \sum_{\gamma \leq \Gamma} 2a_\gamma(m)^2,$$

with no reference to the zero-counting density. The subtracted finite sum inherits the 25-digit accuracy of the ordinates (effect $< 10^{-20}$); the closed form is evaluated by `mpmath` at 40 digits through two independent routes that must agree to 25 digits, anchored at $L(2, \chi_4) = G$ (Catalan), at a directly summed series for $L(2, \chi_3)$, at $(L'/L)(1, \chi_q)$ expressed through generalized Stieltjes constants, and at the closed form for σ_0^2 . This is high-precision floating point, *not* interval arithmetic; we budget a defensive slack of 10^{-15} for it, absorbed into the error radius below. The script aborts if any $V_T \leq 0$; in fact V_T ranges from 0.0030943 ($q = 3$, $m = 0$) to 139.71 ($q = 4$, $m = 100$); see Table 5.

(ii) *The unseen product is a Gaussian with a proved quartic correction.* Since $a_\gamma = 2M/\sqrt{M^2 + \gamma^2}$ is strictly decreasing in γ , every unseen ordinate satisfies $2a_\gamma t \leq 2a_\Gamma t \leq 1$ whenever $t \leq t_{\max} := 1/(2a_\Gamma)$. On $|z| \leq 1$ we use

$$0 \leq -\log J_0(z) - \frac{z^2}{4} \leq c_4 z^4, \quad c_4 = -\log J_0(1) - \frac{1}{4} = 0.0176210647\dots, \quad (32)$$

with the *sharp* constant: from the product $J_0(z) = \prod_n (1 - z^2/j_{0,n}^2)$ over the Bessel zeros, $-\log J_0(z) = \sum_{k \geq 1} (\sigma_k/k) z^{2k}$ with Rayleigh sums $\sigma_k = \sum_n j_{0,n}^{-2k} > 0$, valid for $|z| < j_{0,1} = 2.40\dots$; $\sigma_1 = \frac{1}{4}$ (Rayleigh), and $z^{2k} \leq z^4$ for $k \geq 2$, $|z| \leq 1$, so the middle member of (32) equals $\sum_{k \geq 2} (\sigma_k/k) z^{2k} \in [0, c_4 z^4]$ with $c_4 = \sum_{k \geq 2} \sigma_k/k$, attained at $z = 1$. (The script uses the rounded-up $c'_4 = 0.018$ and re-verifies (32) on a grid.) Summing over the unseen zeros: for $0 \leq t \leq t_{\max}$,

$$\Psi(t) = \exp\left(-\frac{1}{2} V_T t^2 + \varepsilon(t)\right), \quad -c'_4 S_4 t^4 \leq \varepsilon(t) \leq 0, \quad S_4 := \sum_{\gamma > \Gamma} (2a_\gamma)^4 \leq (2a_\Gamma)^2 \cdot 2V_T,$$

the last bound because $(2a_\gamma)^2 \leq (2a_\Gamma)^2$ beyond Γ and $\sum_{\gamma > \Gamma} (2a_\gamma)^2 = 2V_T$. The sign information is exploited: the Gaussian model *overstates* $|\Psi|$, and with $w(t) := c'_4 S_4 t^4$ the factor $e^{\varepsilon(t)} \in [e^{-w}, 1]$ is enclosed by its midpoint,

$$\left| e^{\varepsilon(t)} - \frac{1+e^{-w(t)}}{2} \right| \leq \frac{1-e^{-w(t)}}{2},$$

which halves the tail-model radius relative to the symmetric bound $e^w - 1$.

(iii) *Enclosure*. With $G(t) := e^{-V_T t^2/2}$ and a truncation point $T_1 \leq t_{\max}$,

$$\delta_q(m) = \frac{1}{2} + \frac{1}{\pi} \underbrace{\left(\int_0^{T_1} P G \frac{1+e^{-w}}{2} \frac{\sin t}{t} dt \right)}_{I, \text{ computed}} + R, \quad |R| \leq E_1 + E_2,$$

$$E_1 = \int_0^{T_1} |P| G \frac{1-e^{-w}}{2} dt \quad (\text{tail-model radius, from the midpoint enclosure of } e^\varepsilon),$$

$$E_2 = \frac{(0.8)^K}{\sqrt{\prod_{i \leq K} 2a_{\gamma_i}}} \cdot \frac{T_1^{-K/2}}{K/2} \geq \left| \int_{T_1}^\infty \Phi \frac{\sin t}{t} dt \right| \quad (\text{truncation; envelope on the first } K \text{ ordinates},$$

all other factors bounded by 1, $|\sin t/t| \leq 1/t$). The script chooses $K \in \{24, \dots, 72\}$ and $T_1 \leq t_{\max}$ so that $E_2 \leq \pi \times 10^{-15}$; this holds at every (q, m) computed. What remains *estimated* rather than certified, stated openly: the quadrature error of I and of E_1 (composite Simpson on 2^{17} subintervals in IEEE double precision; a step-doubling Richardson comparison is below 10^{-12} at every (q, m) , and is budgeted as $E_3 = 10^{-12}$; E_1 , itself a bound of size $\lesssim 5 \times 10^{-7}$, is inflated by 50% against its own quadrature error), the double-precision rounding of the 511-fold product ($\lesssim 10^{-13}$ relative), and the 10^{-15} variance slack of (i), together budgeted as $E_4 = 2 \times 10^{-11}$ plus the computed slack sensitivity. The radius of the quoted intervals is $(E_1 + E_2 + E_3 + E_4)/\pi$, dominated everywhere by the proved bound E_1 . These enclosures are rigorous *modulo* exactly the items just listed (double-precision quadrature with Richardson control and the anchored 40-digit evaluation of σ_m^2); no interval arithmetic was used, and we do not claim otherwise.

Enclosure-derived values. Table 1 prints the midpoints of these enclosures rounded to six decimals; ten of the fourteen sixth decimals differ, by one to three units, from the earlier Gaussianized-tail computation (last column of Table 5), a drift quantitatively traced to the estimated tail variance that computation used: re-running the pipeline with the old smoothed-tail variance in place of V_T reproduces the deviations (e.g. -3.2×10^{-6} of the observed -3.5×10^{-6} at $(3, 1)$, the small remainder being the older code's coarser quadrature). Two sixth decimals are *not resolved* by the enclosures and are daggered in Table 1: at $(3, 1)$ the interval $[0.83688043, 0.83688053]$ straddles the rounding boundary 0.8368805 (the value is 0.836880 or 0.836881; the midpoint 0.83688048 sits within 2.5×10^{-9} of the boundary, so no computation at this zero height can be expected to settle it), and at $(3, 60)$ the interval $[0.51971743, 0.51971753]$ straddles 0.5197175 (the value is 0.519717 or 0.519718). The twelve remaining sixth decimals, including $(4, 100)$, are determined by the enclosures. The classical anchors are unaffected: our intervals contain 0.9959279... and 0.9990632... at $m = 0$, consistent with $[1, 9]$ to all six printed decimals.

The two computations, honestly described. The deterministic Gil–Pelaez inversion and the Monte Carlo evaluation of Section 7 share their inputs: the identical 511- resp. 537-ordinate

q	m	V_T	$\delta_q(m)$ lower	$\delta_q(m)$ upper	width	old – mid
4	0	0.0034836	0.9959278380	0.9959280127	1.7×10^{-7}	+0.07
4	1	0.0313527	0.7976303685	0.7976304296	6.1×10^{-8}	–2.40
4	2	0.0870906	0.6945210810	0.6945211213	4.0×10^{-8}	–1.10
4	3	0.1706968	0.6458510415	0.6458510746	3.3×10^{-8}	–1.06
4	8	1.0067182	0.5718700860	0.5718701185	3.3×10^{-8}	–1.10
4	20	5.8541890	0.5384552785	0.5384553290	5.1×10^{-8}	–0.30
4	100	139.71245	0.5137782134	0.5137783837	1.7×10^{-7}	–0.30
3	0	0.0030943	0.9990630500	0.9990633364	2.9×10^{-7}	–0.19
3	1	0.0278484	0.8368804319	0.8368805308	9.9×10^{-8}	–3.48
3	2	0.0773564	0.7234728791	0.7234729425	6.3×10^{-8}	–2.91
3	3	0.1516181	0.6665542399	0.6665542896	5.0×10^{-8}	–2.26
3	8	0.8942043	0.5785355754	0.5785356134	3.8×10^{-8}	–1.59
3	20	5.2001295	0.5407681867	0.5407682369	5.0×10^{-8}	–1.21
3	60	45.202127	0.5197174290	0.5197175306	1.0×10^{-7}	–1.48

Table 5: Validated enclosures $[\delta_{\text{lo}}, \delta_{\text{hi}}]$ for the logarithmic densities — explicit analytic error bounds, with quadrature error estimated by Richardson step-doubling rather than certified — under GRH + LI and completeness of the ordinate lists to height Γ (511 ordinates for χ_4 , 537 for χ_3). V_T is the certified variance of the unseen-zero tail. The last column (“old – mid”) is the deviation of the earlier Gaussianized-tail six-decimal value from the interval midpoint, in units of 10^{-6} ; entries exceeding 1 in absolute value mark a changed sixth decimal (ten of the fourteen). Table 1 prints the midpoints. All widths are below 3×10^{-7} , against the design target 5×10^{-6} .

lists and (in their original form) the identical Gaussianized tail. Their mutual agreement (largest deviation 5.5×10^{-4} , i.e. 2.2 binomial standard errors at the actual sample counts of Table 1) therefore validated only the *evaluation* step — quadrature against sampling — and could not have detected an error in the shared tail model. They are independent as implementations, not as models. The enclosures of this subsection close that gap from the model side: the tail variance is now the certified V_T of (i), the non-Gaussianity of the tail is bounded by the proved E_1 , and a Monte Carlo rerun with the certified V_T in the tail agrees with the enclosure midpoints within half a standard error at each tested point (2×10^6 samples at $(q, m) = (4, 0), (4, 1), (3, 1)$).

General real primitive modulus. Theorem 1 is stated and proved in Section 2 for an arbitrary real primitive character $\chi \bmod q$ with $\chi(-1) = (-1)^a$, $a \in \{0, 1\}$, under GRH(χ), LI(χ), and the no-central-zero hypothesis

$$\text{NCZ}(\chi) : \quad L(\tfrac{1}{2}, \chi) \neq 0$$

(there $R_m(x) = -\sum_{p \leq x} \chi(p)p^m \log p$, nonresidues minus residues; $E_m = (2m+1)x^{-(m+1/2)}R_m$; $a_\gamma(m) = 2M/\sqrt{M^2 + \gamma^2}$ over the positive ordinates of $L(s, \chi)$; and parts (i)–(iii) hold with $C(\chi) = \sum_{\gamma_\chi > 0} 2\gamma_\chi^{-2}$). Here we record what the generalization changes relative to the mod-4

case, and the general form of the variance identity: with $\xi(s, \chi) = (q/\pi)^{(s+a)/2} \Gamma(\frac{s+a}{2}) L(s, \chi)$,

$$\sigma_m^2(\chi) = 4M \frac{\xi'}{\xi}(m+1, \chi), \quad C(\chi) = (\log \xi)''(\tfrac{1}{2}, \chi).$$

What changes and what does not. *Unchanged:* the prime-square bias term — the $k = 2$ prime powers contribute $\sum_{p \leq \sqrt{x}, p \nmid q} p^{2m} \log p = x^{m+1/2}/(2m+1) + o(x^{m+1/2})$ because $\chi(p)^2 = 1$ for every $p \nmid q$, and the at most $\omega(q)$ primes $p \mid q$ contribute $O_{q,m}(1)$ in total (each term is $p^{2m} \log p \leq (1+q^{2m}) \log q$, a constant; see the proof of Lemma 3), so the normalized mean is 1 for every real χ ; the Chernoff/Chebyshev bounds of (iii), which use only the arcsine moment generating function. *Changed:* the gamma factor and conductor in ξ (parity a enters the variance asymptotic only through the constant term $2a-1$ of Proposition 3); all implied constants of the explicit-formula lemmas, which now depend on q , and the first-ordinate constants ($|\rho+m| \geq \gamma_1(\chi)$ in Lemma 2; numerically $\gamma_1(\chi_4) = 6.0209\dots$, $\gamma_1(\chi_3) = 8.0397\dots$, and $\gamma_1(\chi) > 0$ is exactly NCZ under GRH); for $a = 0$ the pole of $1/s$ in the Perron contour meets the trivial zero of $L(s, \chi)$ at $s = 0$, and the classical formula carries an extra $-\log x + b(\chi)$ term [17, Ch. 19], absorbed by the $O(\log x)$ floor of (2); and the functional equation input, $\xi(s, \chi) = \xi(1-s, \chi)$, i.e. root number $+1$, which holds for every real primitive χ by the evaluation of the sign of the quadratic Gauss sum ($\tau(\chi) = \sqrt{q}$ for $a = 0$, $i\sqrt{q}$ for $a = 1$; [17, Ch. 2] for prime modulus, the general case reducing to it via the factorization of Gauss sums over coprime moduli — for the moduli 3, 4 used here the two-term sums are evaluated directly in the verification script). On the no-central-zero hypothesis itself: for χ_4 it is the alternating series $L(\frac{1}{2}, \chi_4) = \sum_k (-1)^k (2k+1)^{-1/2} > 0$; for χ_3 it is *not* an alternating-series statement (the zero terms $\chi_3(3k) = 0$ interleave), but it is still unconditional: the Dirichlet series $\sum \chi_3(n) n^{-1/2}$ converges by Dirichlet's test (bounded character sums) and equals $L(\frac{1}{2}, \chi_3)$ by analytic continuation, and regrouping the convergent series in blocks of three gives

$$L(\tfrac{1}{2}, \chi_3) = \sum_{k \geq 0} \left[\frac{1}{\sqrt{3k+1}} - \frac{1}{\sqrt{3k+2}} \right] \geq 1 - \frac{1}{\sqrt{2}} > 0$$

(every block positive; numerically $0.4808675\dots$). For general q we do not prove NCZ: under GRH one has $L(\frac{1}{2}, \chi) \geq 0$ (a negative central value would force a real zero in $(\frac{1}{2}, 1)$), so NCZ is then exactly central nonvanishing, which is Chowla's conjecture for quadratic characters — known in every numerically checked case but open in general; it must be assumed, or verified computationally for the modulus at hand, as part of the hypotheses. Nothing in this paper uses NCZ beyond $q \in \{3, 4\}$, where it is proved above. For the record, the certified constants at the two moduli used here are

$$\begin{aligned} C(\chi_4) &= 0.15602964988964488\dots, & C(\chi_3) &= 0.11340215685275857\dots, \\ \sigma_0^2(\chi_4) &= 0.15556797992358593\dots, & \sigma_0^2(\chi_3) &= 0.11322996985747234\dots \end{aligned}$$

Machine certification in ball arithmetic. The analytic inputs of the enclosures have additionally been machine-certified in ball arithmetic (FLINT/Arb via python-flint; script `certify_arb.py`). Certified as rigorous enclosures: the constant $C(\chi_4) = 0.156029649889644884607682283389 \pm 8.4 \times 10^{-32}$ (via rigorous Cauchy-integral evaluation of $(\log L)''(\frac{1}{2})$); all fourteen variances σ_m^2 (maximum ball radius 4.3×10^{-28}); $L(\frac{1}{2}, \chi_4) > 0$ and $L(\frac{1}{2}, \chi_3) > 0$; a zero of the completed L -function within 10^{-20} of every listed ordinate (511 for χ_4 , 537 for χ_3 , worst certified Hardy-Z

radius 1.2×10^{-55}); and, by rigorous winding-number contour counts, that the lists are *complete* up to height 639.5622 for χ_4 (510 of the 511 listed ordinates; the 511th is certified to exist but lies above the certified-completeness height) and up to 699.4645 for χ_3 (536 of 537). The final Gil–Pelaez quadrature remains floating-point with estimated error, as stated above.

Exact races: segmented sieves to 10^{10} with exact signed 128-bit (integer weights) or compensated long-double (fractional weights) accumulation, cross-checked against independent implementations at overlapping ranges; the $m = 0$ anchors (first flip 26,861; no mod-3 flip below 10^{10}) reproduce the classical values. The finite- x logarithmic measure of a set S reported in the introduction is $(\log(X/3))^{-1} \int_3^X \mathbf{1}_S(t) dt/t$ for modulus 4 (the race is tracked from the first odd prime) and $(\log(X/2))^{-1} \int_2^X \mathbf{1}_S(t) dt/t$ for modulus 3 (the race starts at $p = 2$); intervals on which $D_m = 0$ are assigned to the previous leader. The convention is not cosmetic: at $X = 10^{10}$ the choice of lower endpoint alone shifts the normalization by $\log 3 / \log X \approx 4.8\%$. The density kernel is derived for the log-weighted (θ -form) race; partial summation gives every term of the unlogged race the same $1/\log x$ factor, which cancels under the bias normalization, so the limiting distribution and density coincide — the same passage Rubinstein–Sarnak make at $m = 0$. The explicit-formula kernel derives from a uniform smoothed-Perron framework developed in a companion manuscript (analytic computation of $S_m(x)$ from zeros, exact to 10^9 at $m = 1$), available with all code and data at <https://github.com/raghavs1729/weighted-prime-races> (archived at doi:10.5281/zenodo.21947102). Every headline number in this note was re-computed by a second implementation. For the sieve computations the two implementations are independent throughout; for the densities they shared the ordinate lists and, prior to the enclosures above, the Gaussian tail model, so their agreement tested evaluation error only — the tail model itself is now controlled by the validated enclosures of Table 5 (its variance exactly, by Proposition 2; its non-Gaussianity by the proved bound E_1).

Assumptions and limitations. Densities are conditional on GRH + LI, as throughout this literature. The existence of the limiting distribution, the continuity of $m \mapsto \delta(m)$, and the strict bounds $\frac{1}{2} < \delta(m) < 1$ are proved for all $m \in (-\frac{1}{2}, \infty)$ in Section 2 (the $m > 0$ part is the routine direction of the extension: the only changes are the $(1 + x^m) \log x$ error floor in the explicit formula and the constant $2 \max K + 1$ in the continuity argument). The dissolution law, stated in the introduction as a Gaussian-comparison prediction, is proved with an error term in Theorem 2 (for the density of the limiting variable; the transfer to the race itself uses LI(χ) and holds for every real primitive χ via Theorem 1(ii), Corollary 3 recording the case $q = 4$). The finite- x log-measure for $(q, m) = (3, 2)$ shows the weakest agreement in the study (0.023 at 10^{10}), plausibly a slow almost-periodic phase; we flag it rather than explain it.

Outlook: growing weights and short intervals. Everything in this note keeps m fixed as $x \rightarrow \infty$. In the joint limit $m = m(x) \rightarrow \infty$ the weight localizes the race: writing $p = x(1 - u)$, the weight $p^m = x^m e^{m \log(1-u)}$ is within a bounded factor of its maximum only for $u \ll 1/m$, so the fluctuating part of $\sum_{p \leq x} \chi(p) p^m \log p$ is dominated by the primes in the short window $x - p \ll x/m$. A growing-weight race is in this sense a smoothed race between primes in a short interval at the edge of the range, and the weight aspect should connect to the extensively studied statistics of primes in short intervals — in particular to the Gaussian limit laws, under RH and a quantitative form of LI, of Leung [22] for primes in multiple short intervals, which include short-interval race applications. We state this only as a direction: we prove nothing in any joint regime $m = m(x) \rightarrow \infty$, and even the existence of a limiting distribution there is open.

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